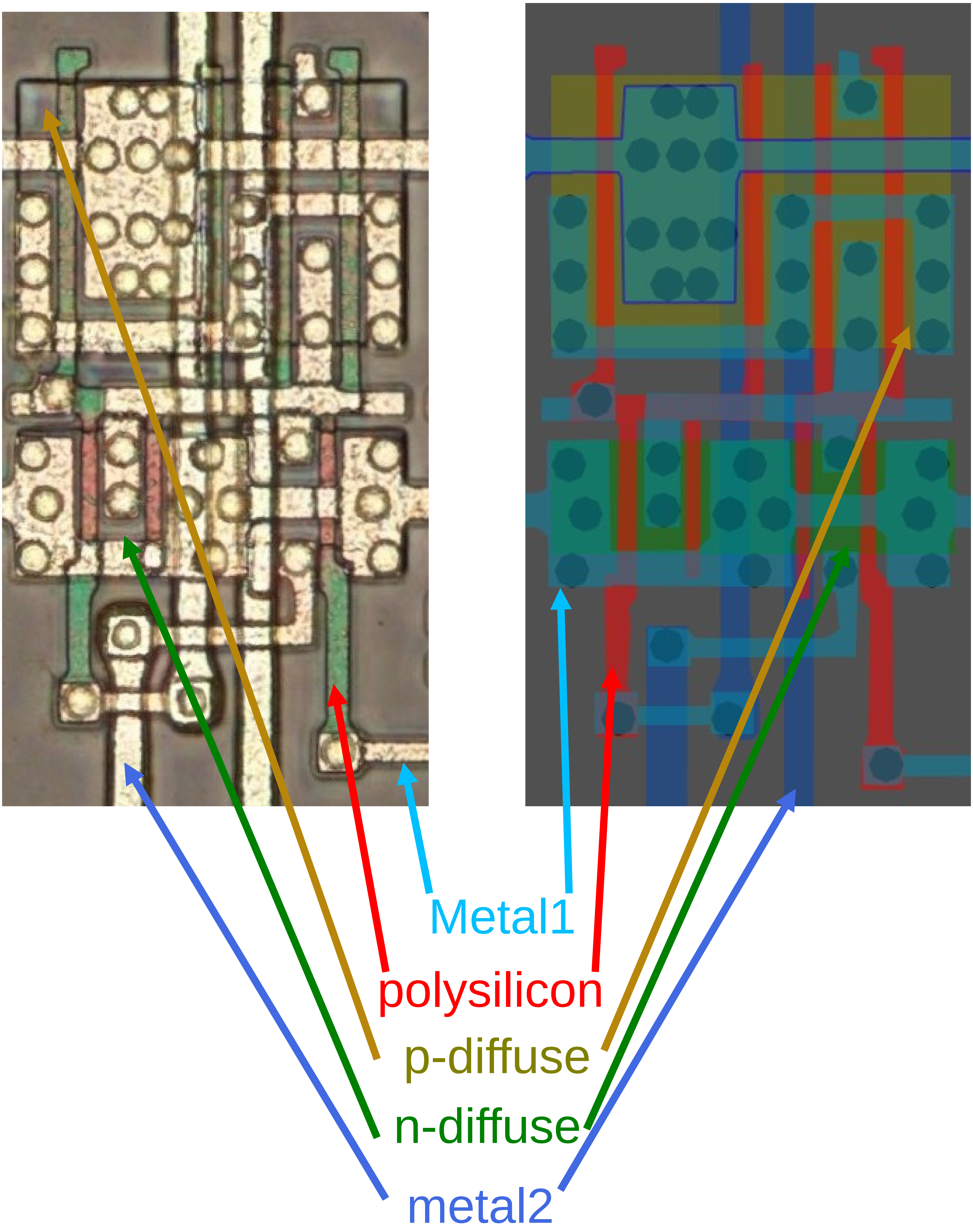
Document subject to change (for now)



Inferred rules for this 1986/87, 1.8 micron standard cell process called ‘AV-OEBB’

(It is entirely possible the node size was something other than 1.8 micron)

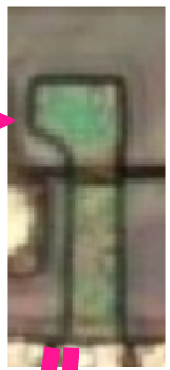
AV-OEBB Series **1.8u CMOS standard cell**

Fujitsu Micro-electronics	AV-OEBB Si-gate CMOS 1.8 µm	P1 6 µm M1 7 µm M2 9 µm	100	n/s	1.2	140 digital	RAM, ROM, PLA, multipliers, ALU	• •	• • •	None

- Cells can be flipped and mirrored with minimal adjustments.
- Gate height can extend up/down y-axis for the purposes of routing.
- Cell via routing seems unique to each cell type. Good for finding them easily
- These polysilicon ends are where m1 *can be* connected

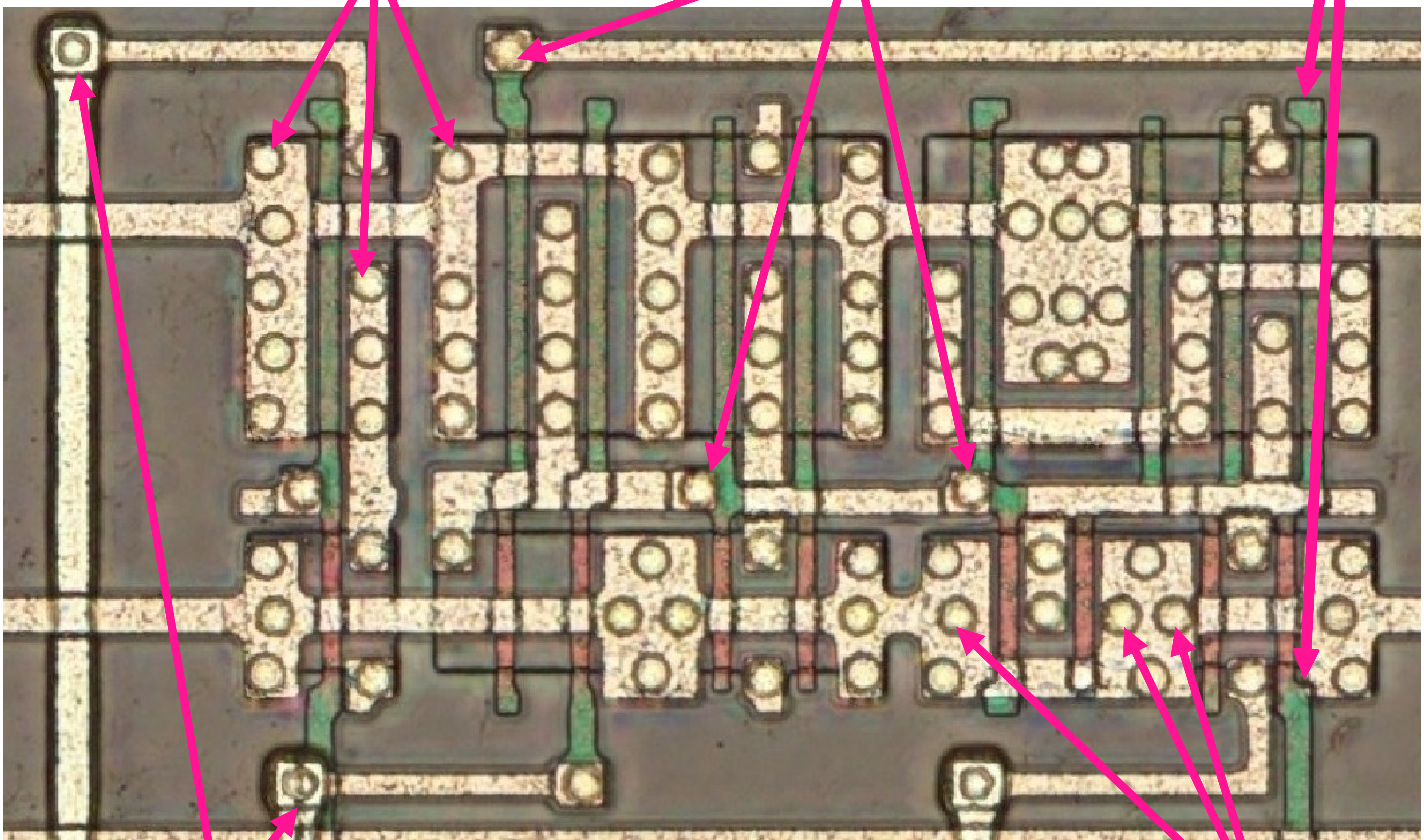
***Tip**: To understand whether the diffuse is PMOS or NMOS,*
you need to identify the polysilicon gate and its color:

The gate is green in PMOS, but red in NMOS.



m1 to P-MOS diffuse via:

m1 to polysilicon via:



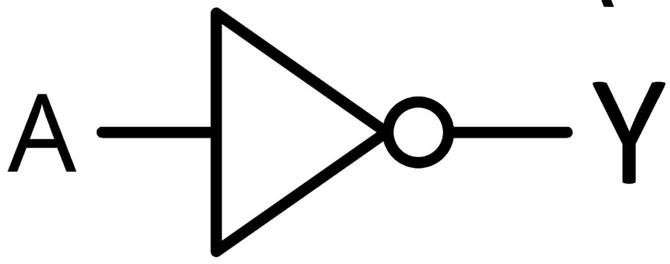
m1 to m2 via:

Generally has smaller vias

m1 to N-MOS diffuse via:

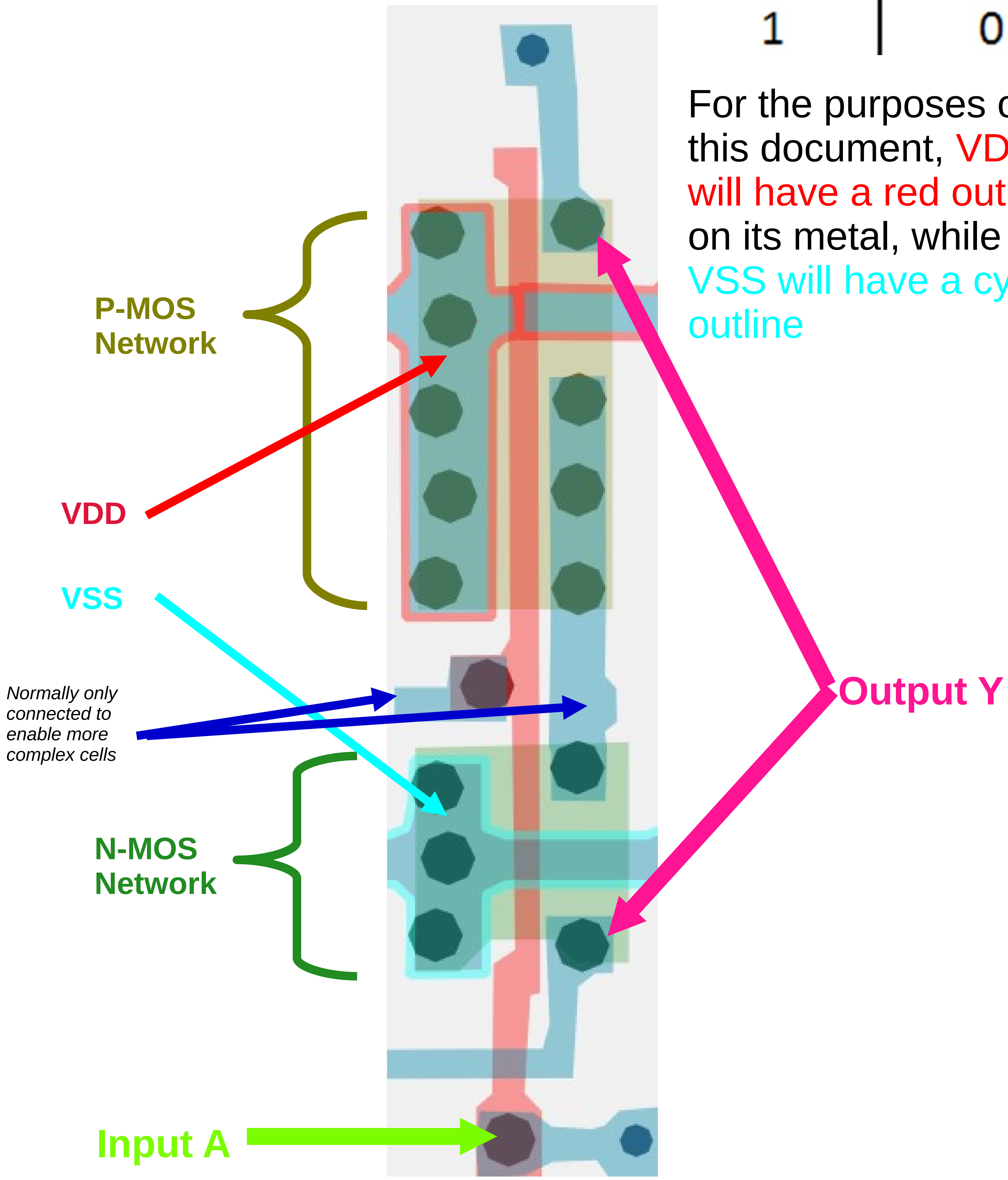
How it all works

(Inverter/NOT example)



Input	Output
A	Y
0	1
1	0

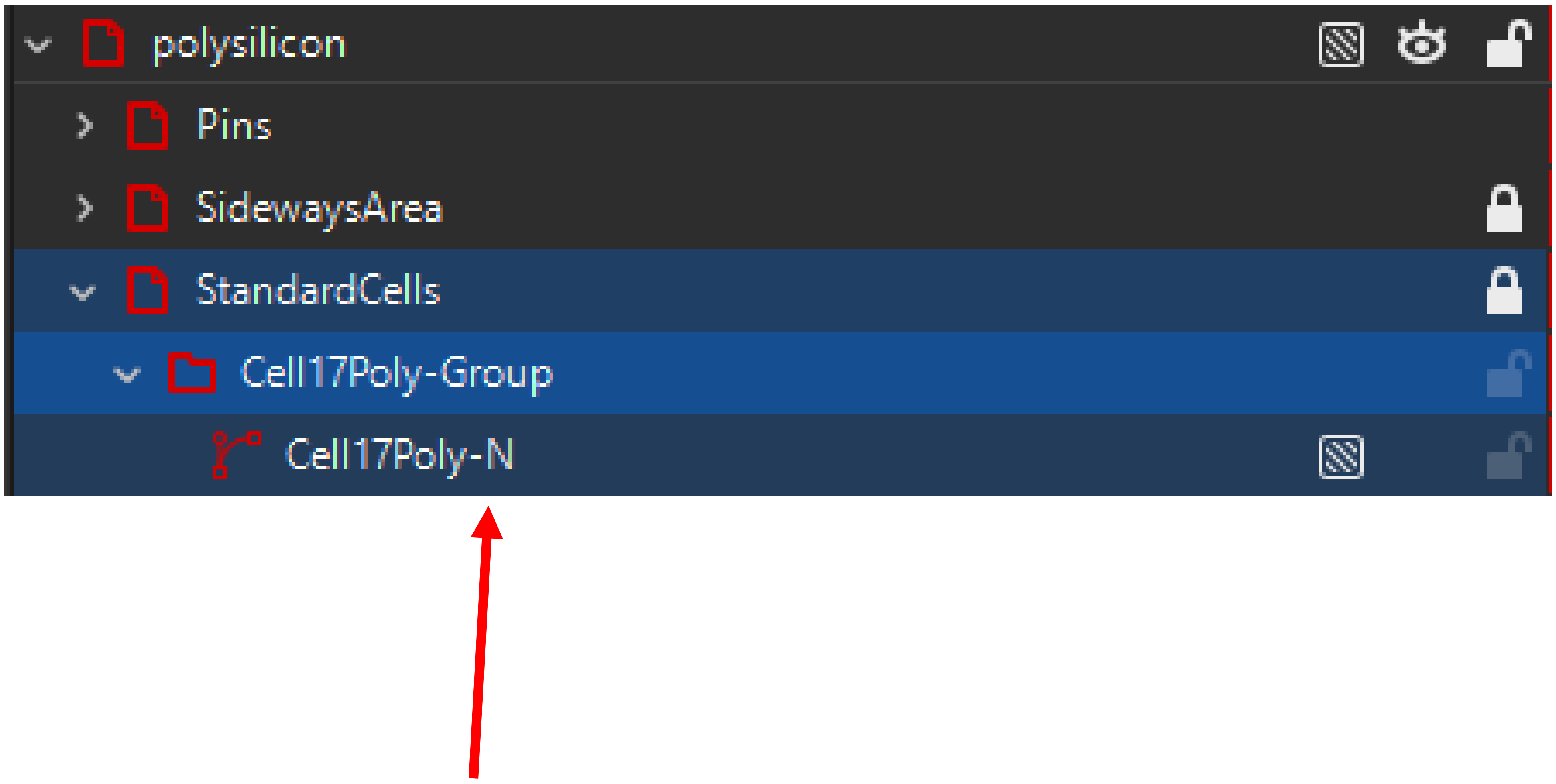
For the purposes of this document, VDD will have a red outline on its metal, while VSS will have a cyan outline



Object naming and how to understand

I have done my best to make object naming as simple as possible while keeping object names practical.

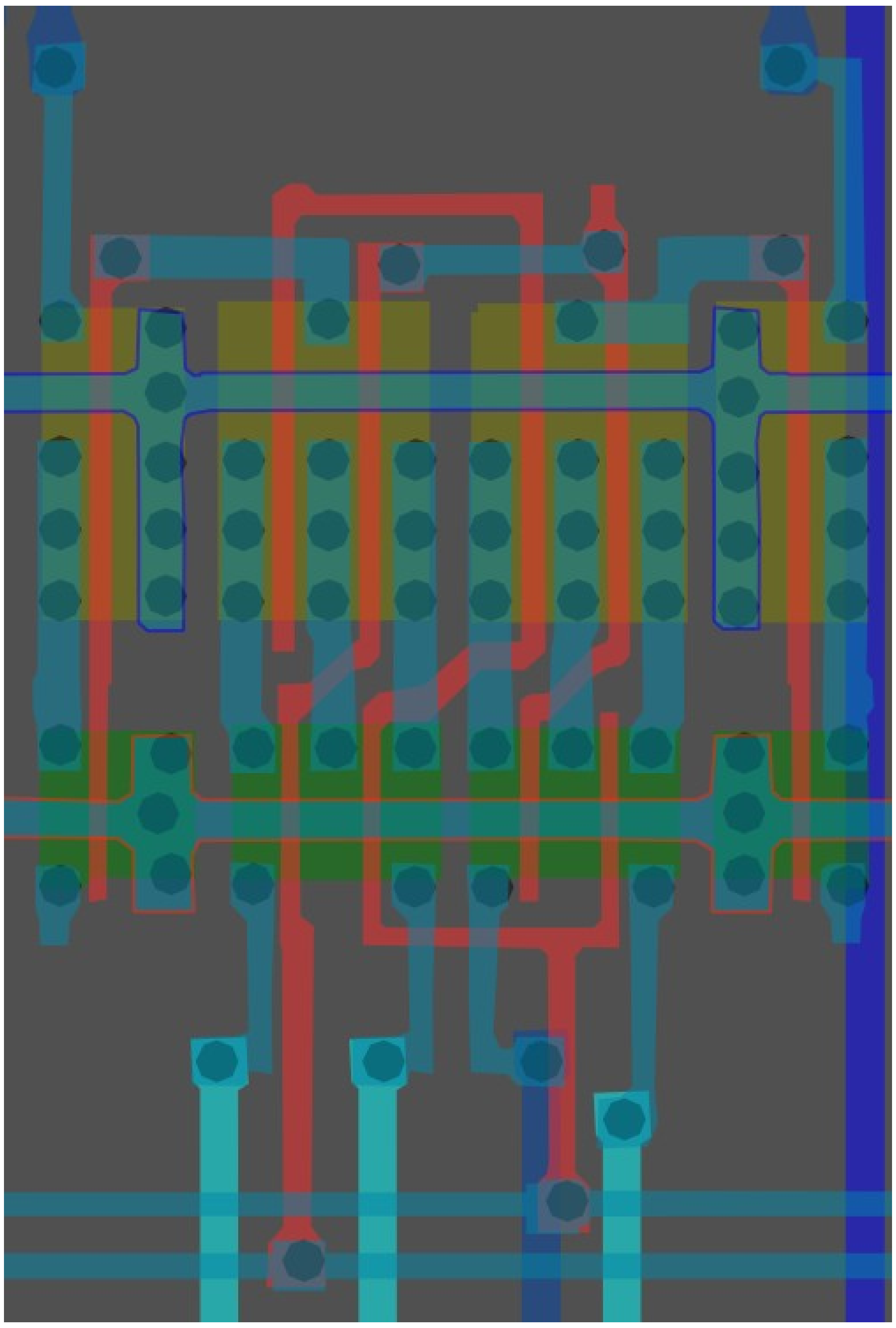
PDi – P-Diffusion
NDi – N-Diffusion
Poly – Polysilicon

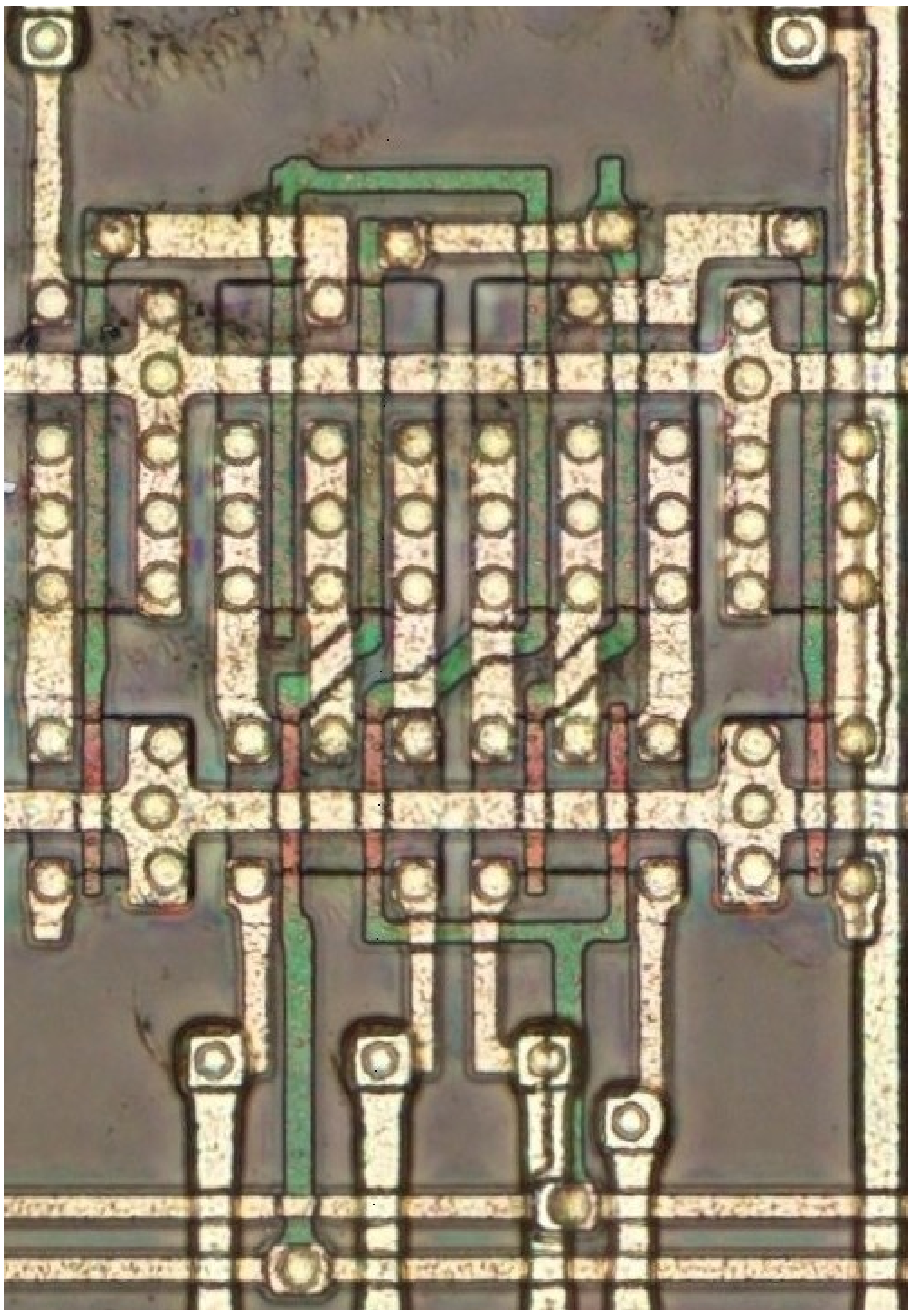


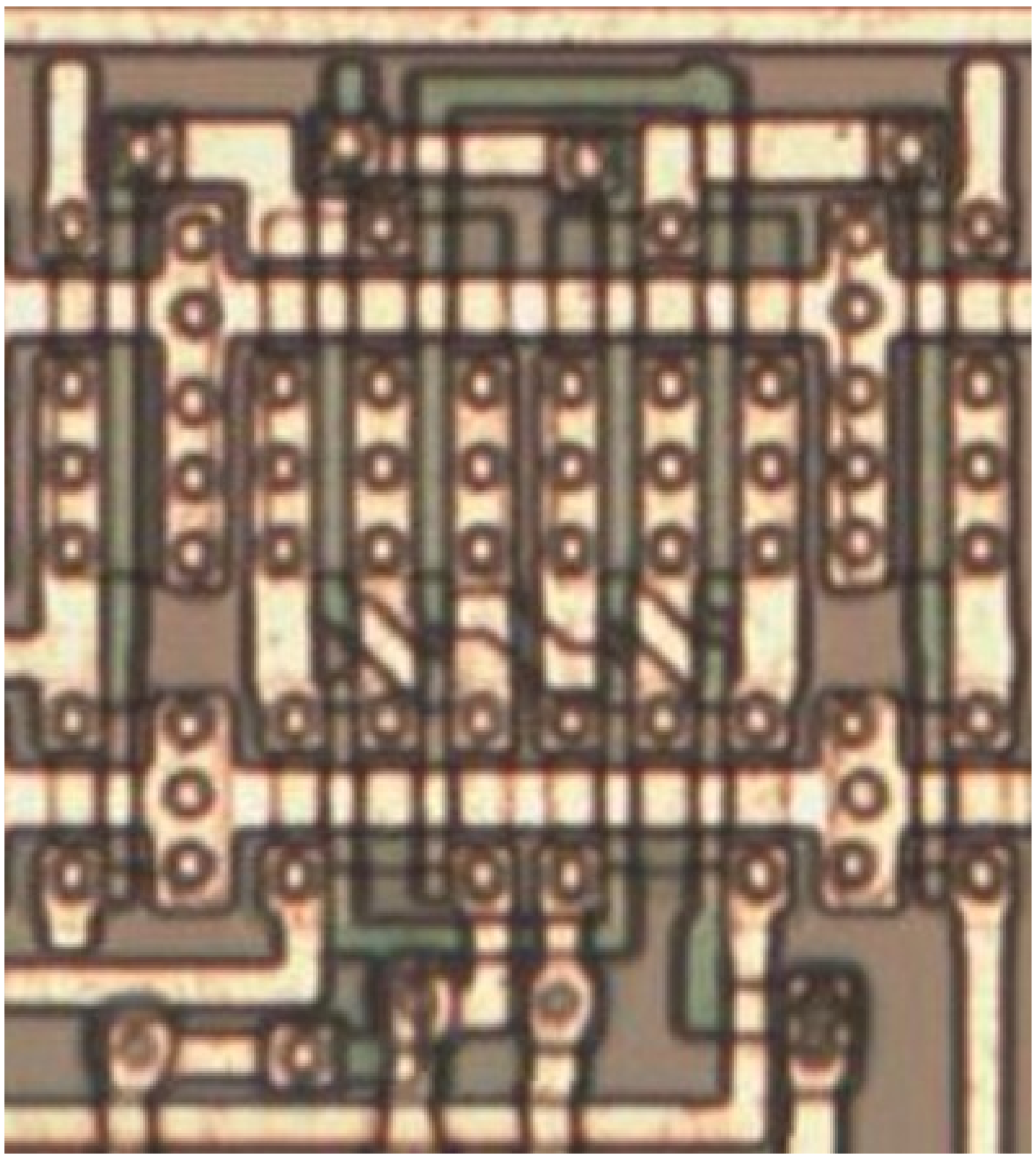
Cell 17 in standard cell area, Polysilicon layer, sub-object N.

#f321a5ff

Cell type 1



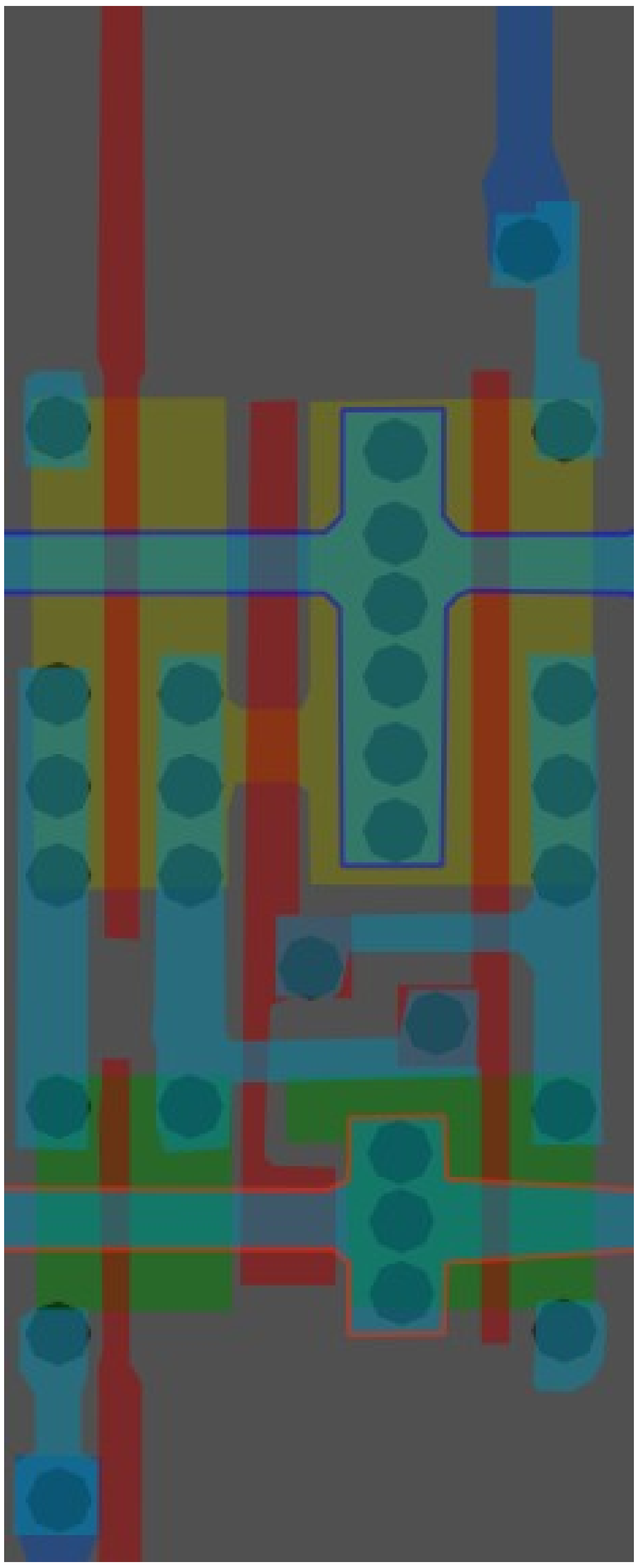




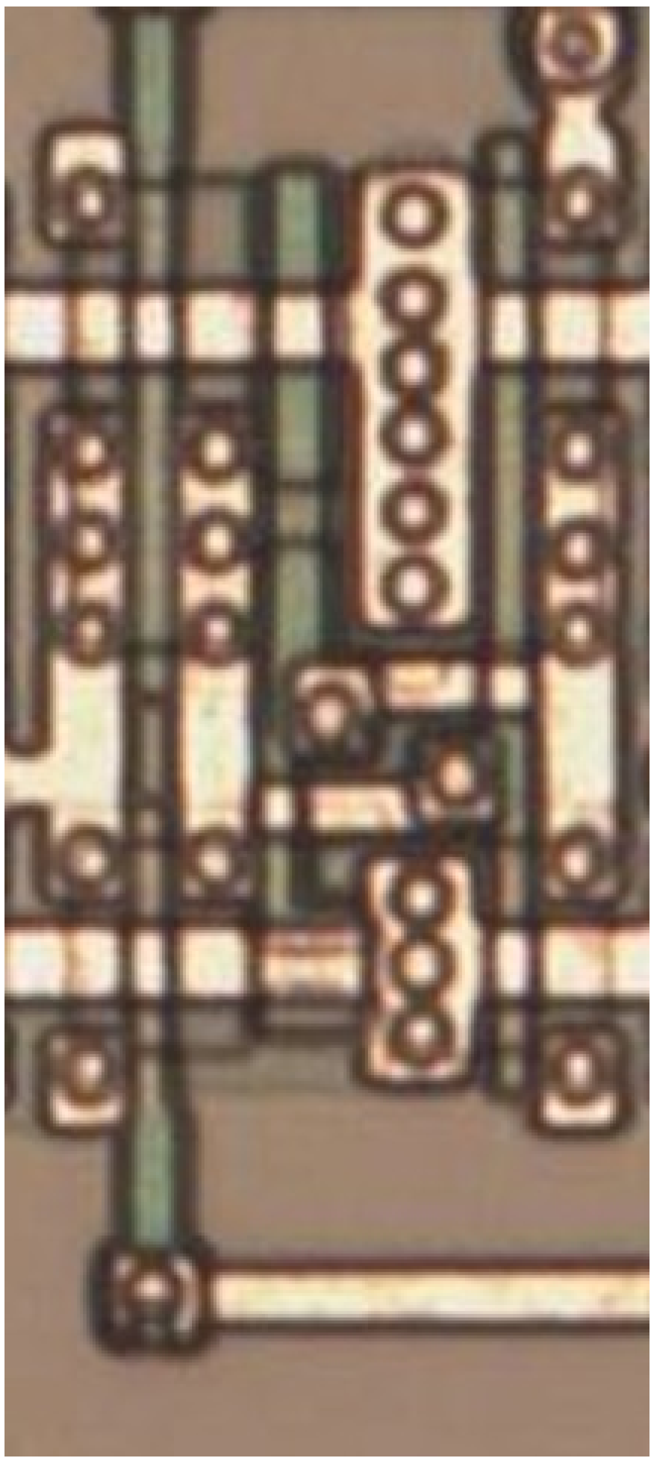
Same cell on LA32 (mirrored) (MB87136A)

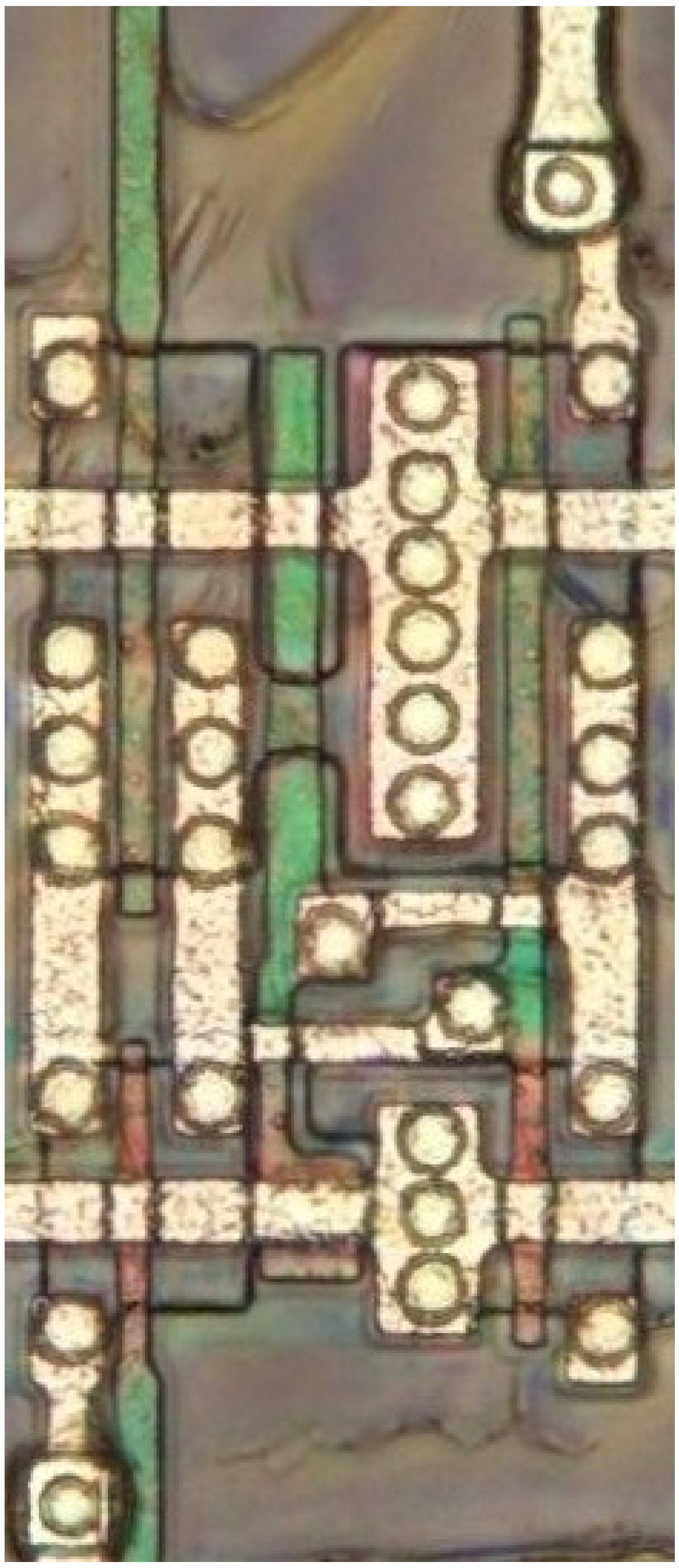
#f34f00ff

Cell 2



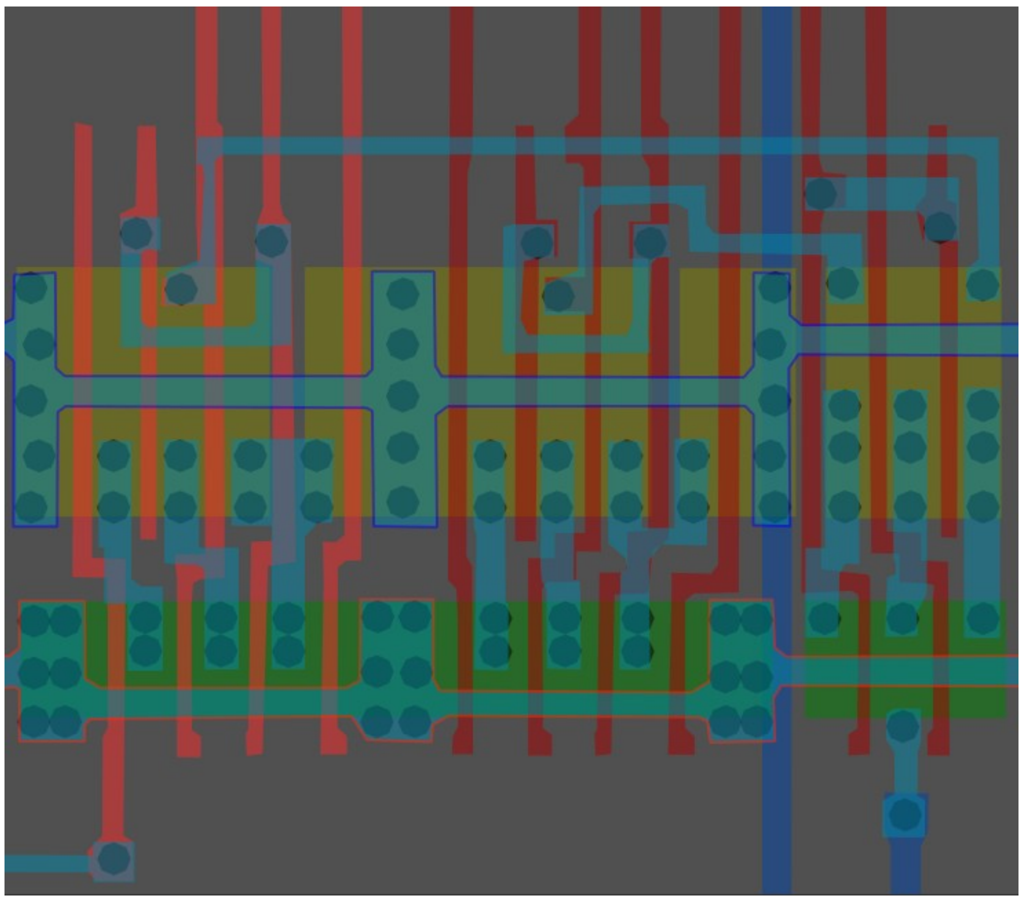
Maybe a transmission gate?

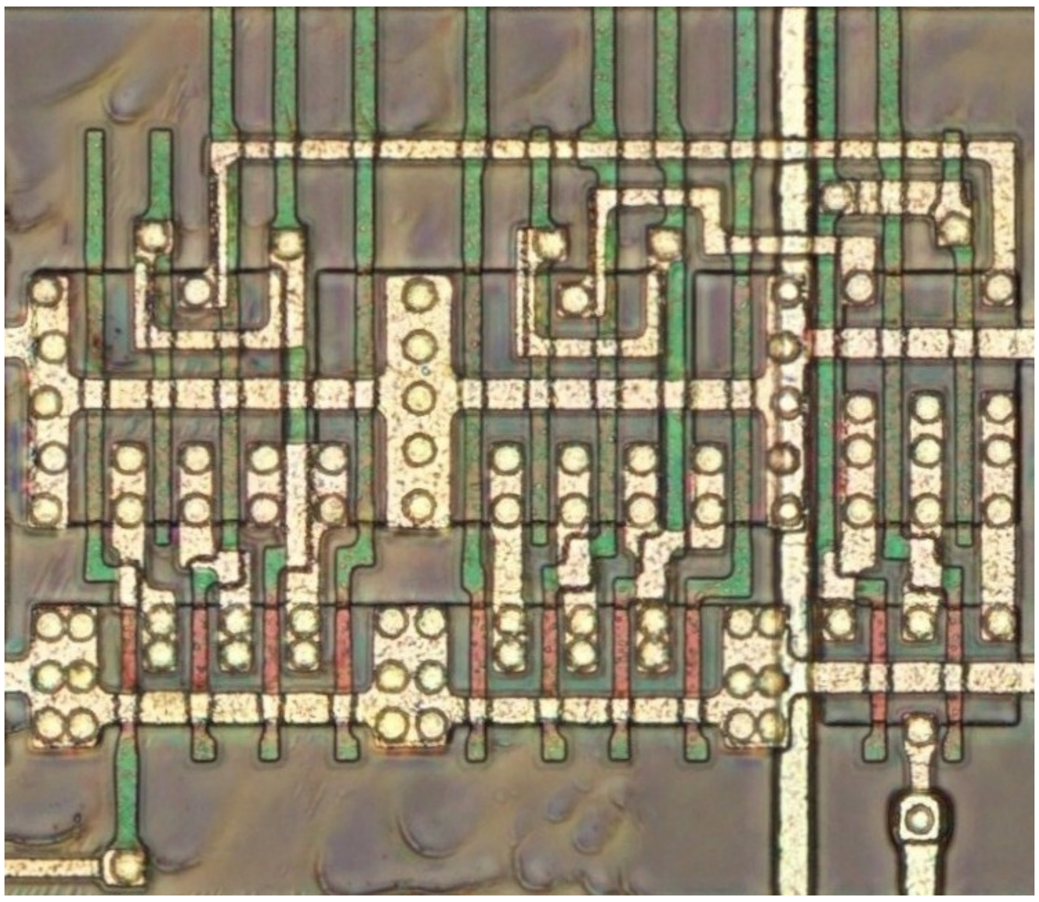




Same cell on LA32 (MB87136A)

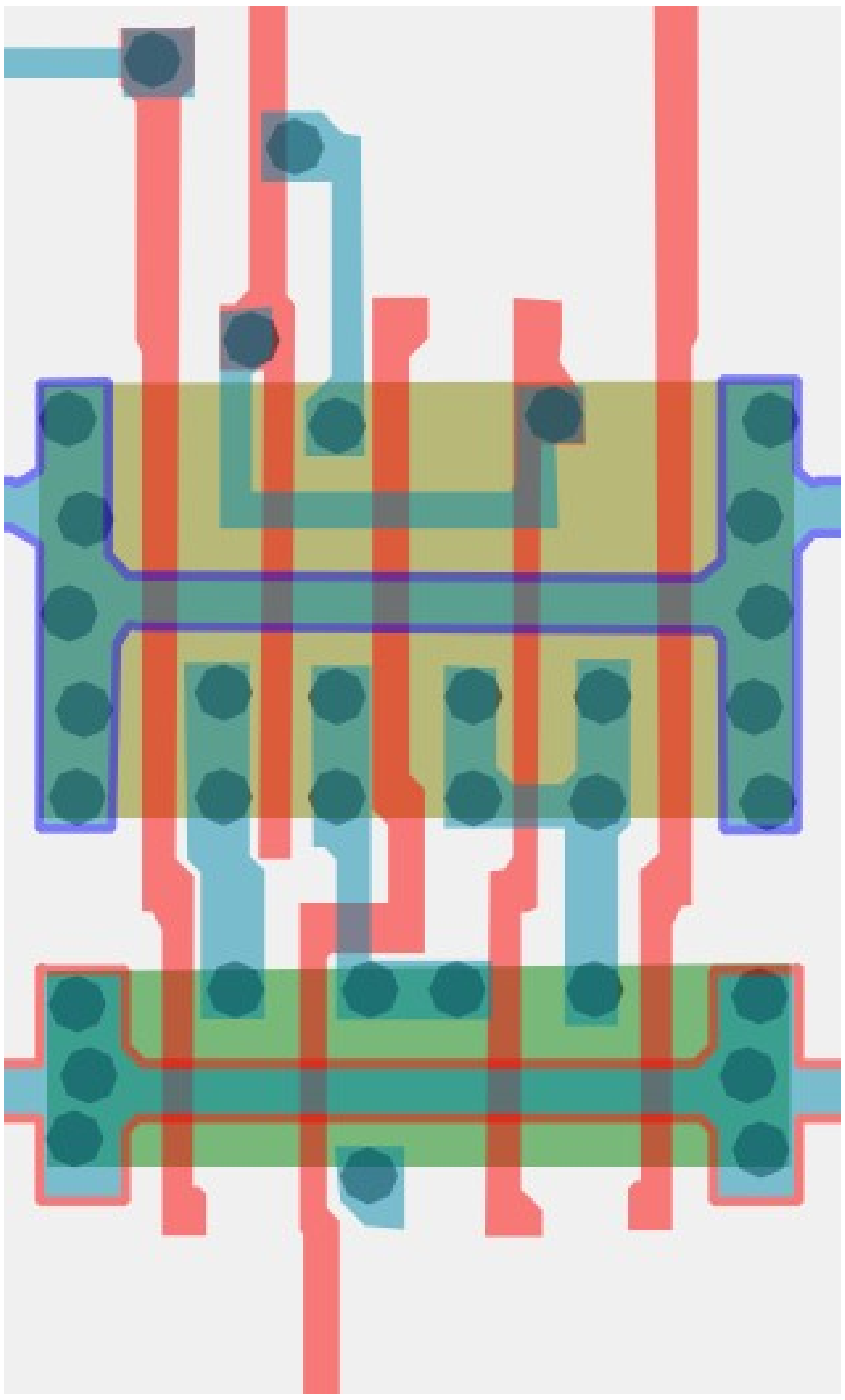
#ffff00ff Cell type 3a

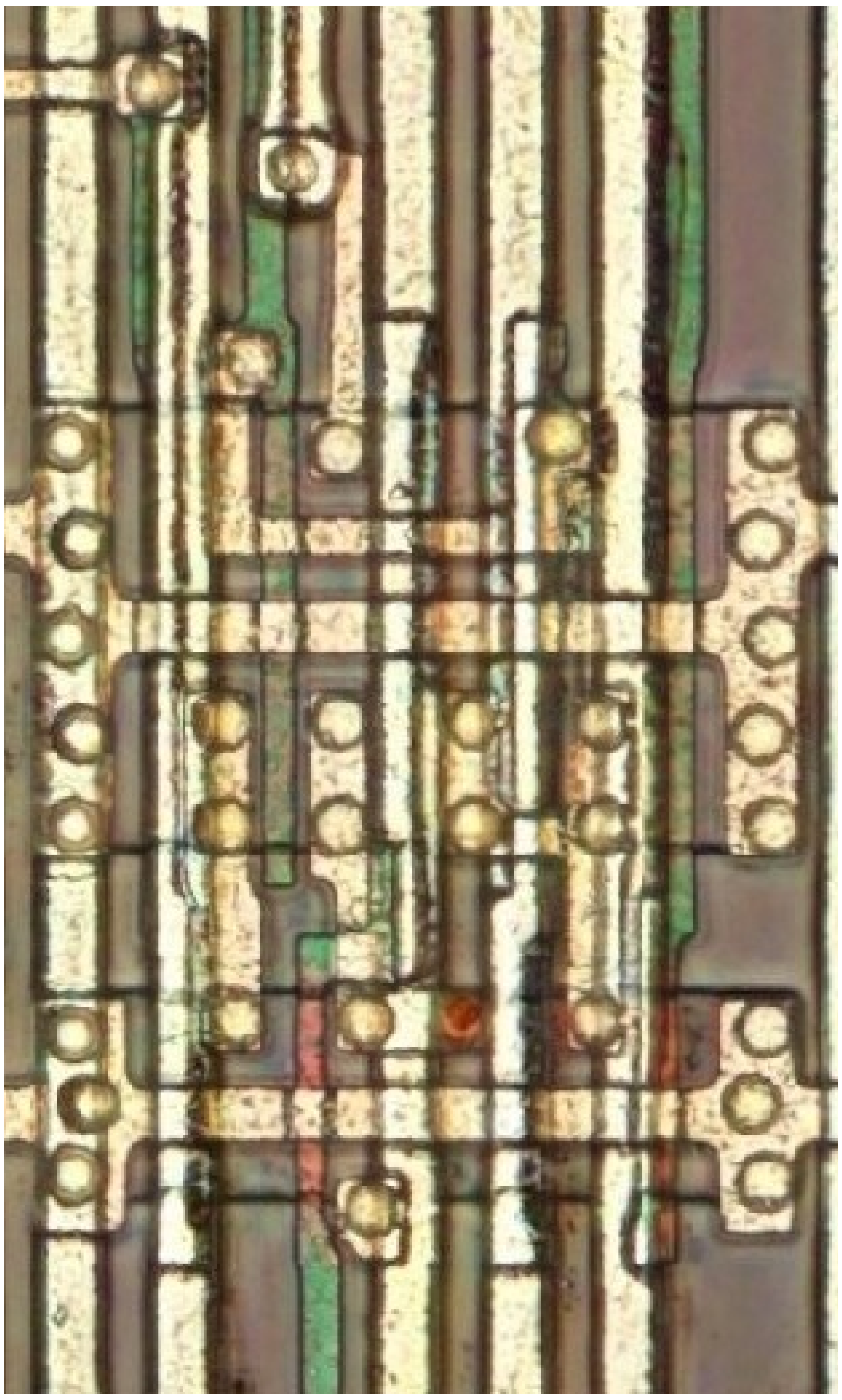




Unable to find on LA32

#b29e31ff Cell type 3b

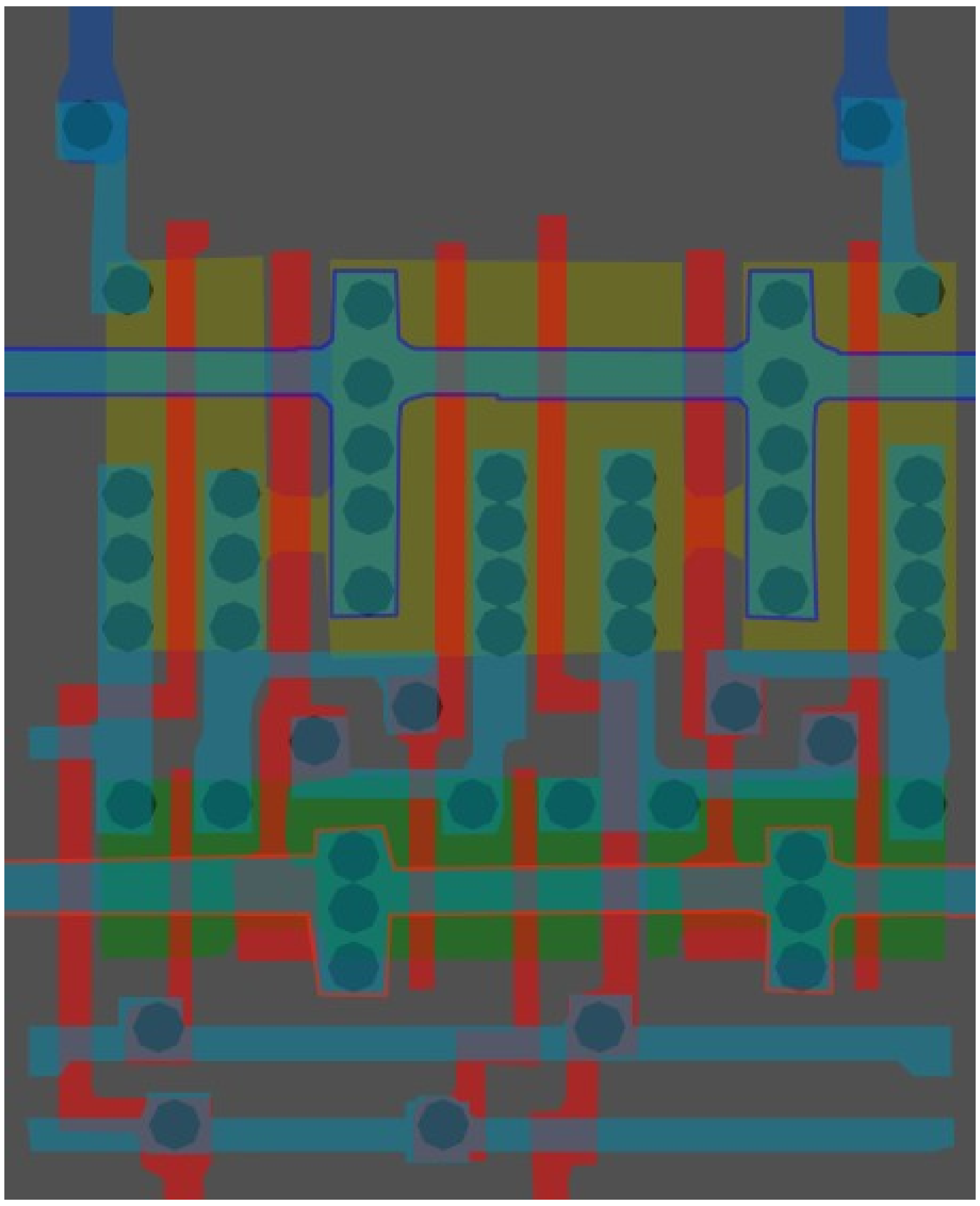


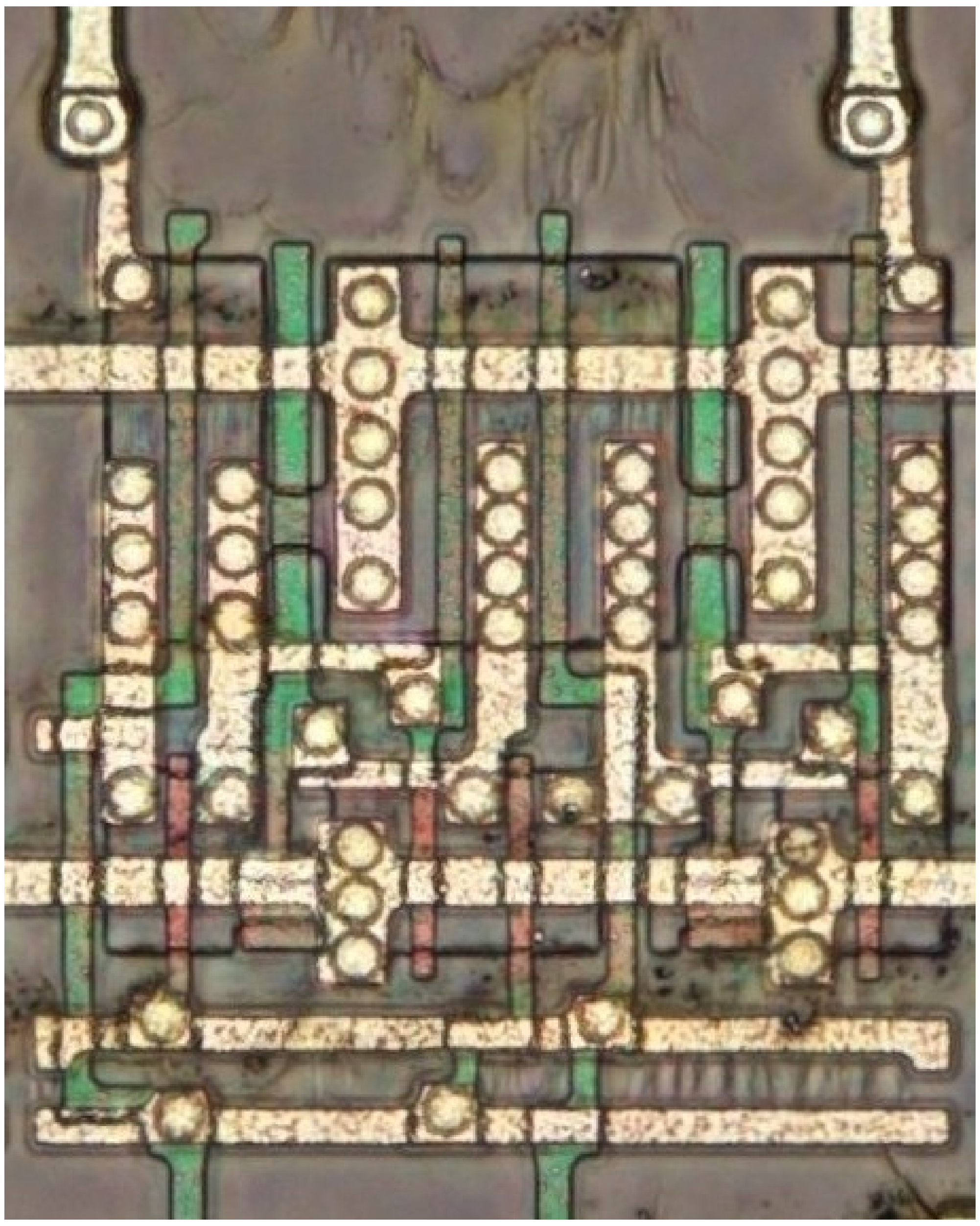


Looks like a small segment of 3a,
but altered slightly

#00ff00ff

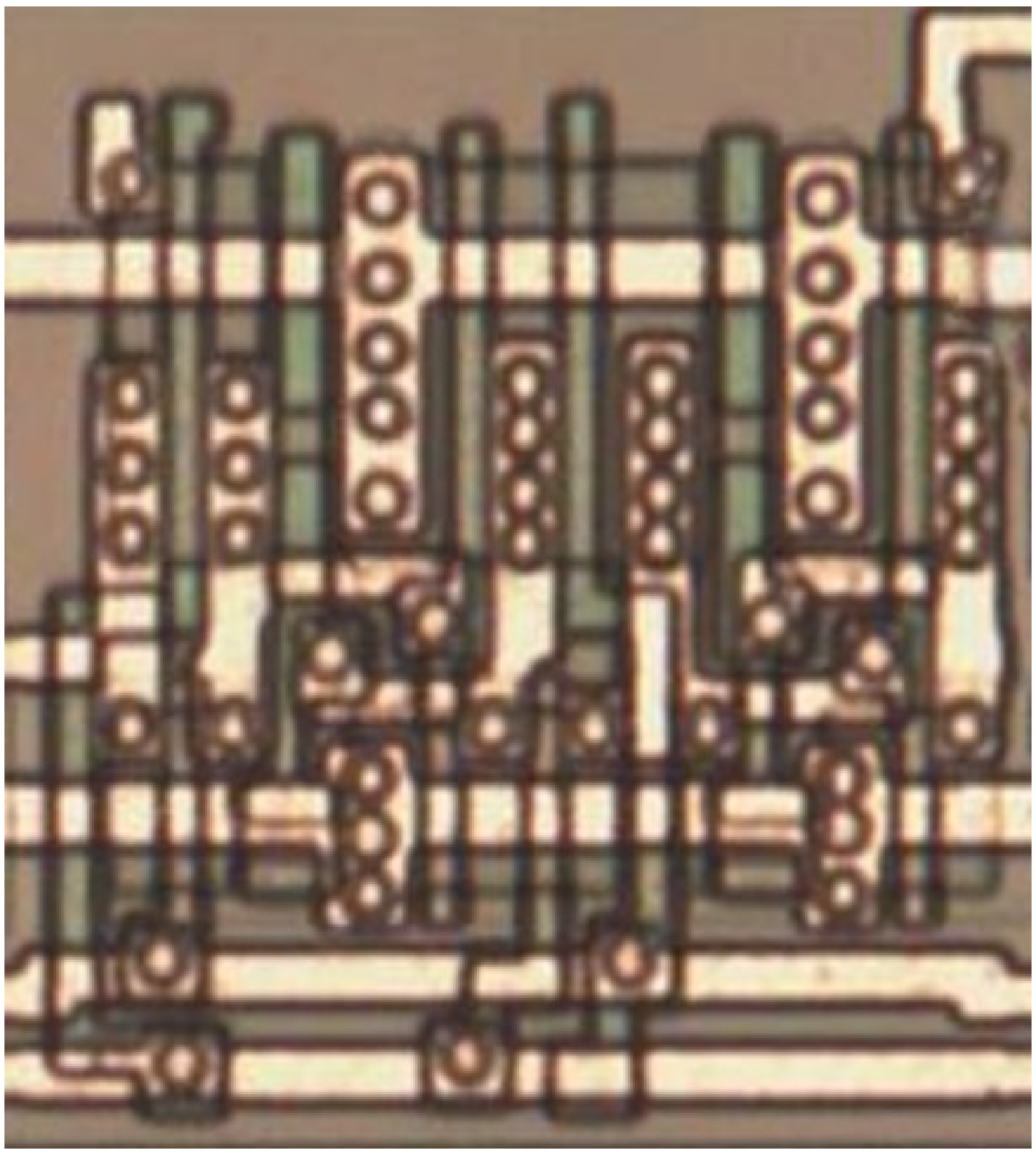
Cell type 4



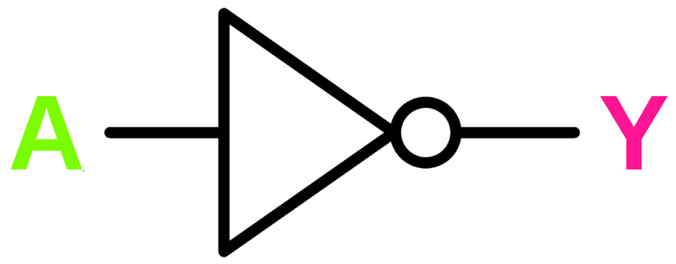


I suspect this is some sort of flip-flop

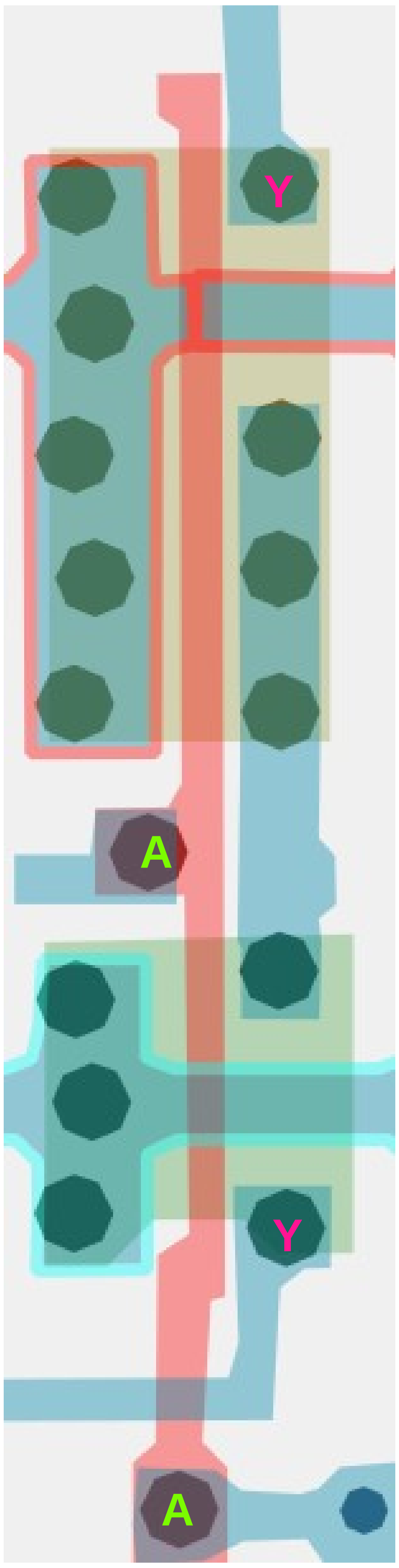
Same cell on LA32 (MB87136A)

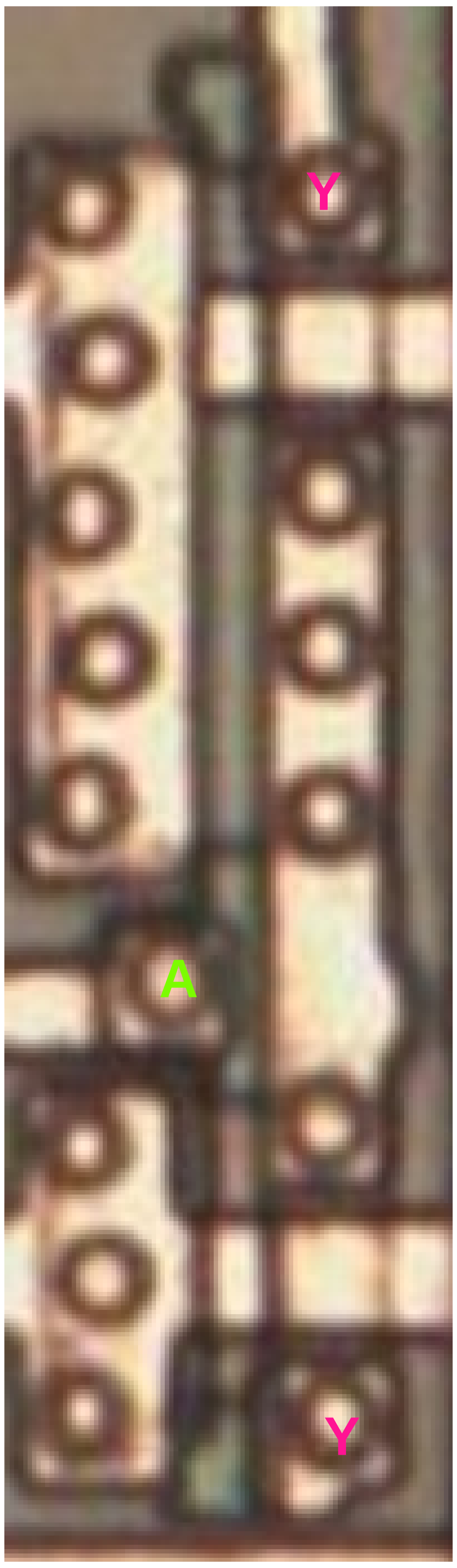


#51215aff Inverter/NOT x1 Cell 5



A	Y
0	1
1	0

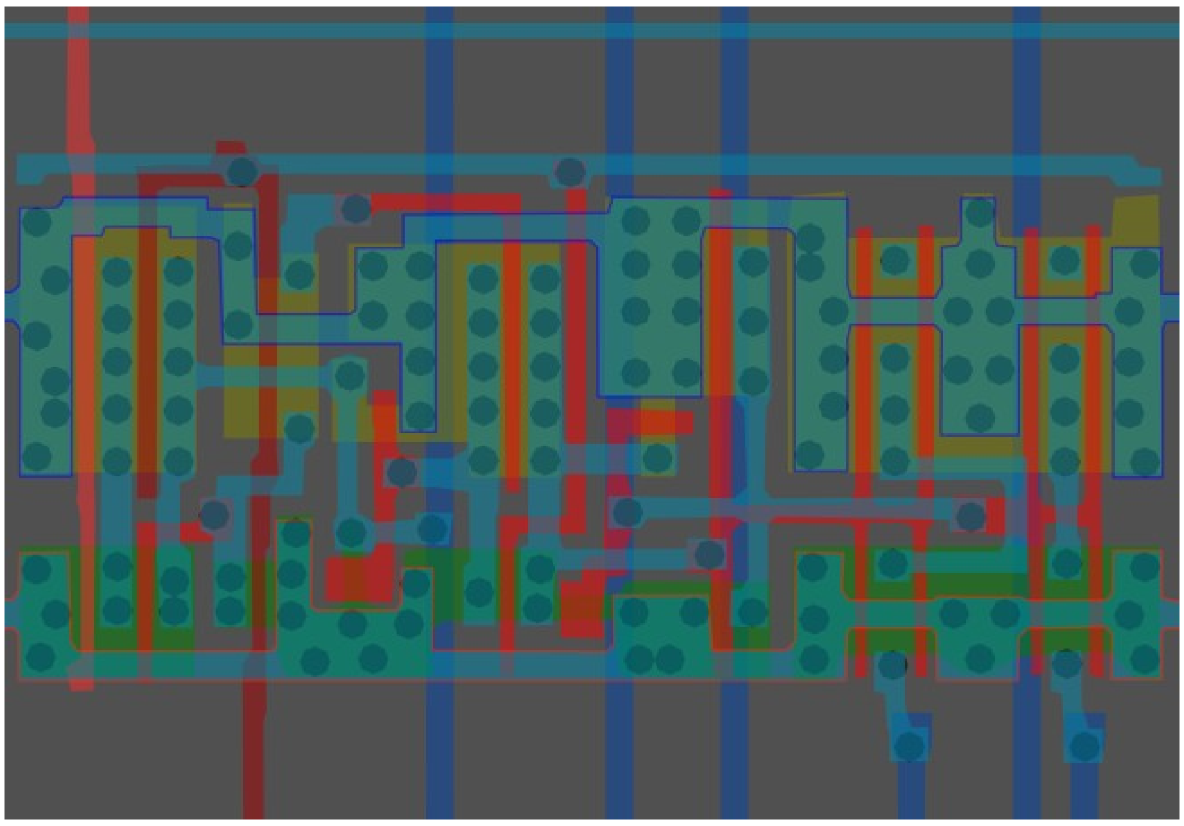


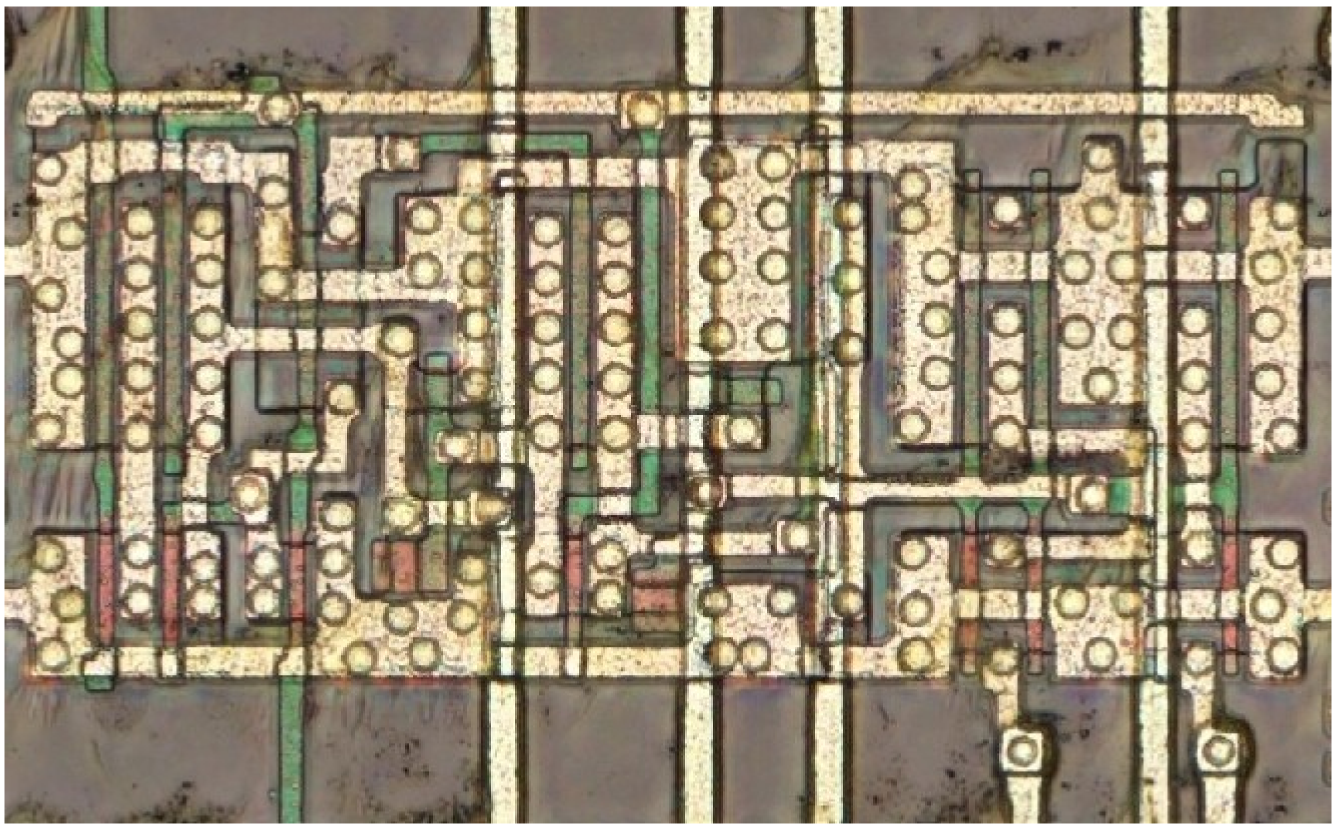


Same cell on LA32 (MB87136A)

#955f00ff

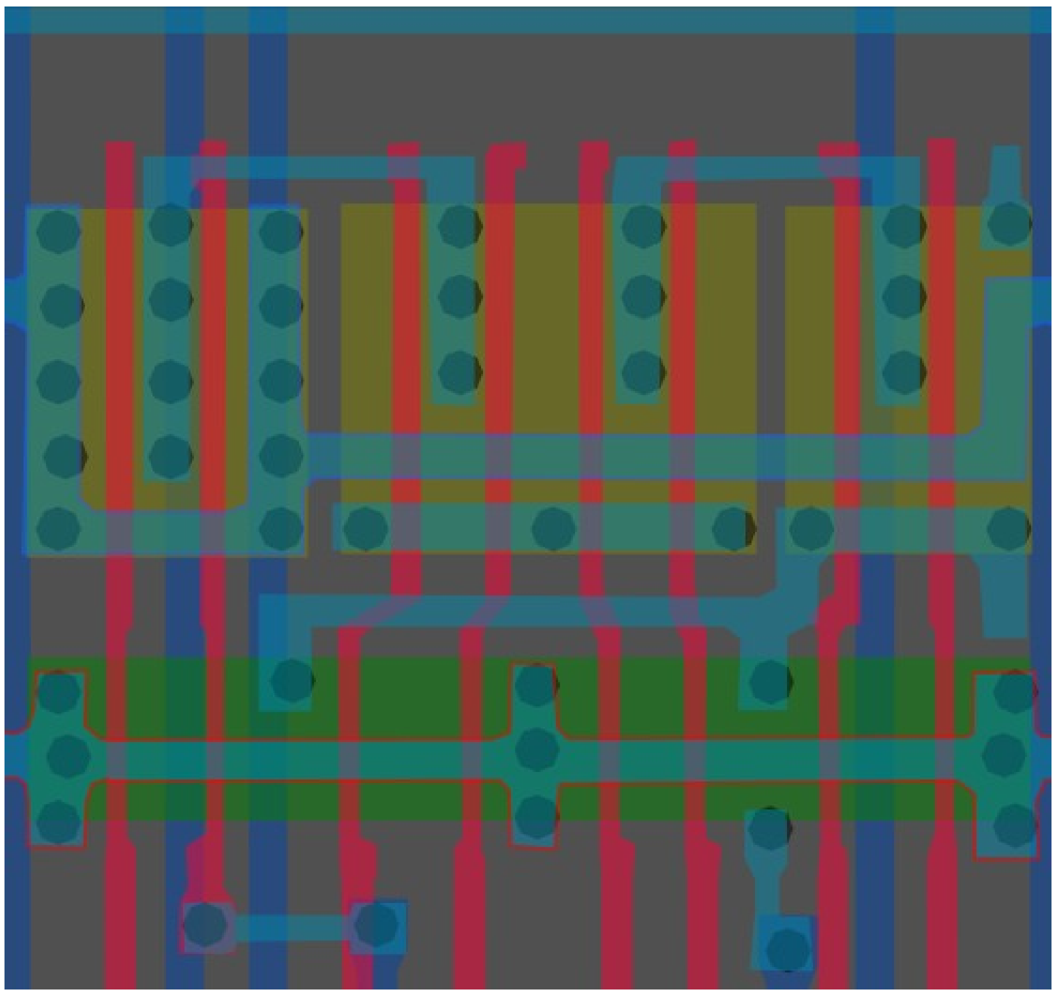
Cell type 6

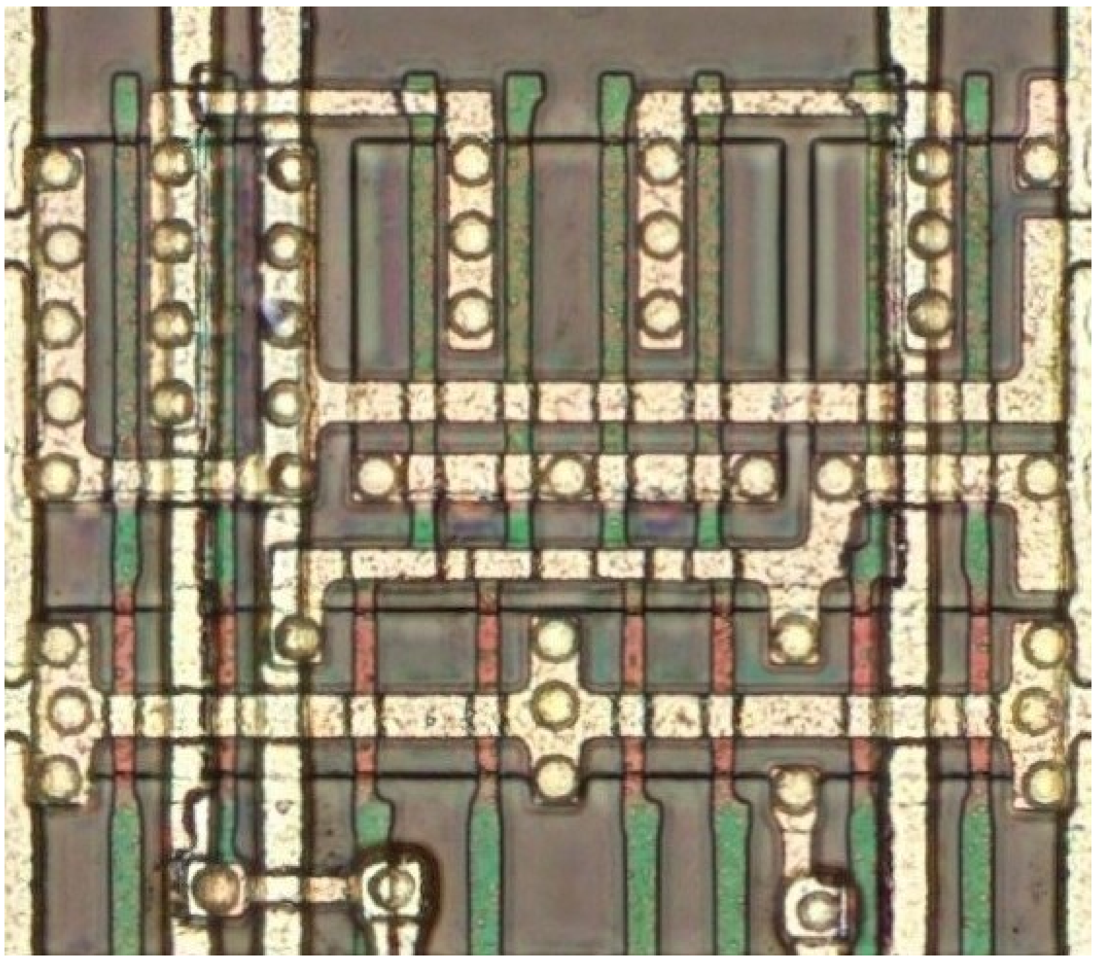




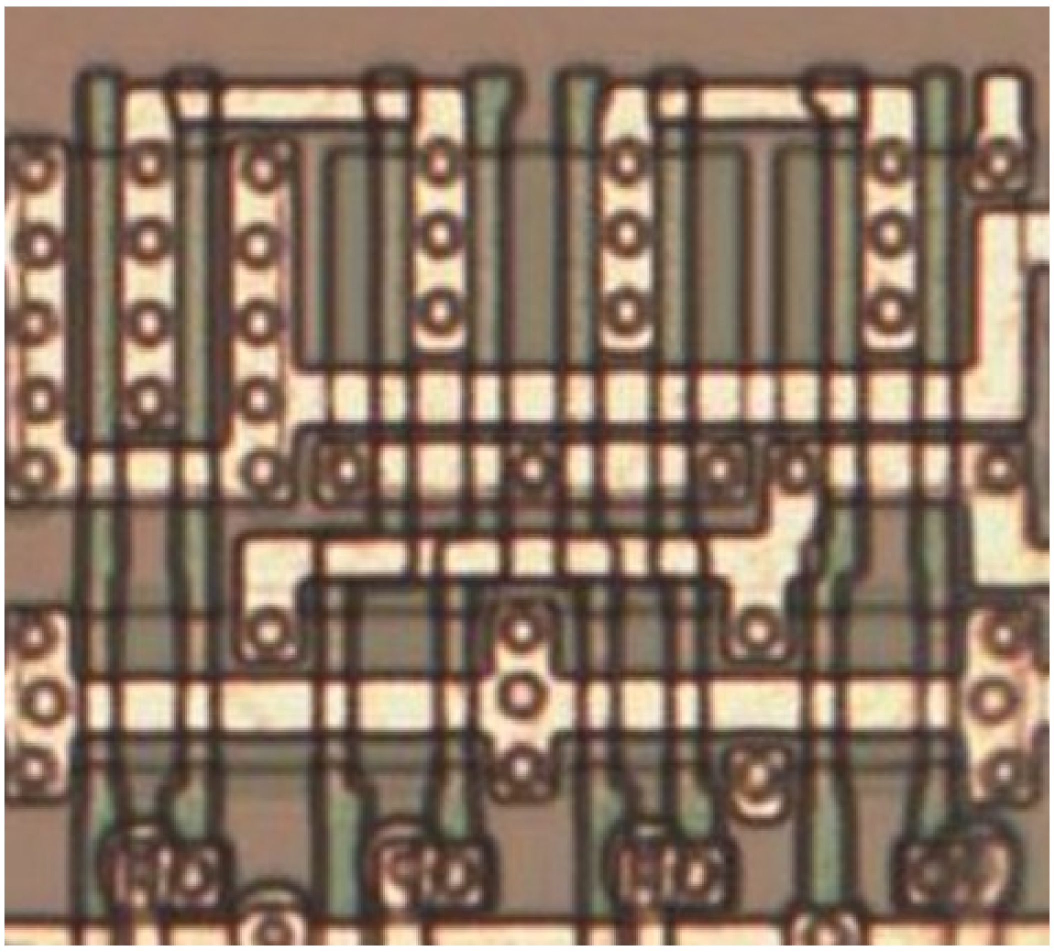
#13215aff

Cell type 7





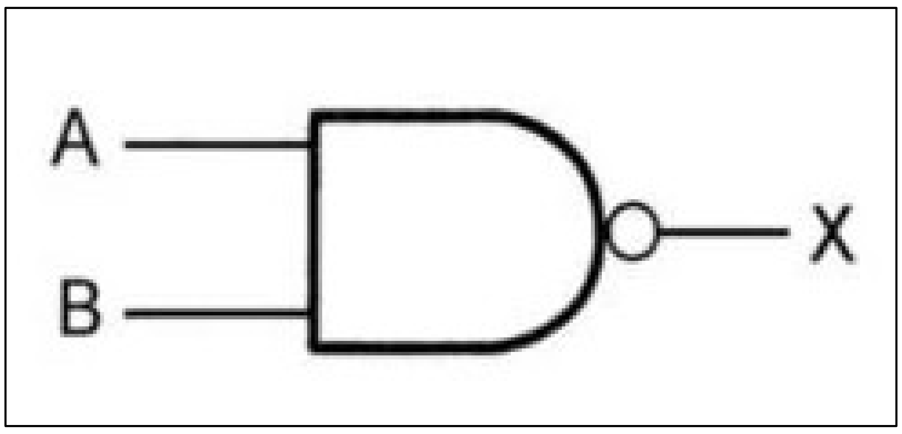
Same cell on LA32 (MB87136A)



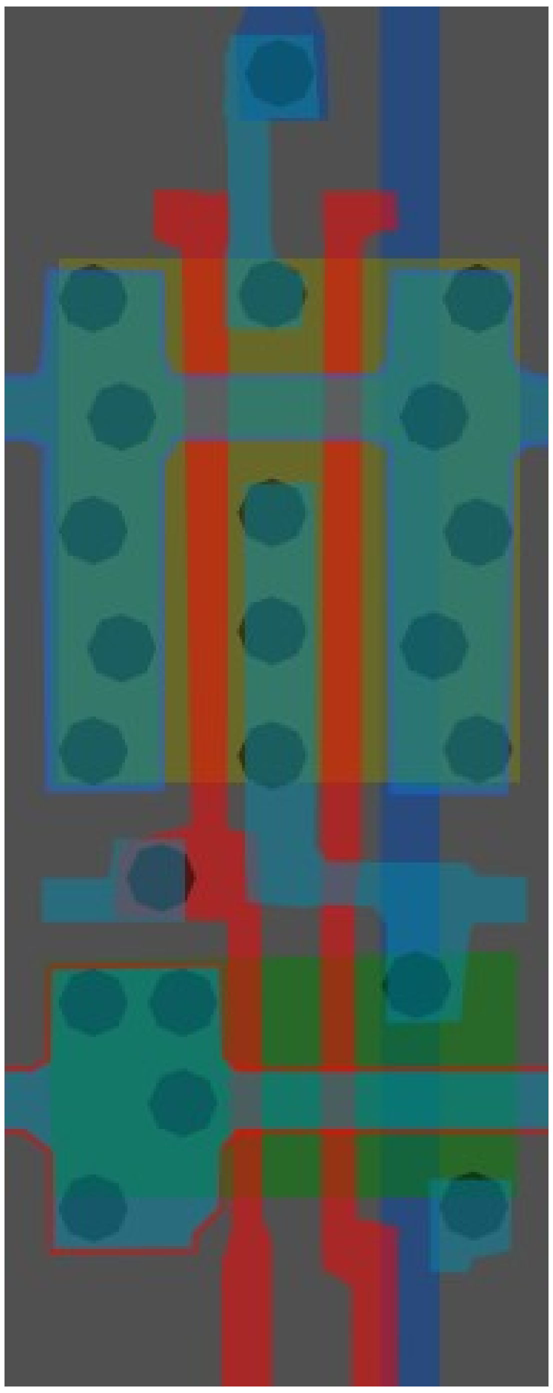
#89ff86ff

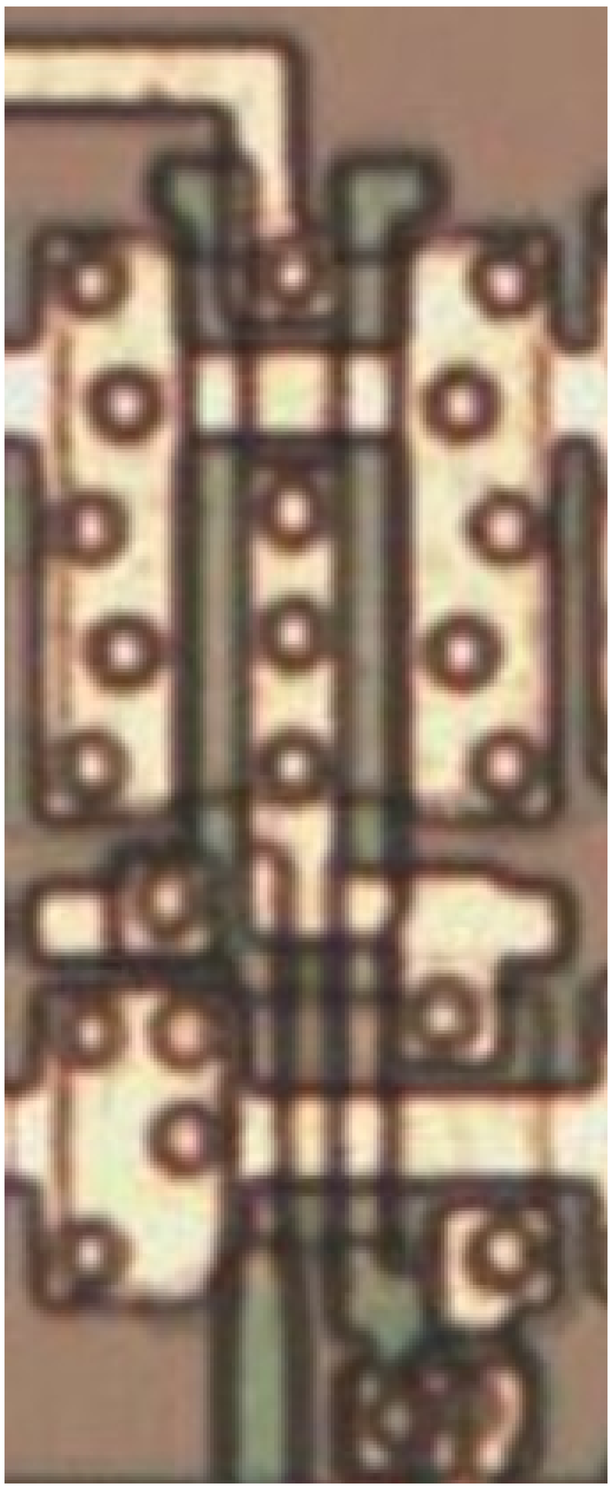
NAND(NOT AND)

Cell 8

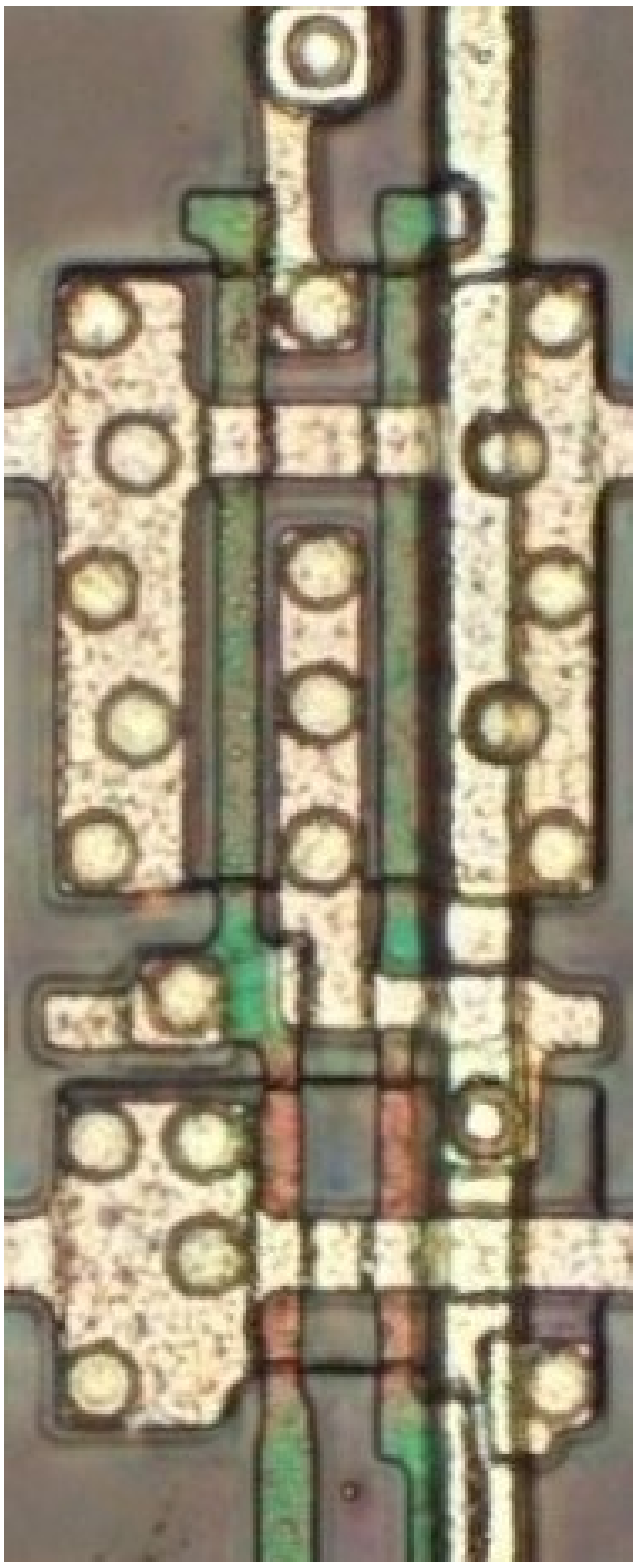


A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

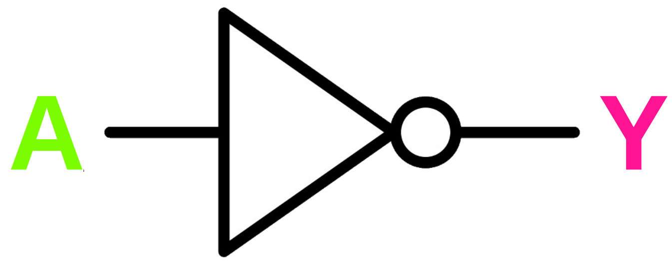


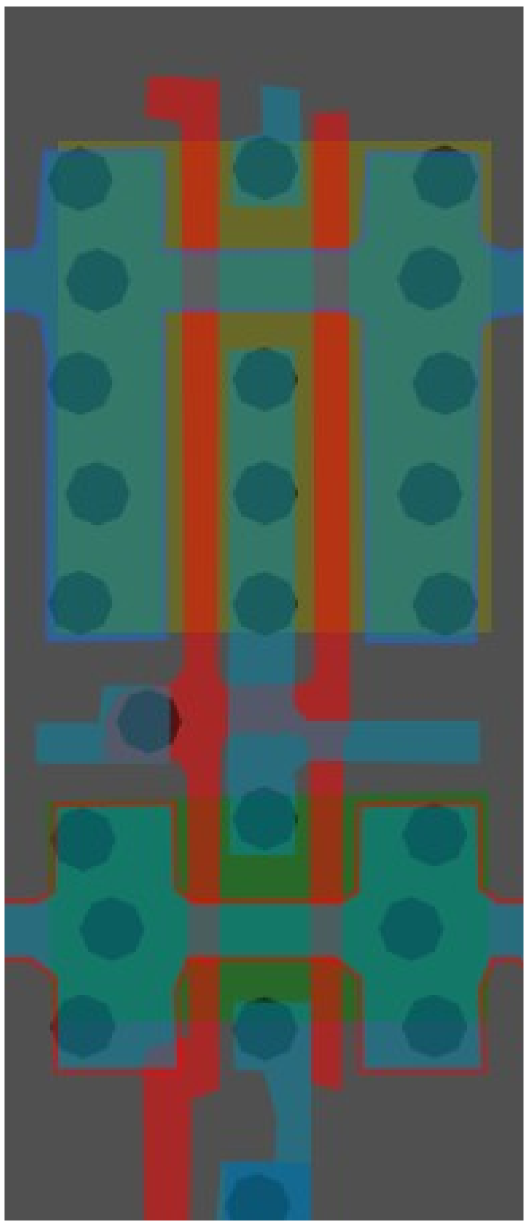


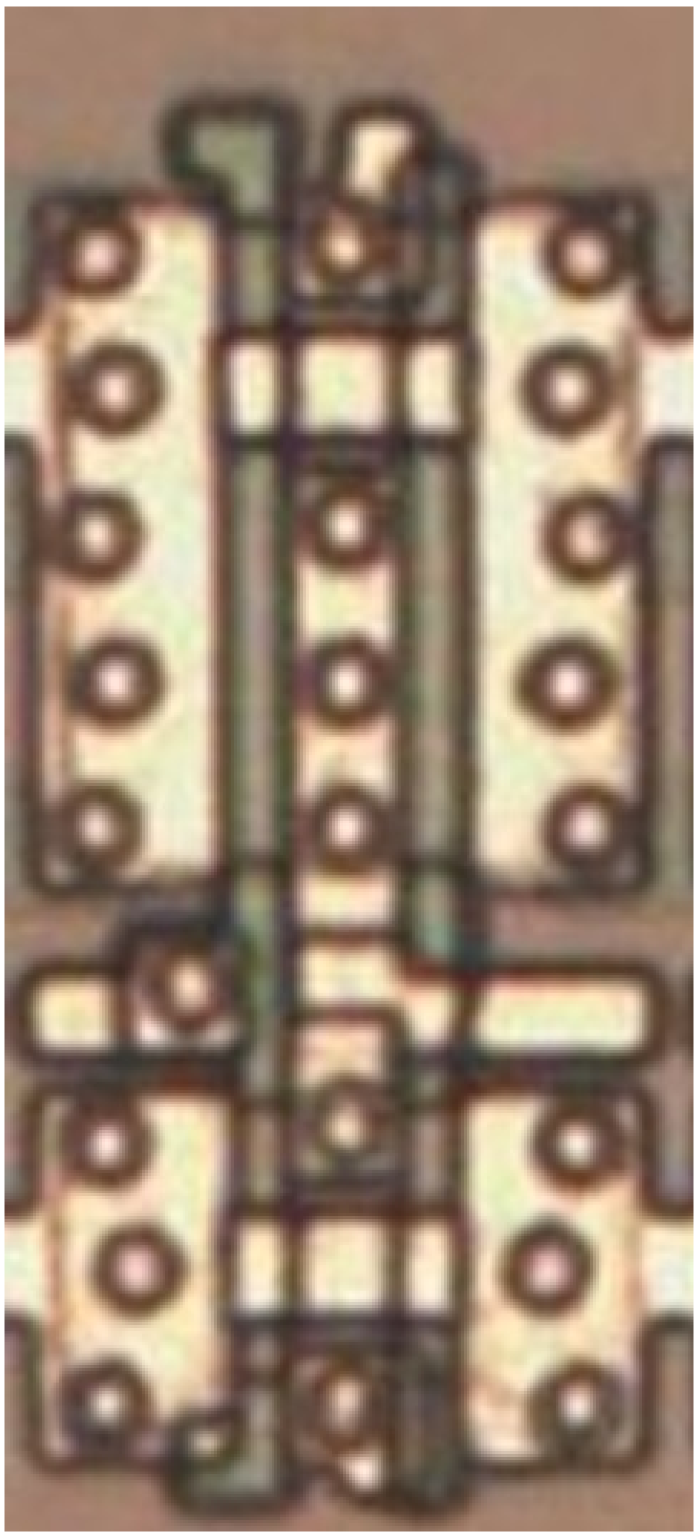
Same cell on LA32 (MB87136A)



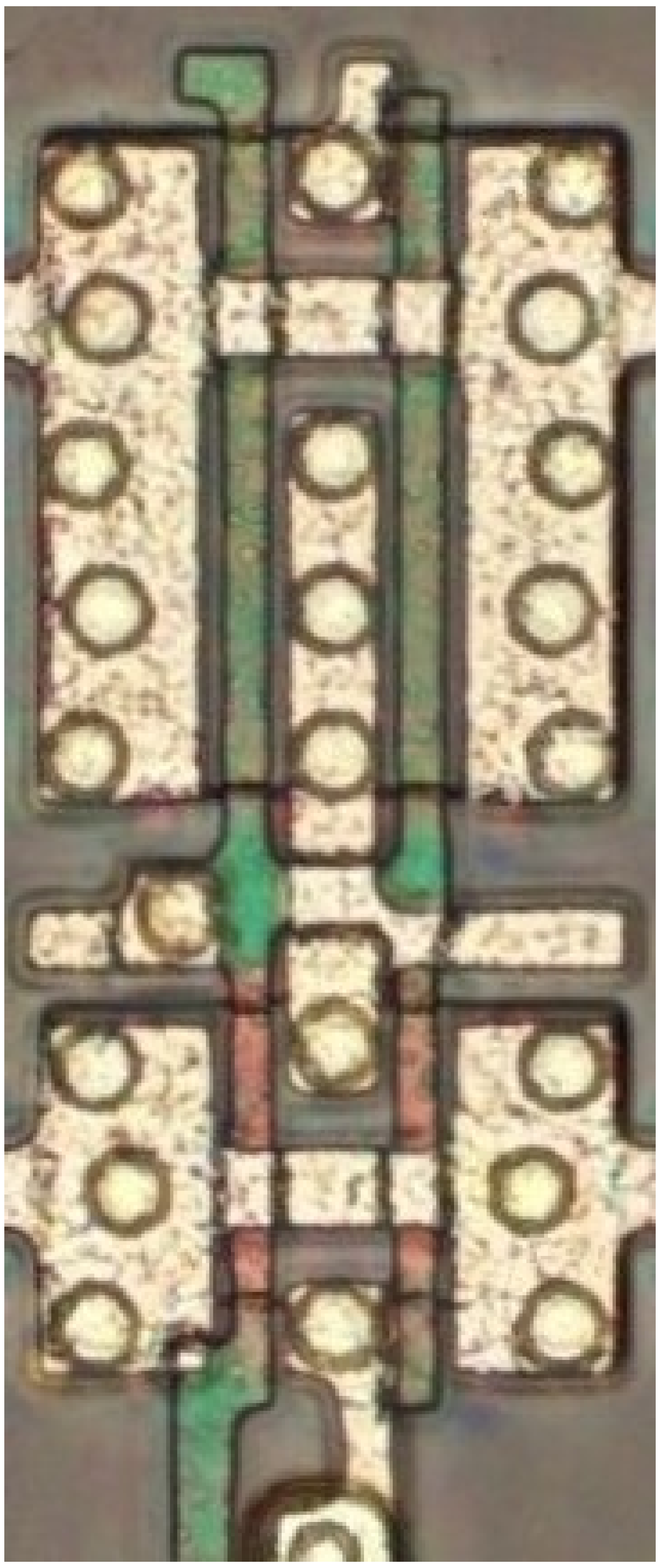
#393c3cff Inverter/NOT X2 Cell 9



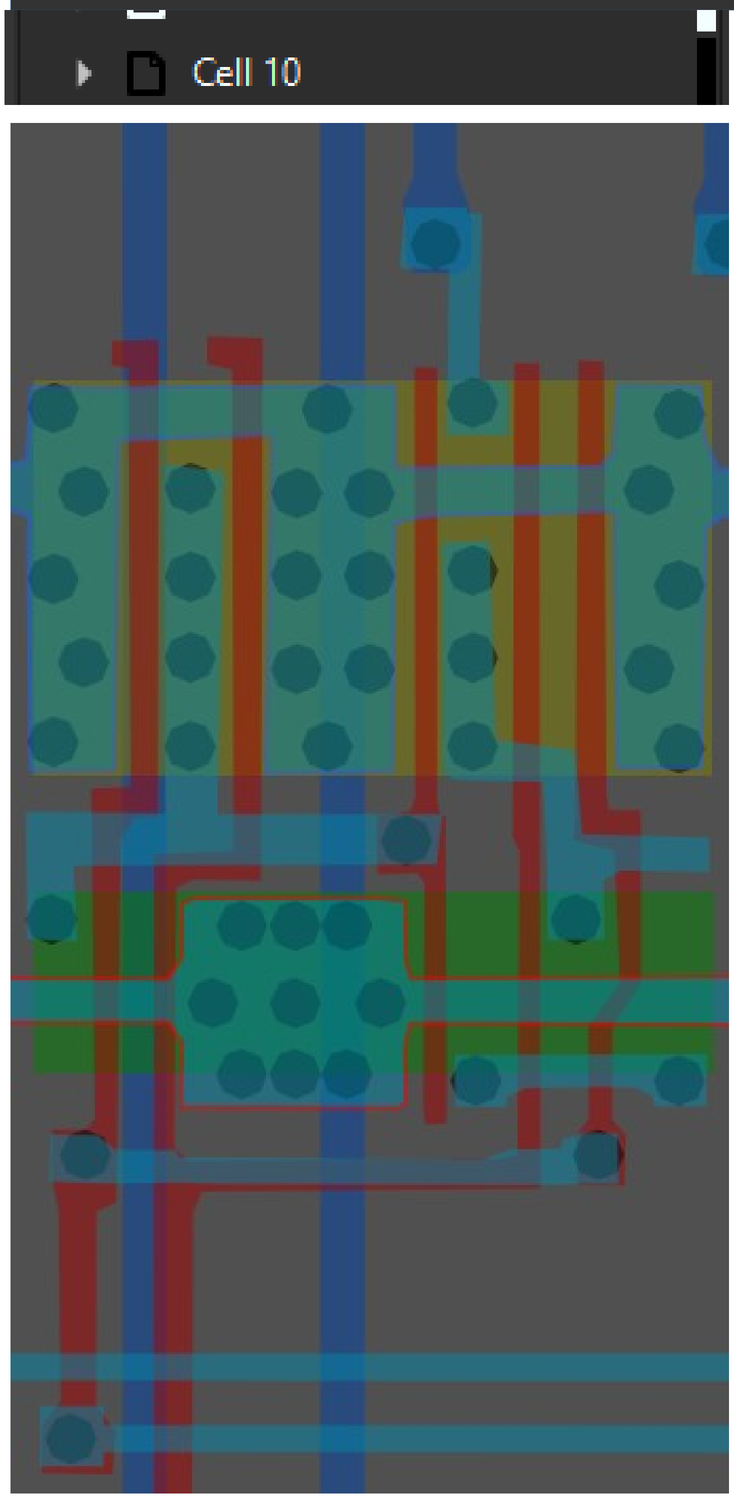


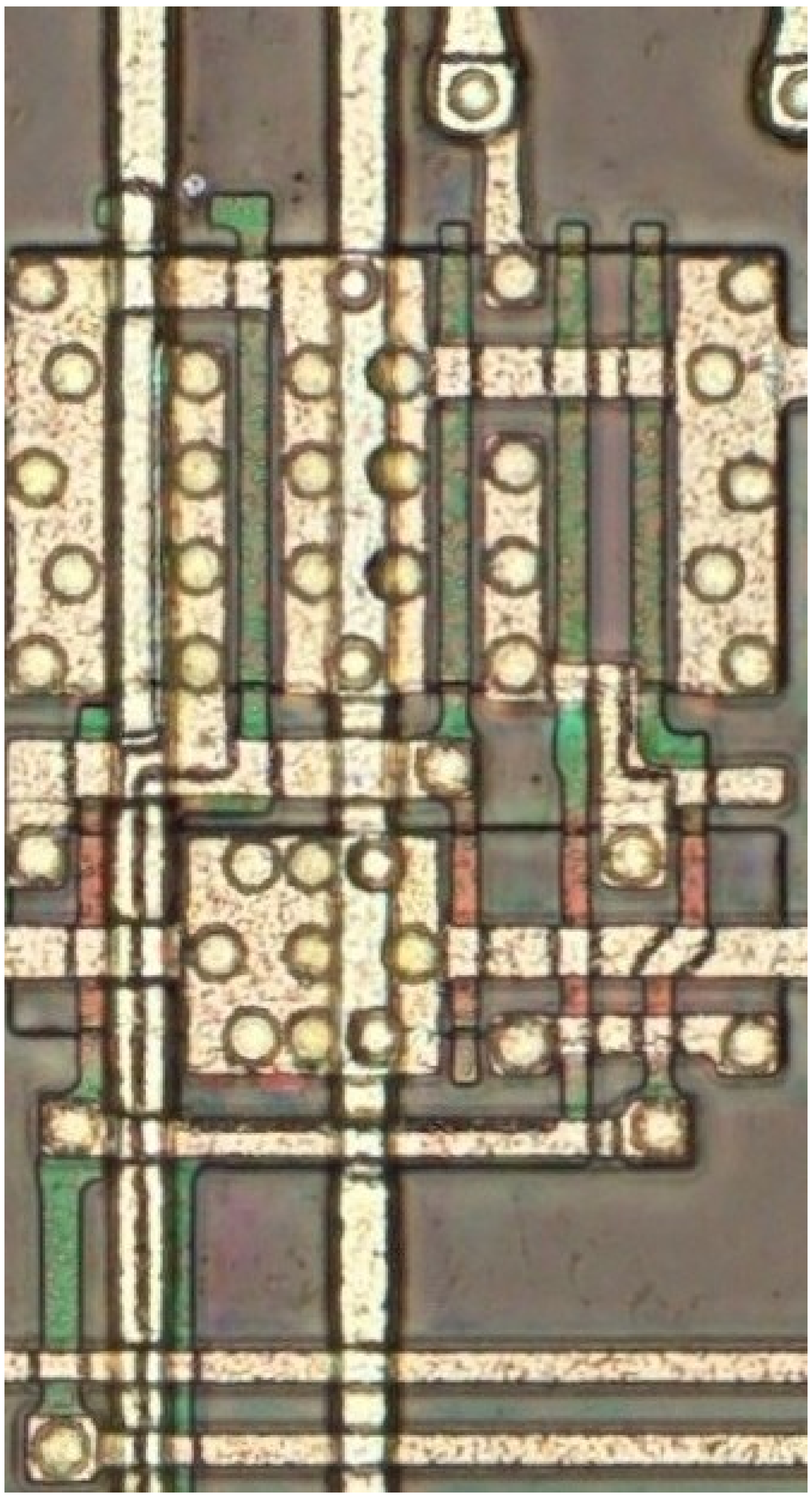


Same cell on
LA32
(MB87136A)

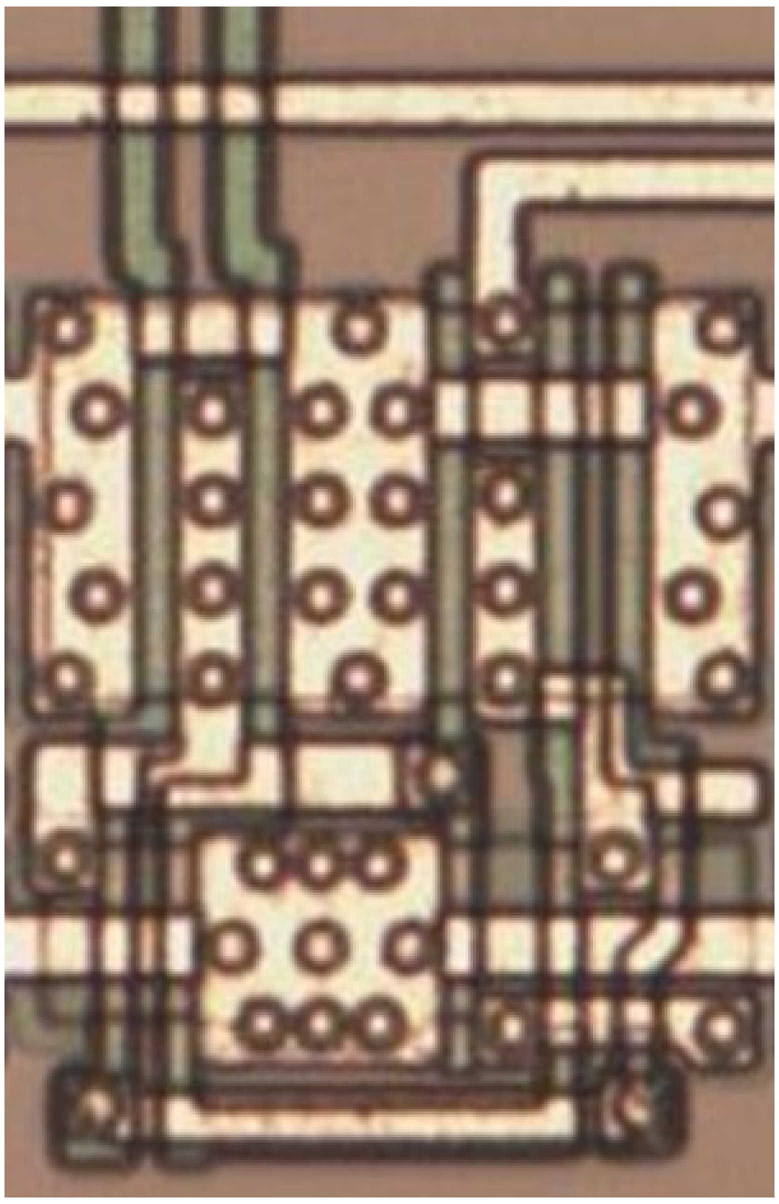


#000000ff Cell type 10

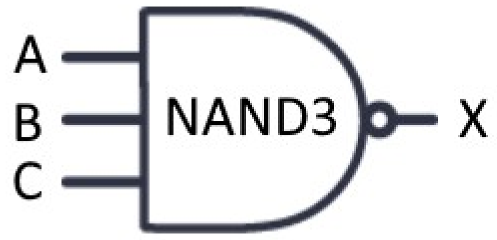




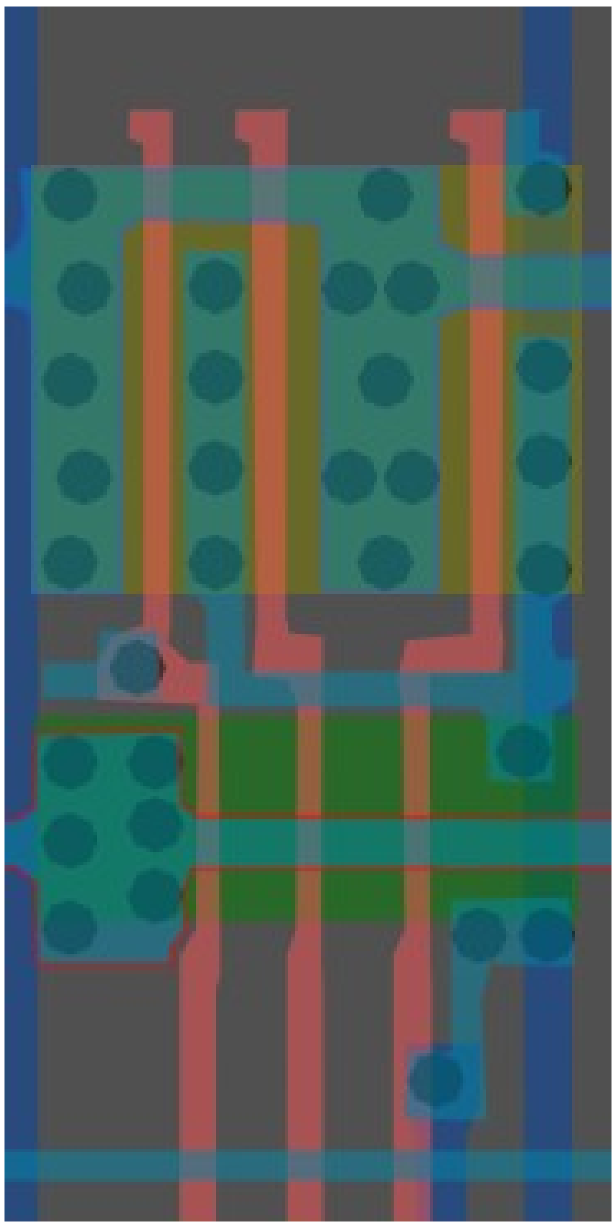
Same cell on LA32 (MB87136A)

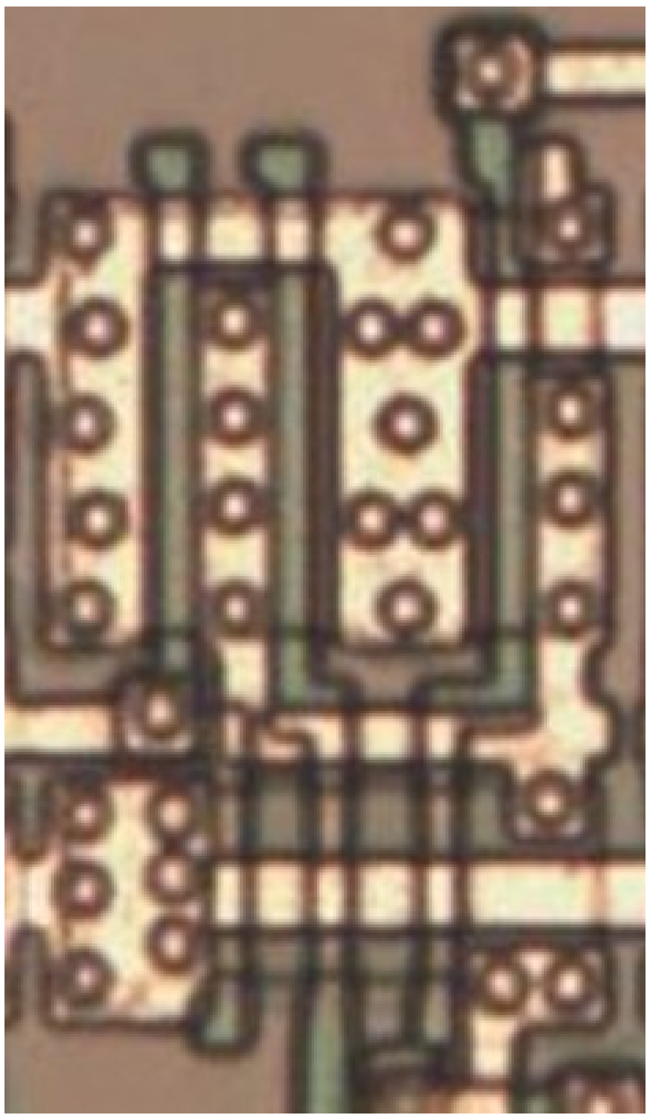


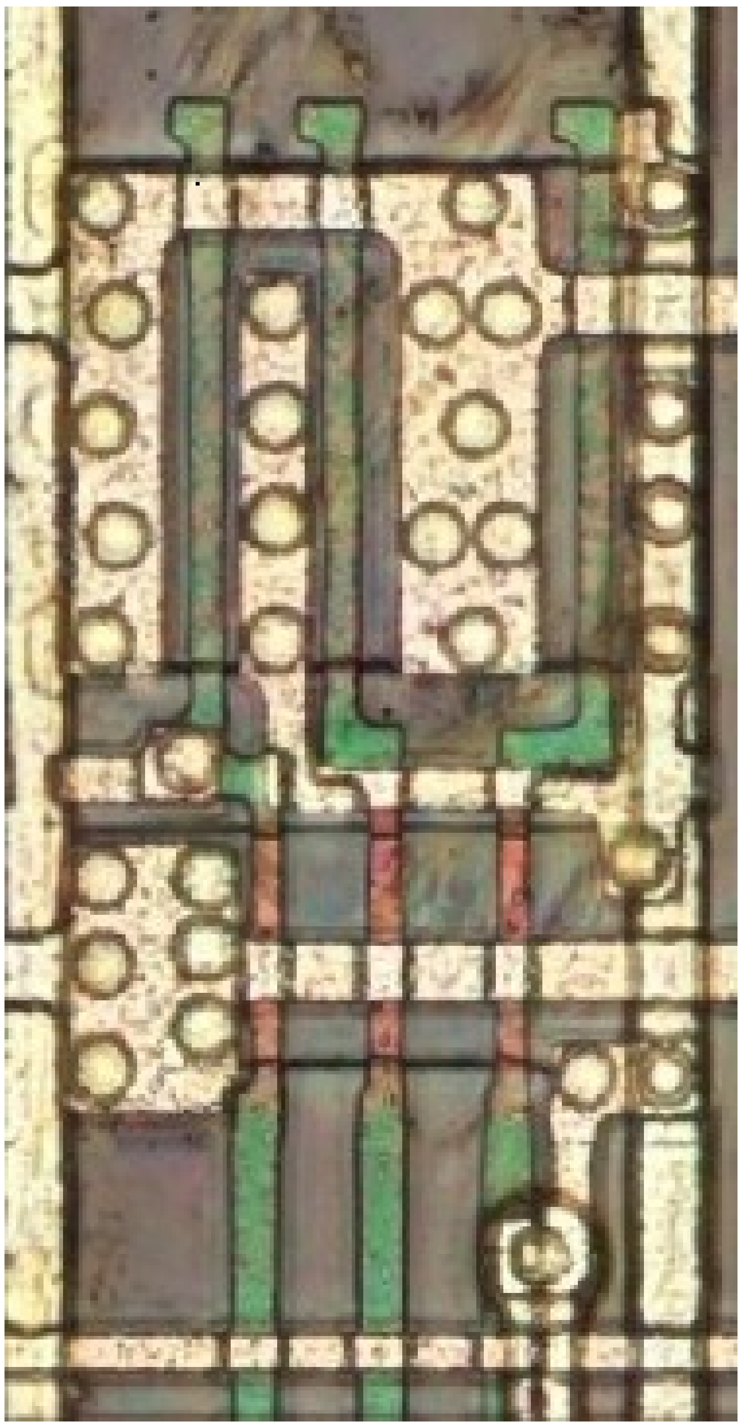
#f3ffffff 3-Input NAND Cell 11



A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

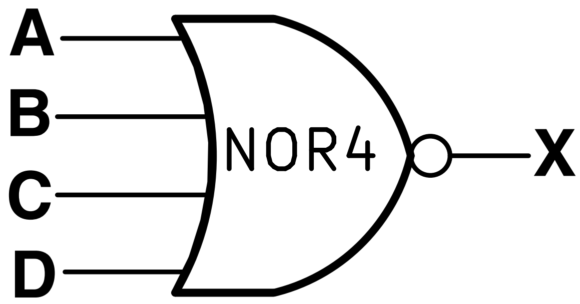




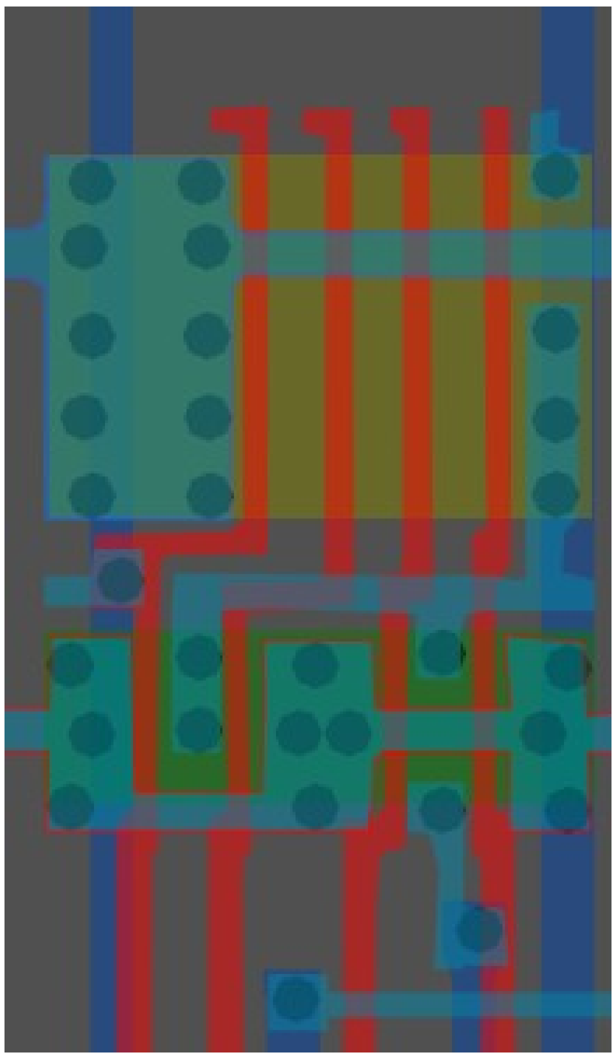


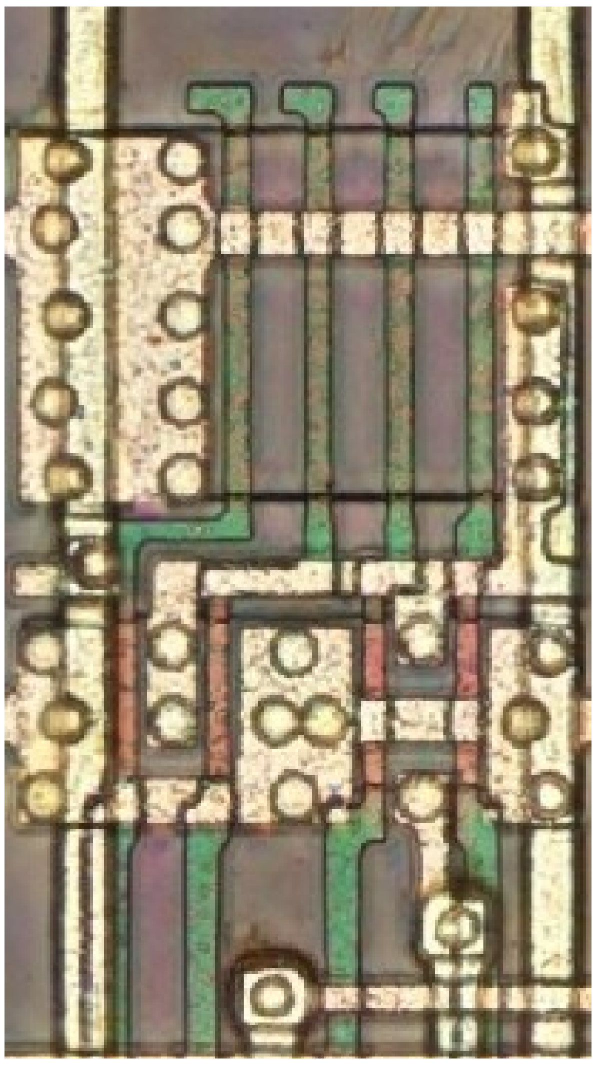
Same cell on
LA32
(MB87136A)

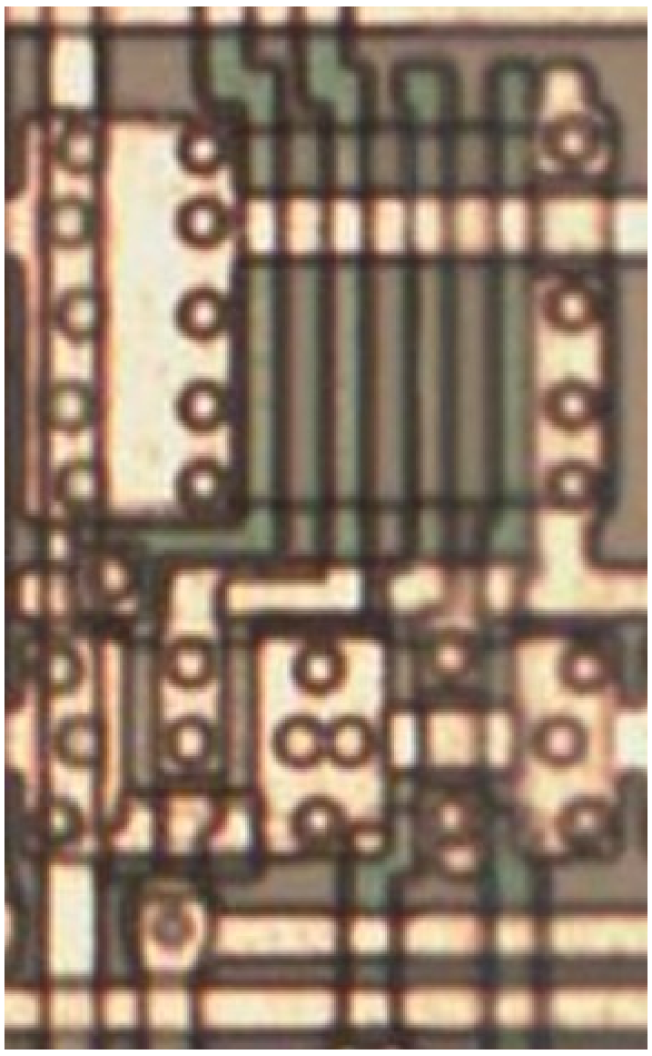
#007affff 4-Input NOR Cell 12



A	B	C	D	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0





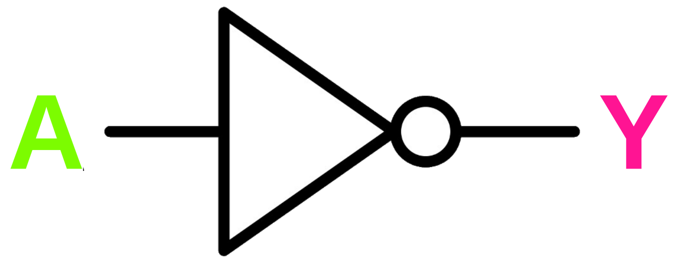


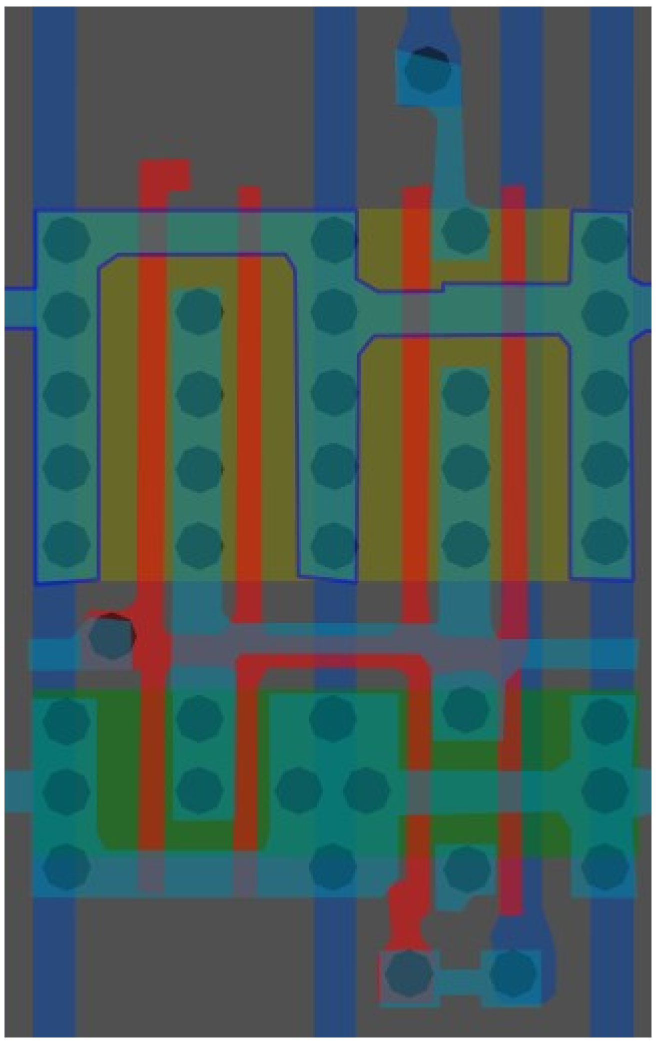
Same cell on LA32 (MB87136A)

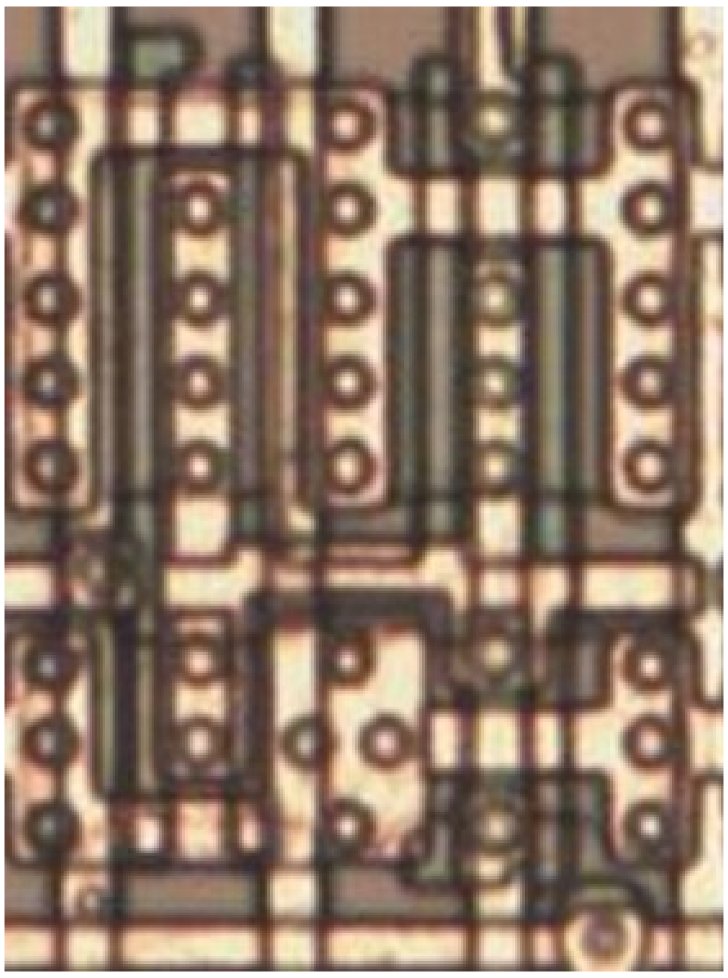
#b9ff43ff

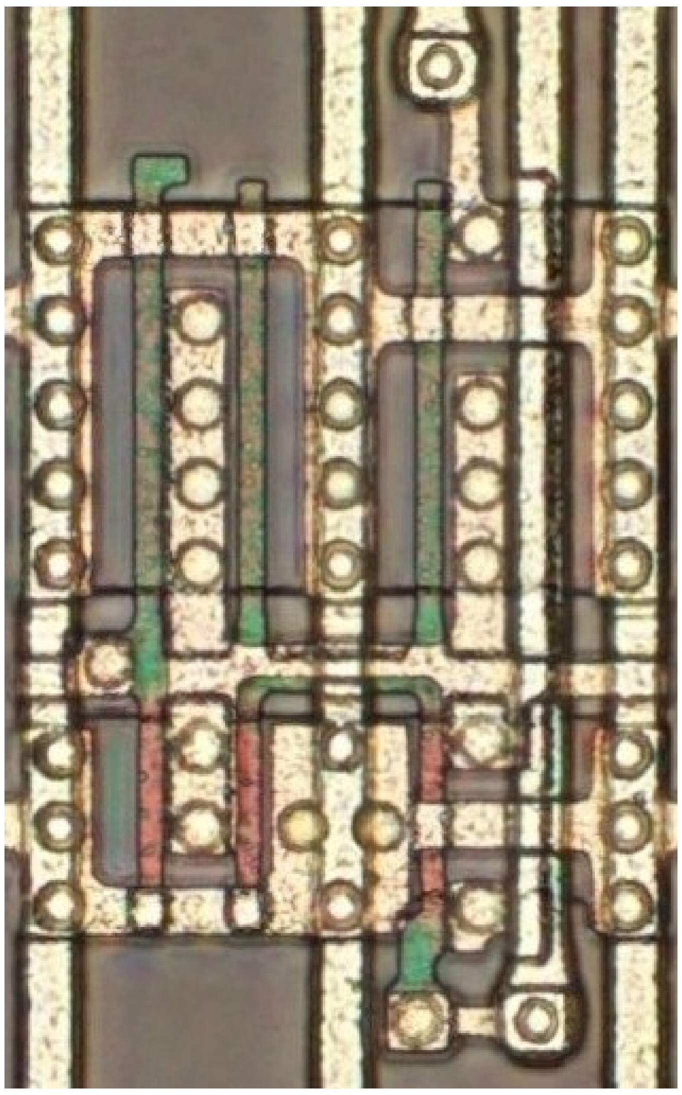
Inverter/NOT x4

Cell 13a





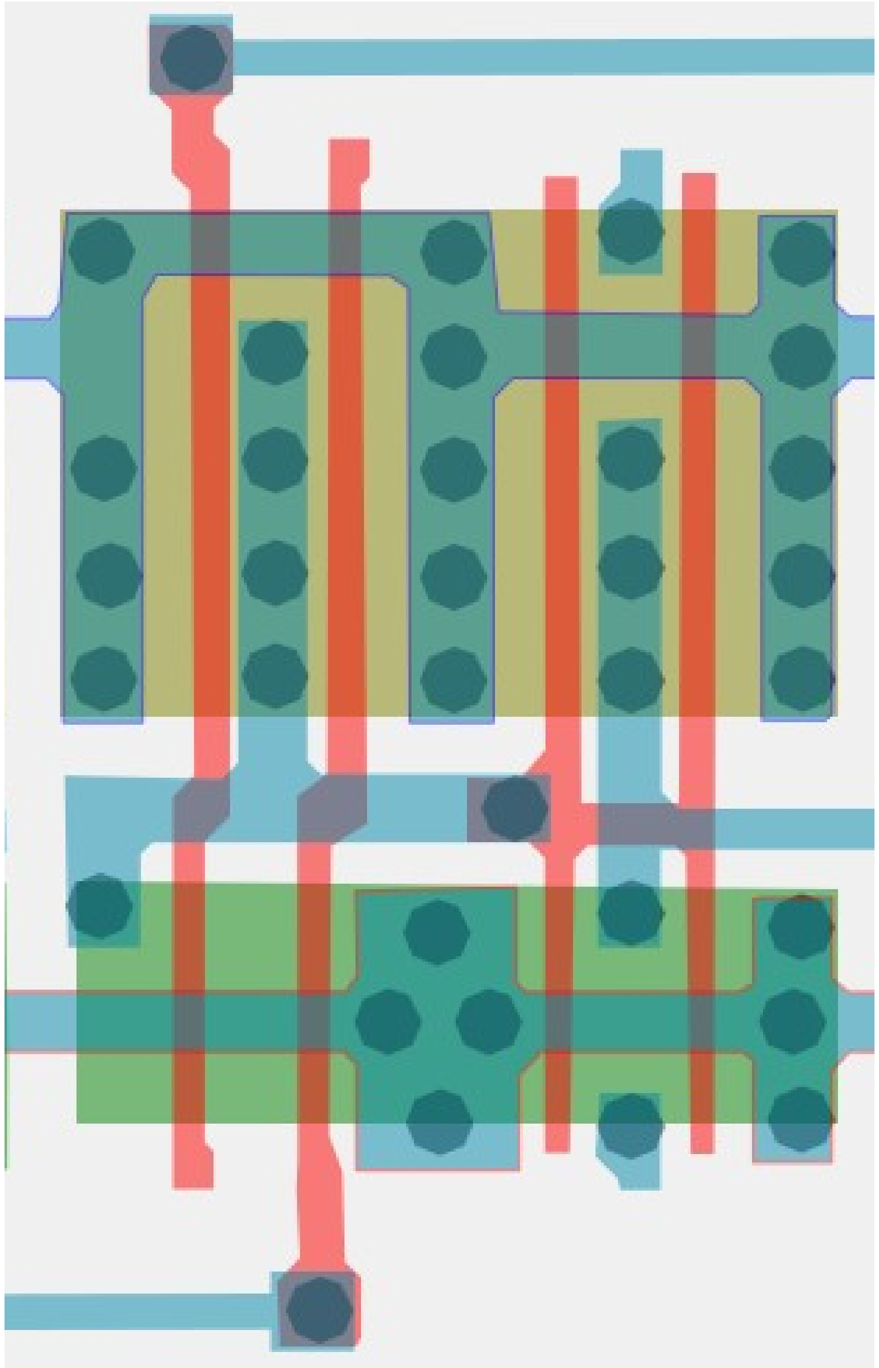


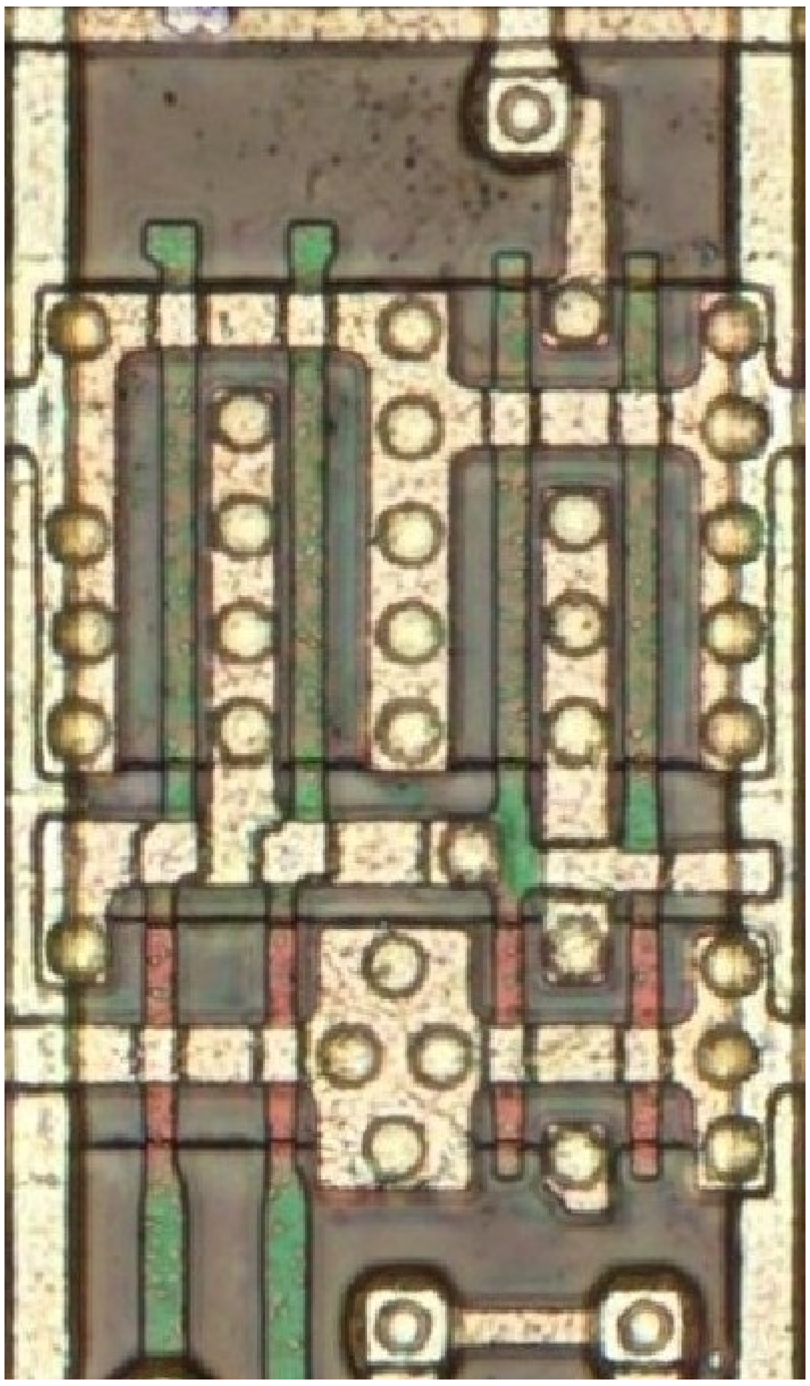


Same cell on
LA32
(MB87136A)

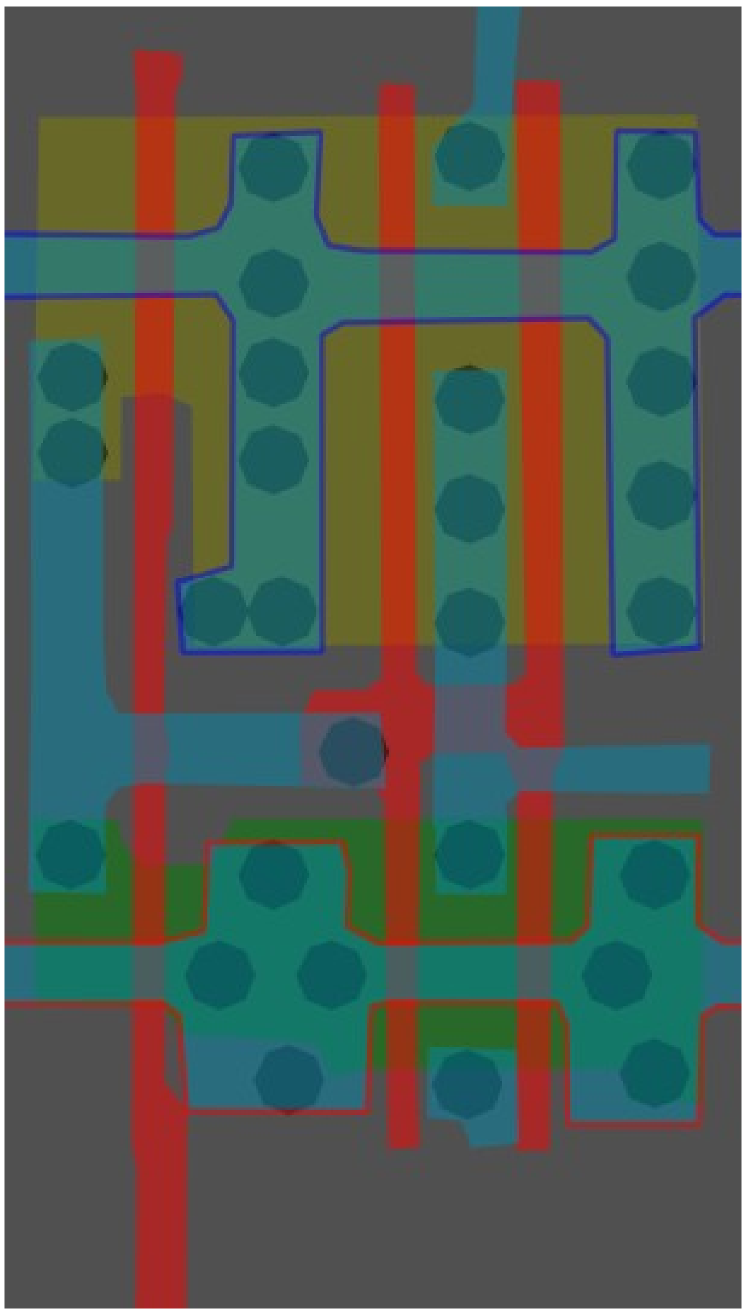
#bfa867ff

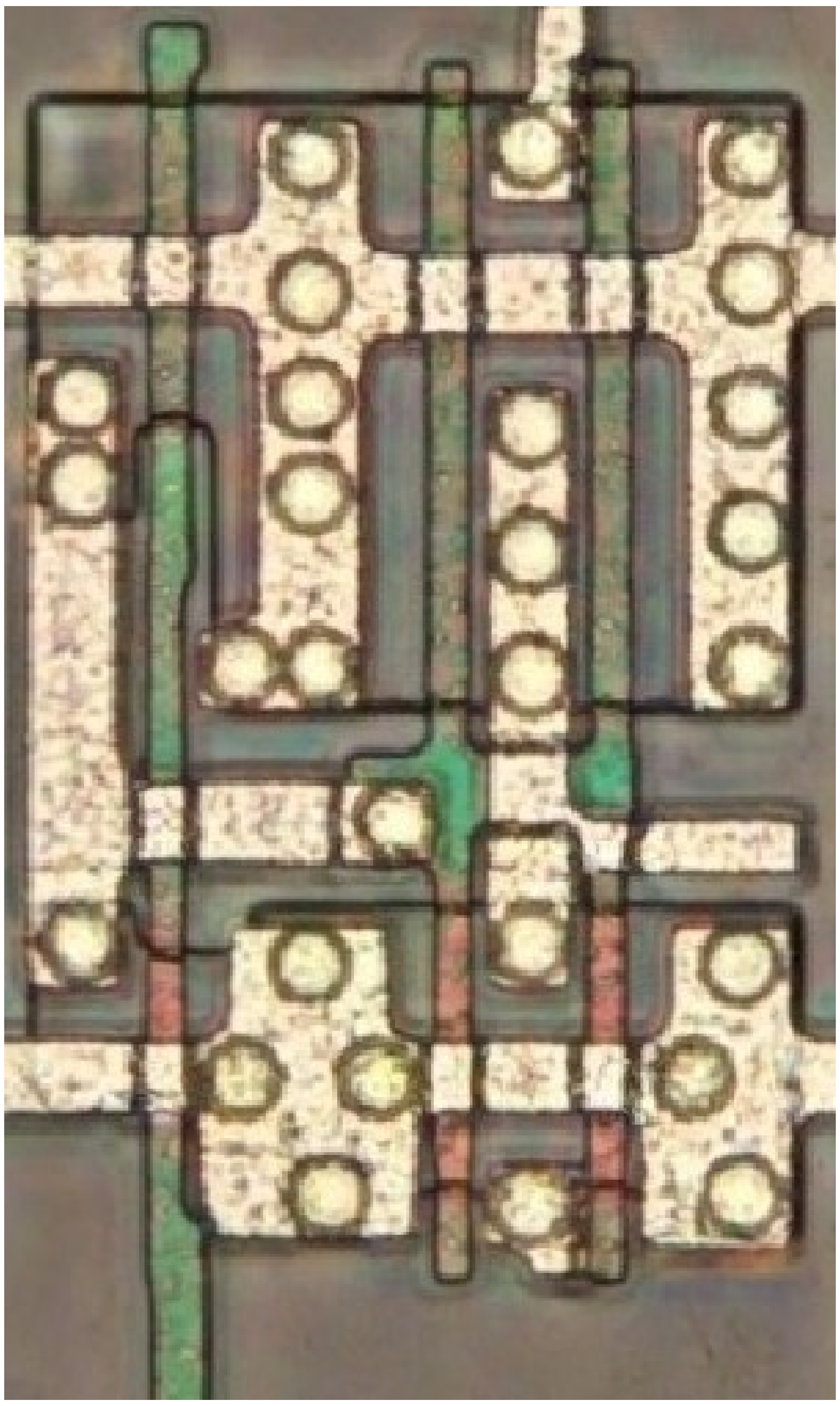
Cell type 13b



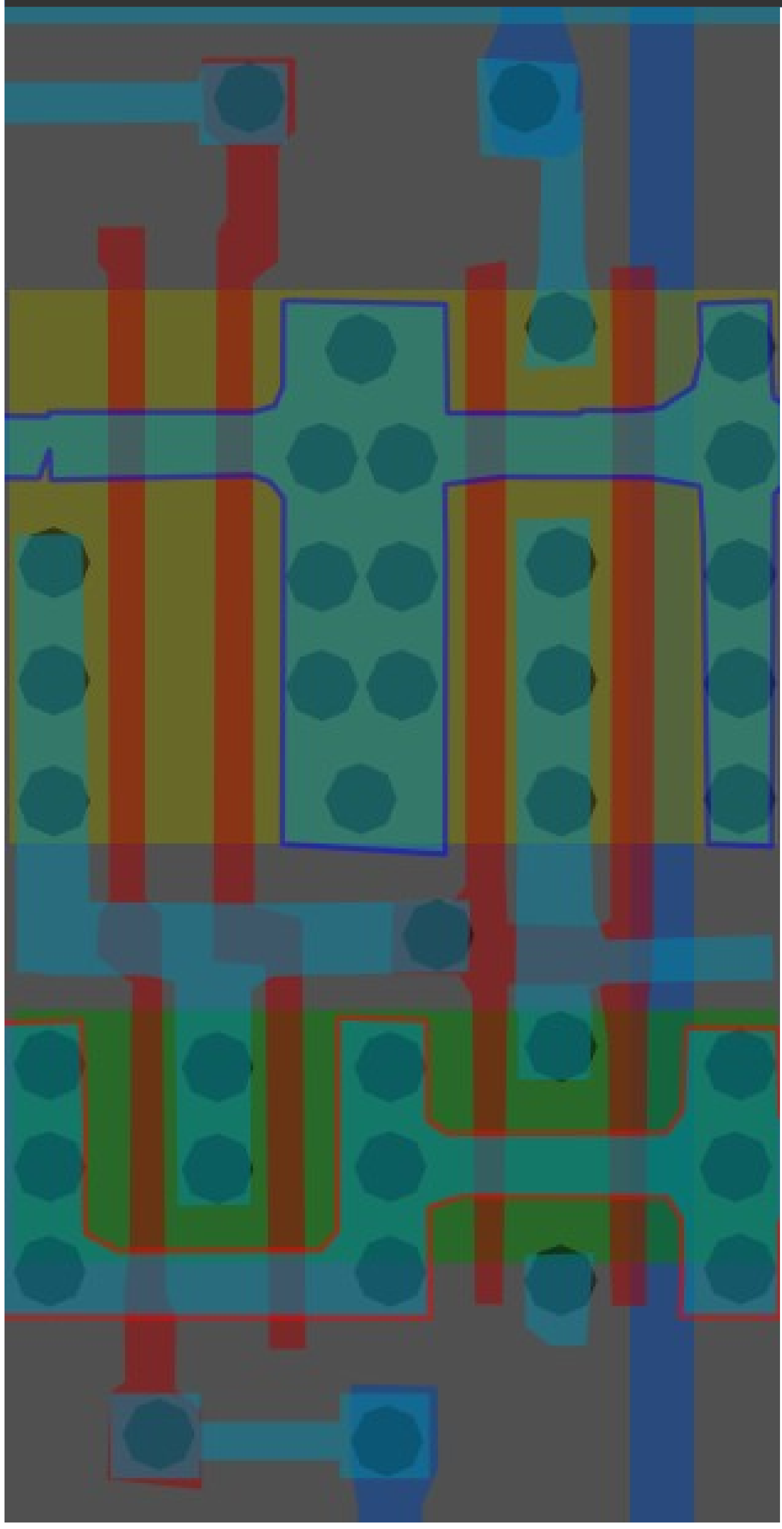


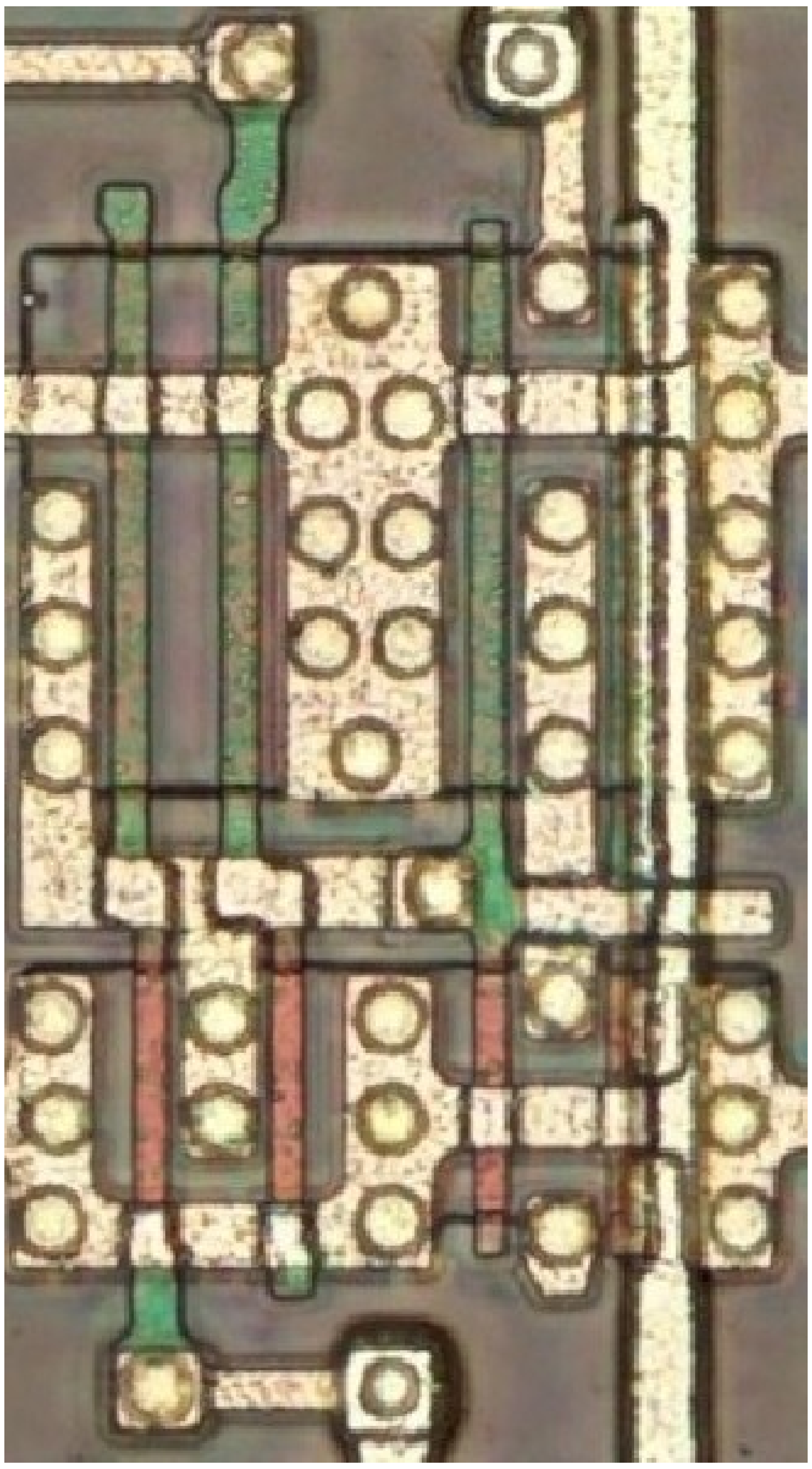
#a03ac0ff **Cell type 13c**



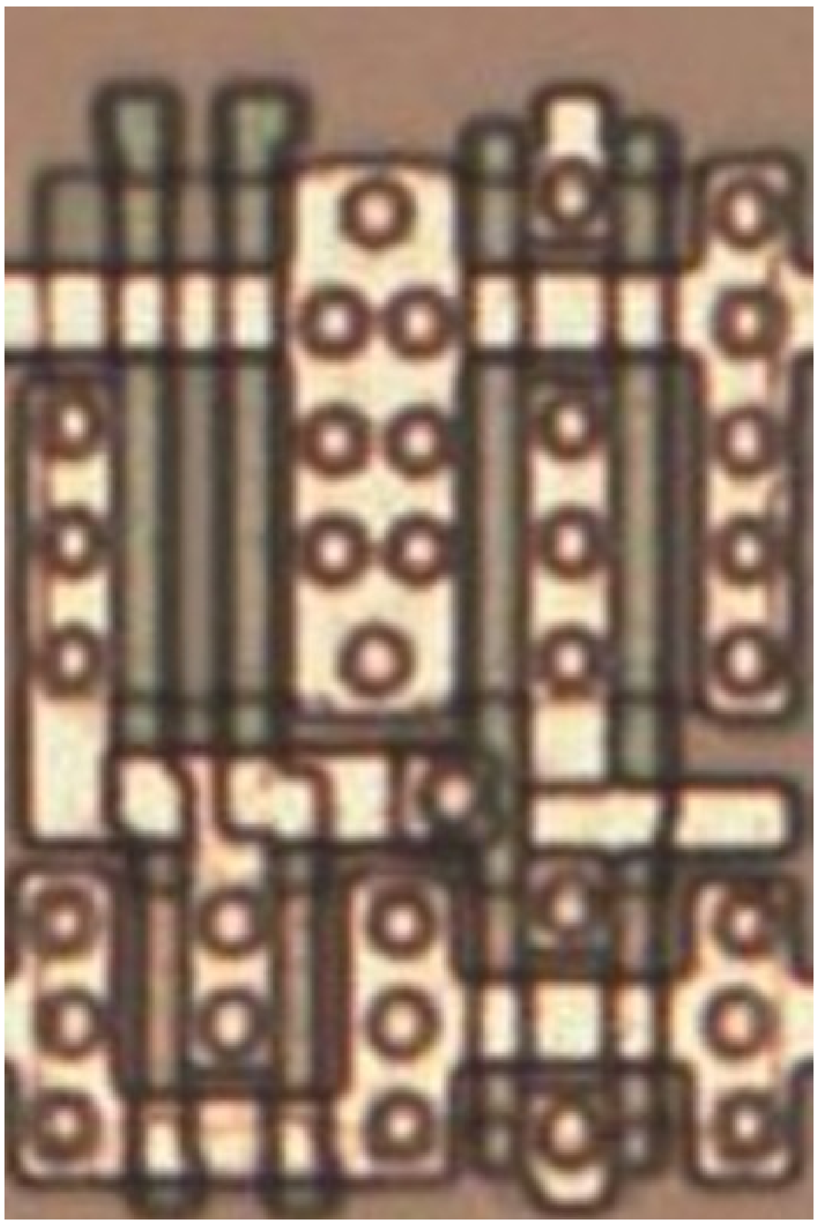


#ffbcacff Cell type 13d



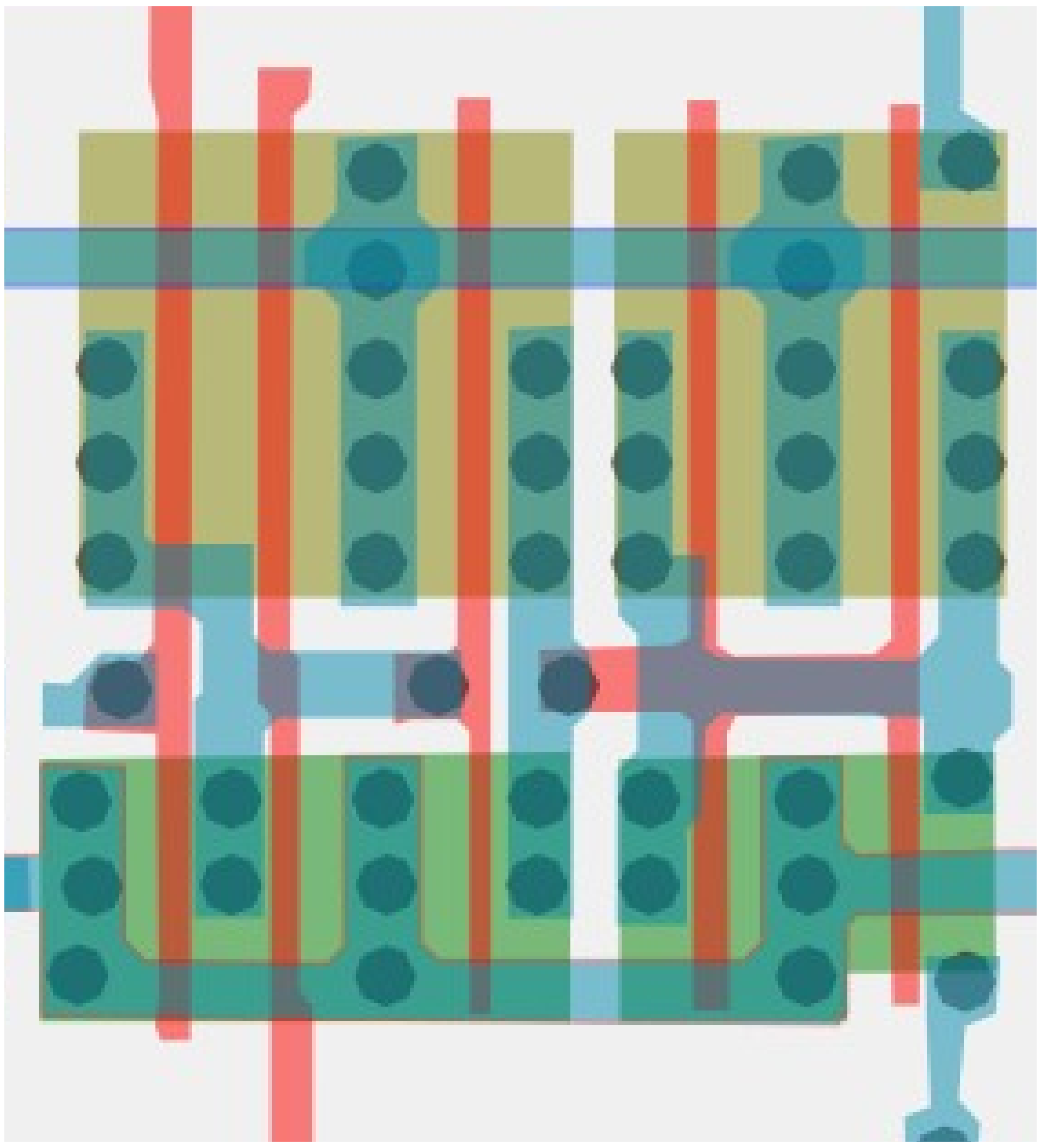


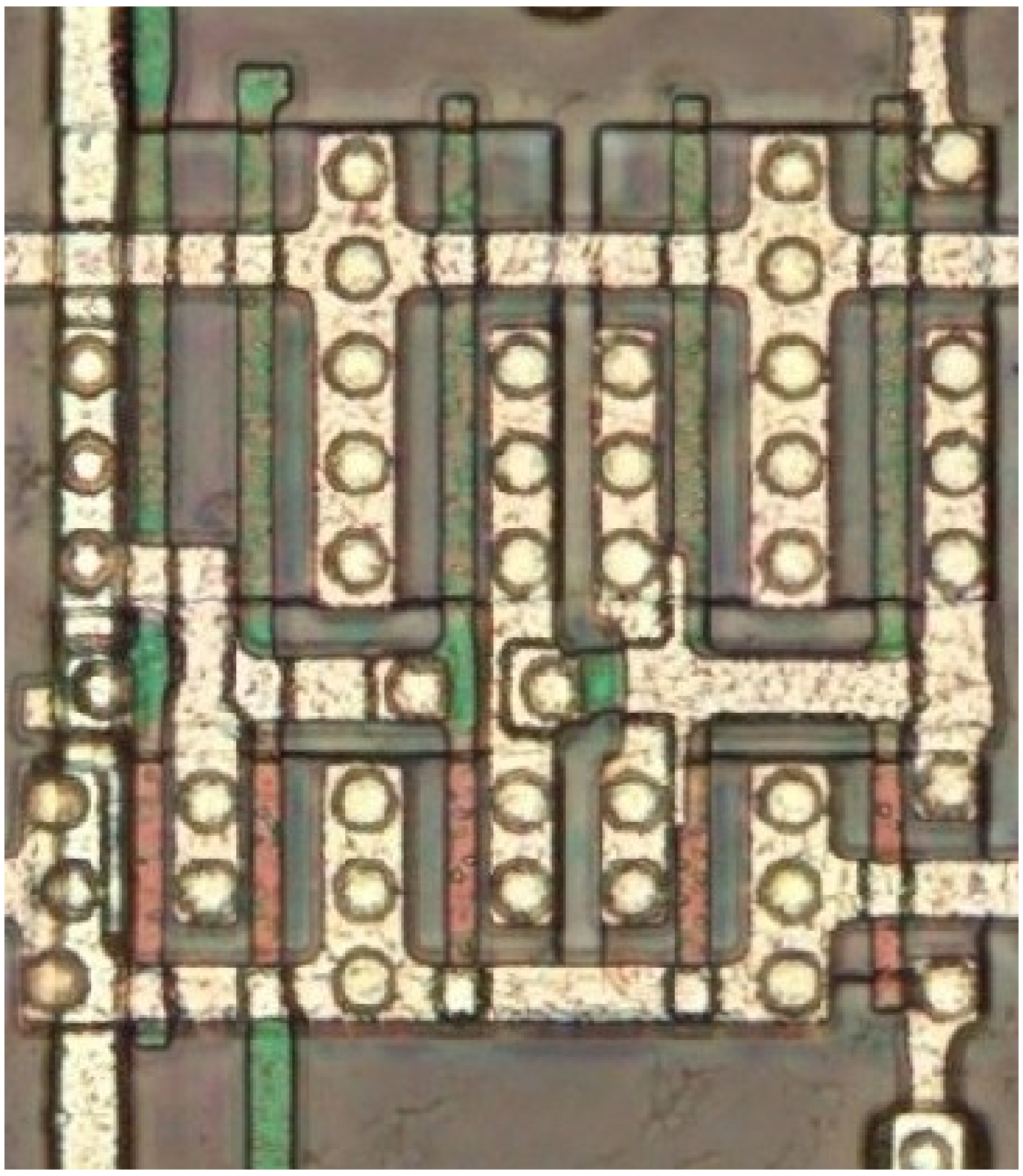
Same cell on
LA32
(MB87136A)



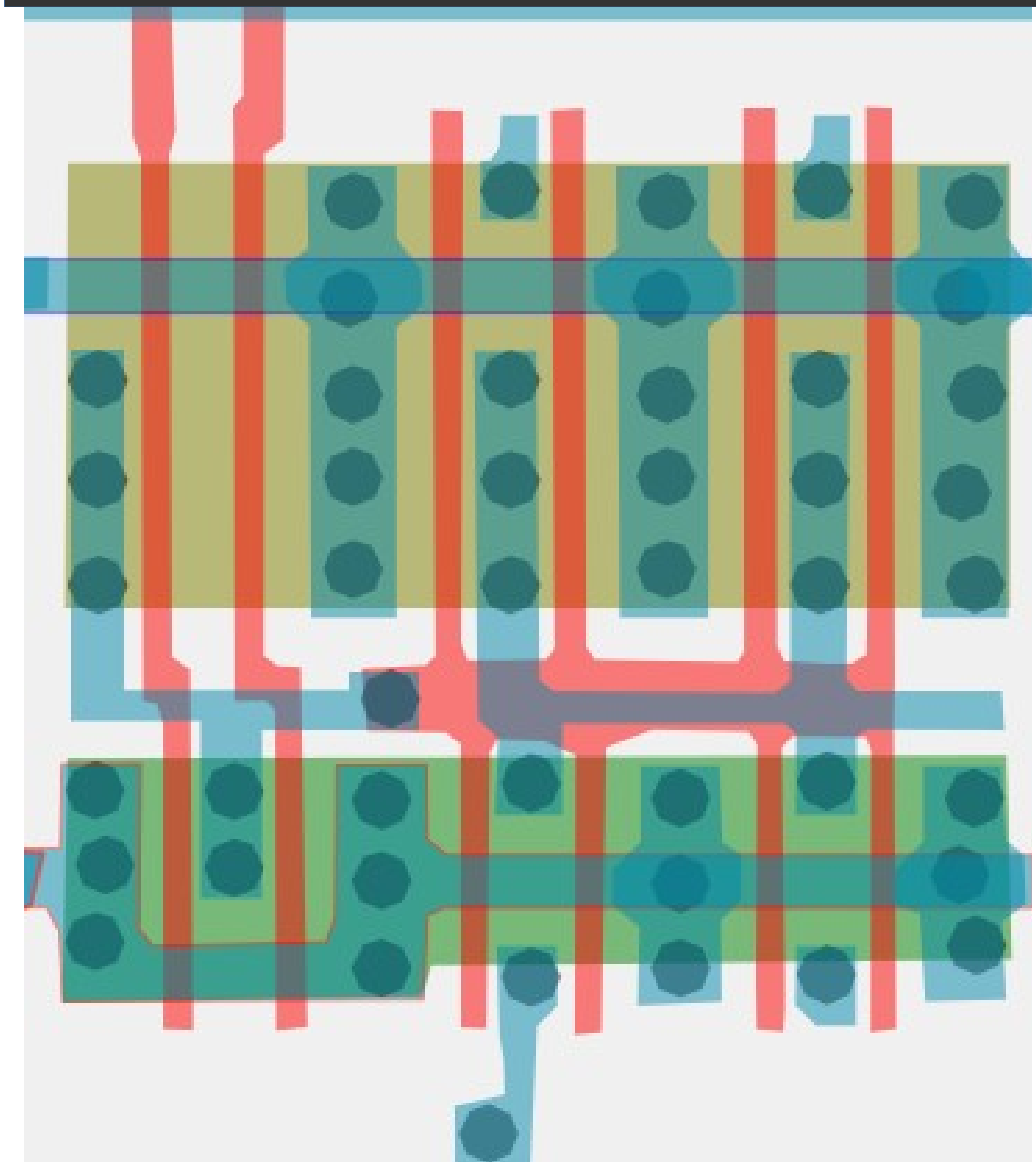
#ff7b94ff

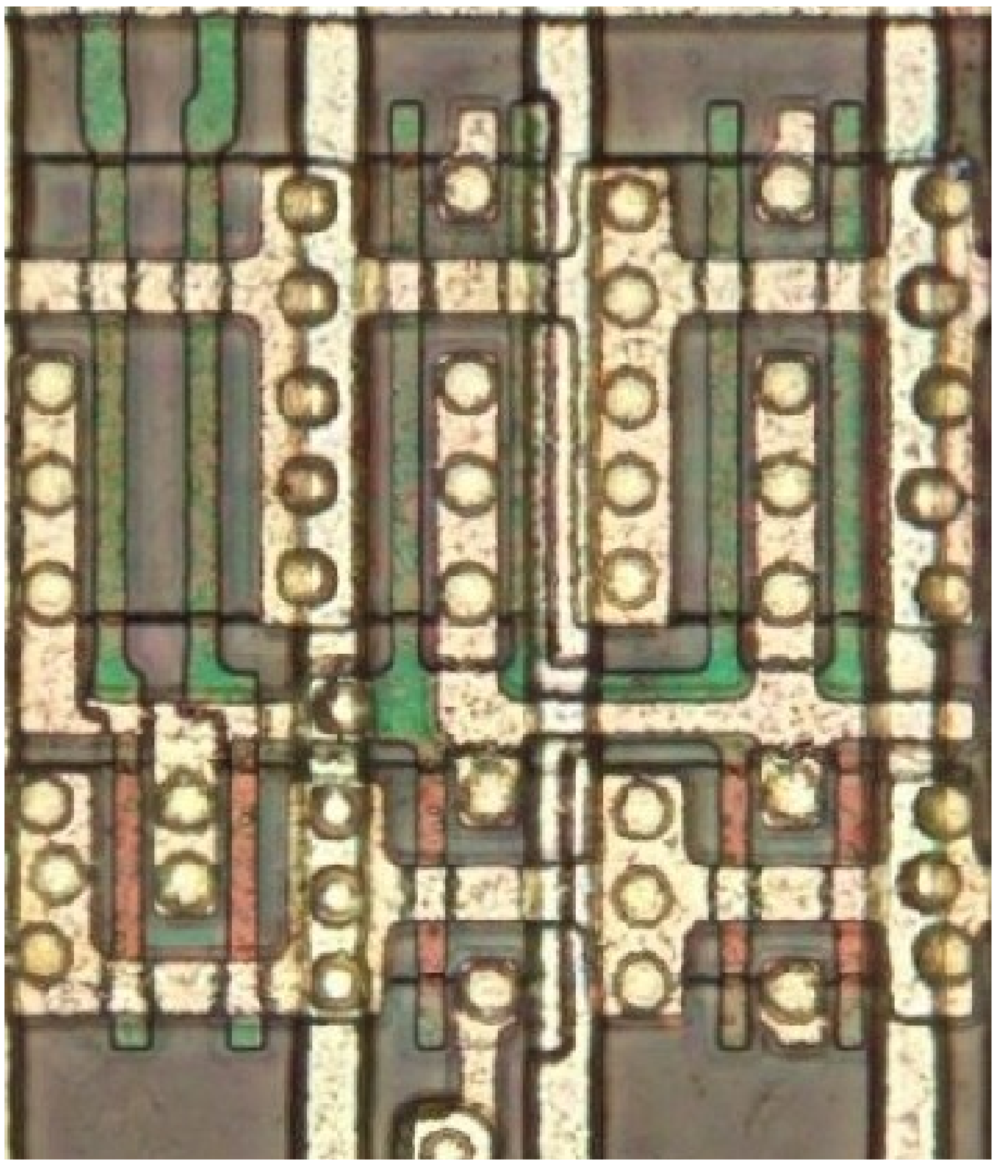
Cell type 13e





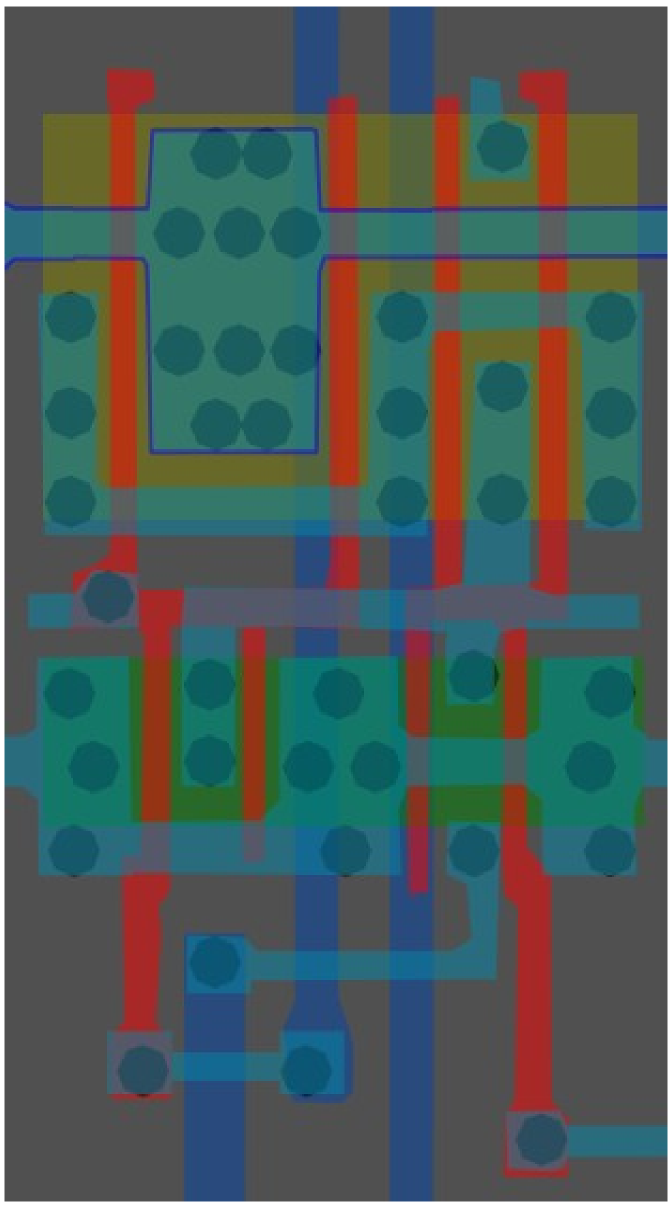
#fba968ff Cell type 13f

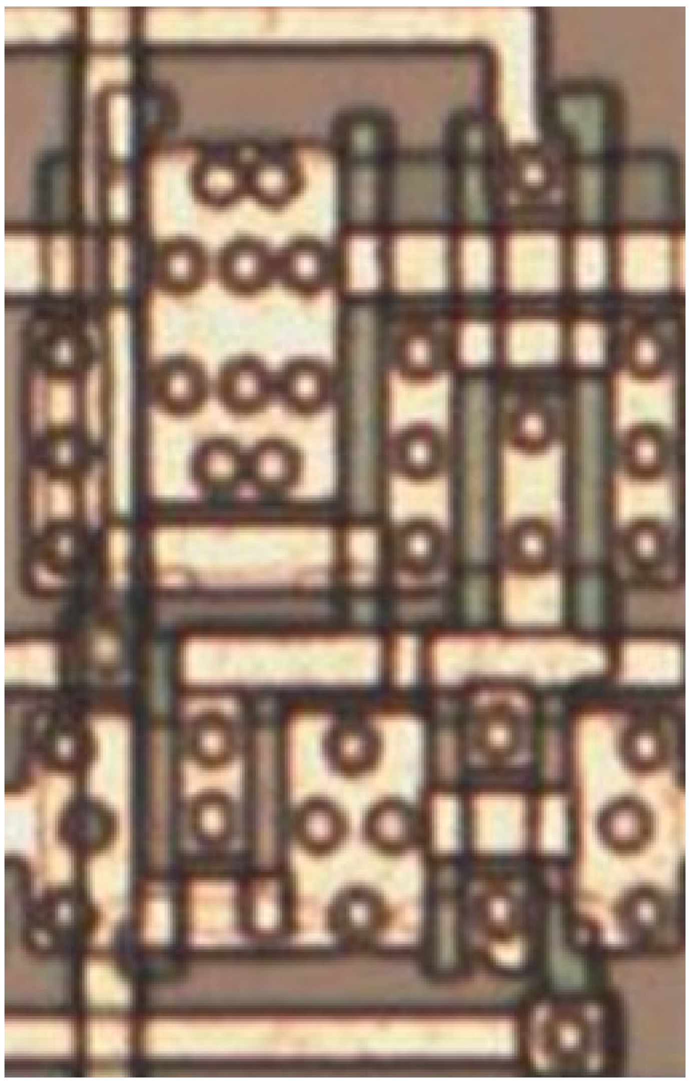




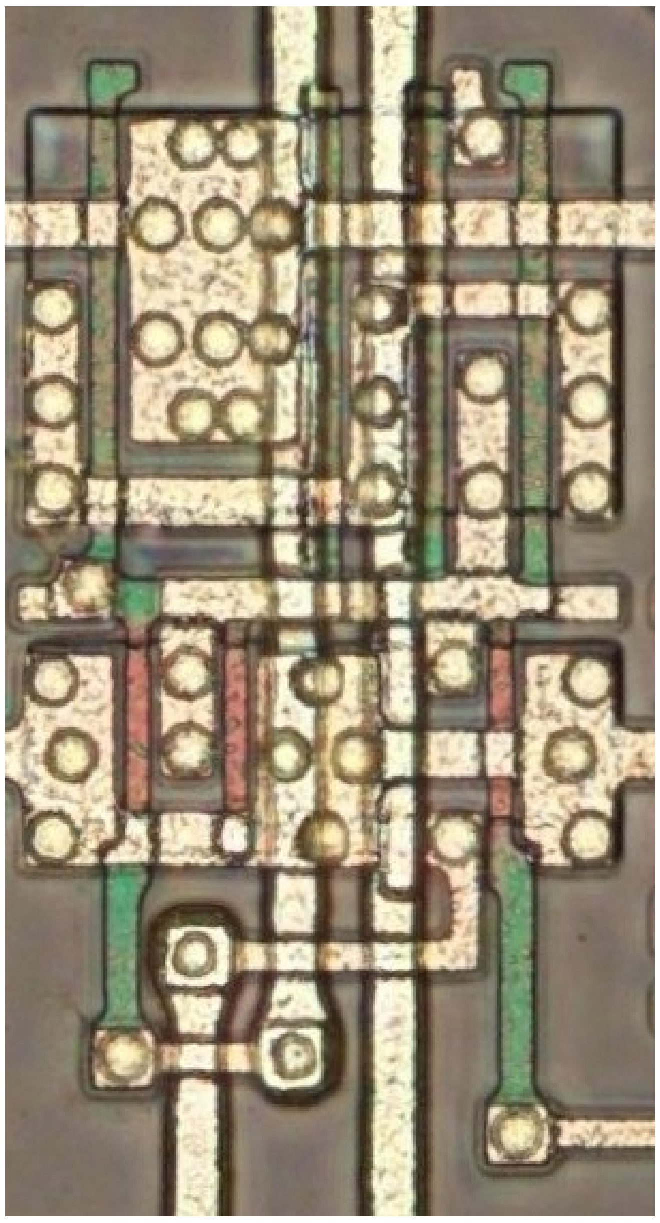
#f3ffa7ff

Cell type 14

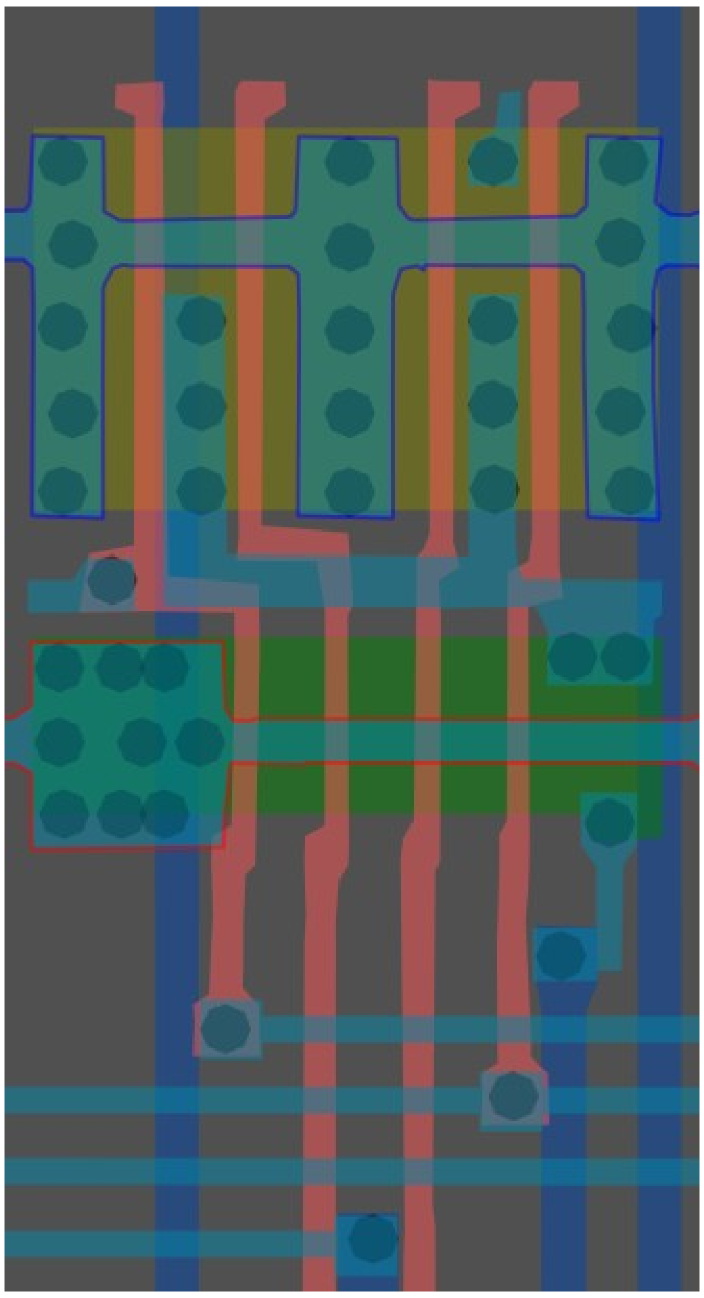


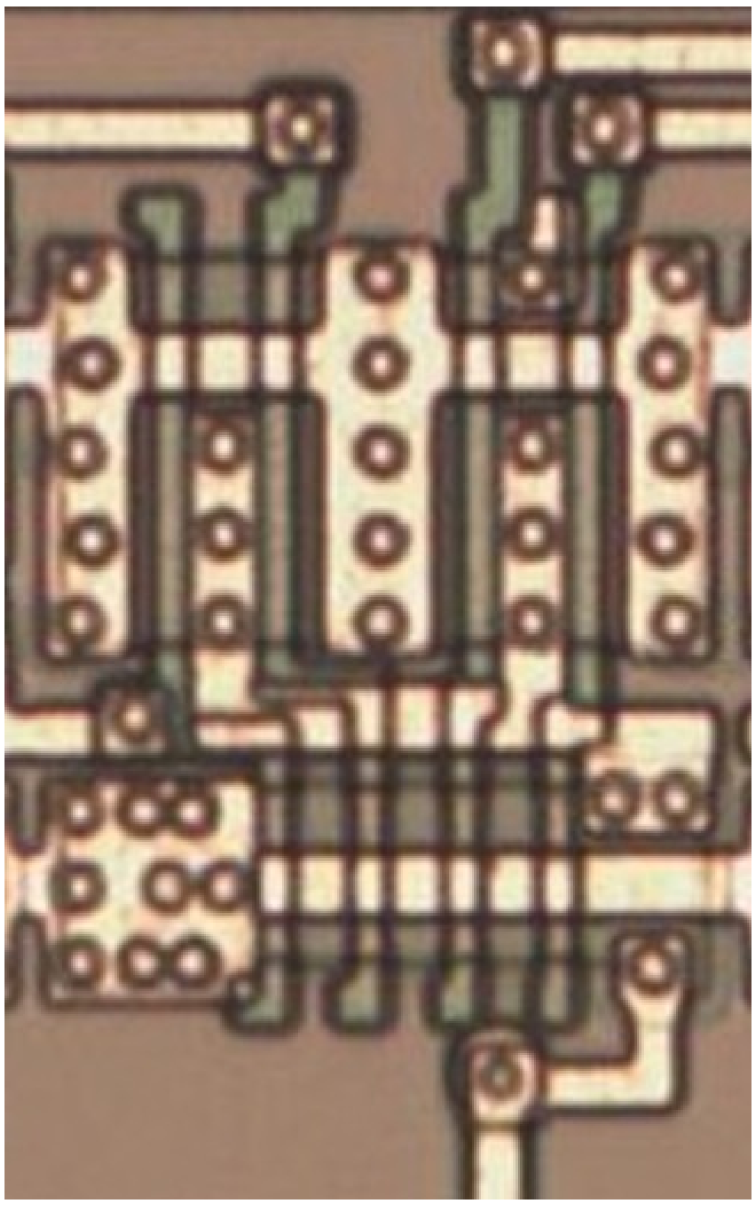


Same cell on
LA32
(MB87136A)

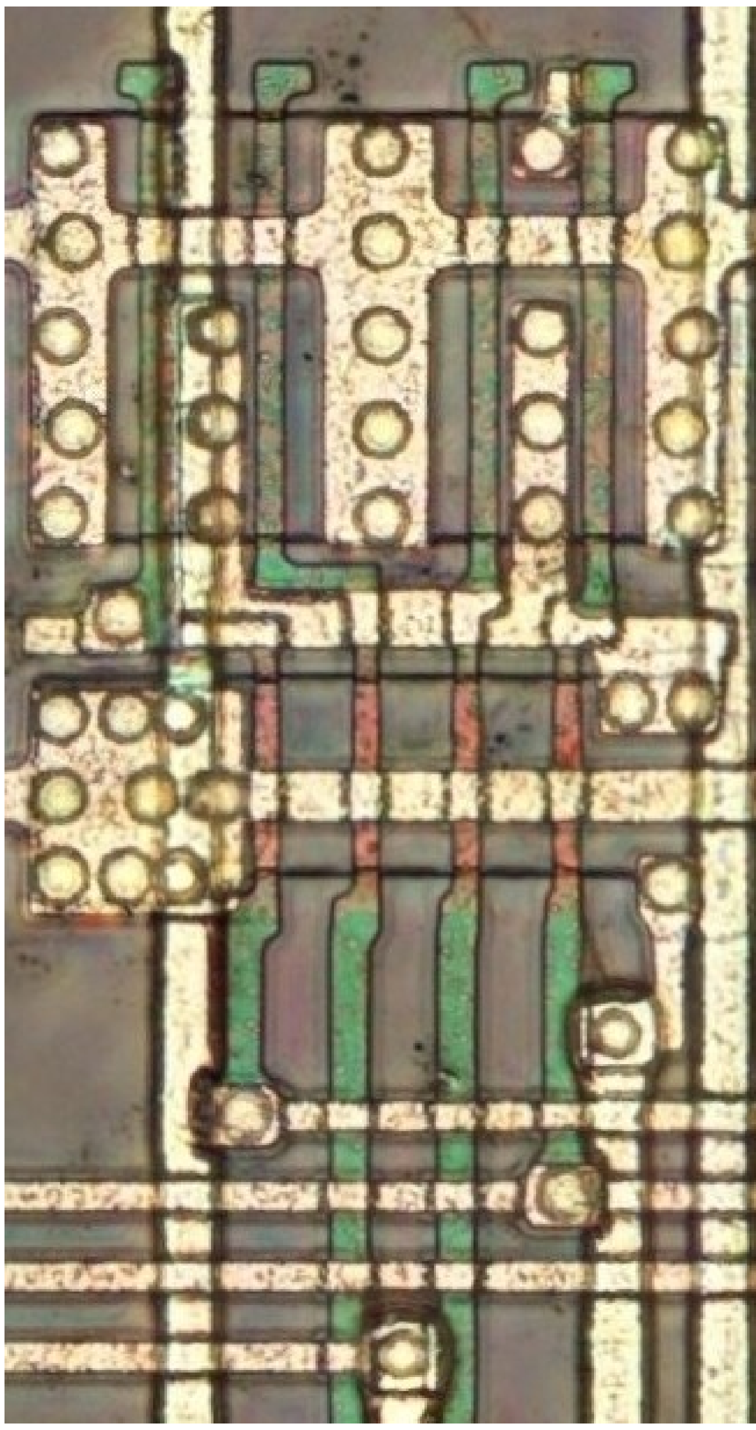


#00ffffff | 4-Input NAND | Cell 15



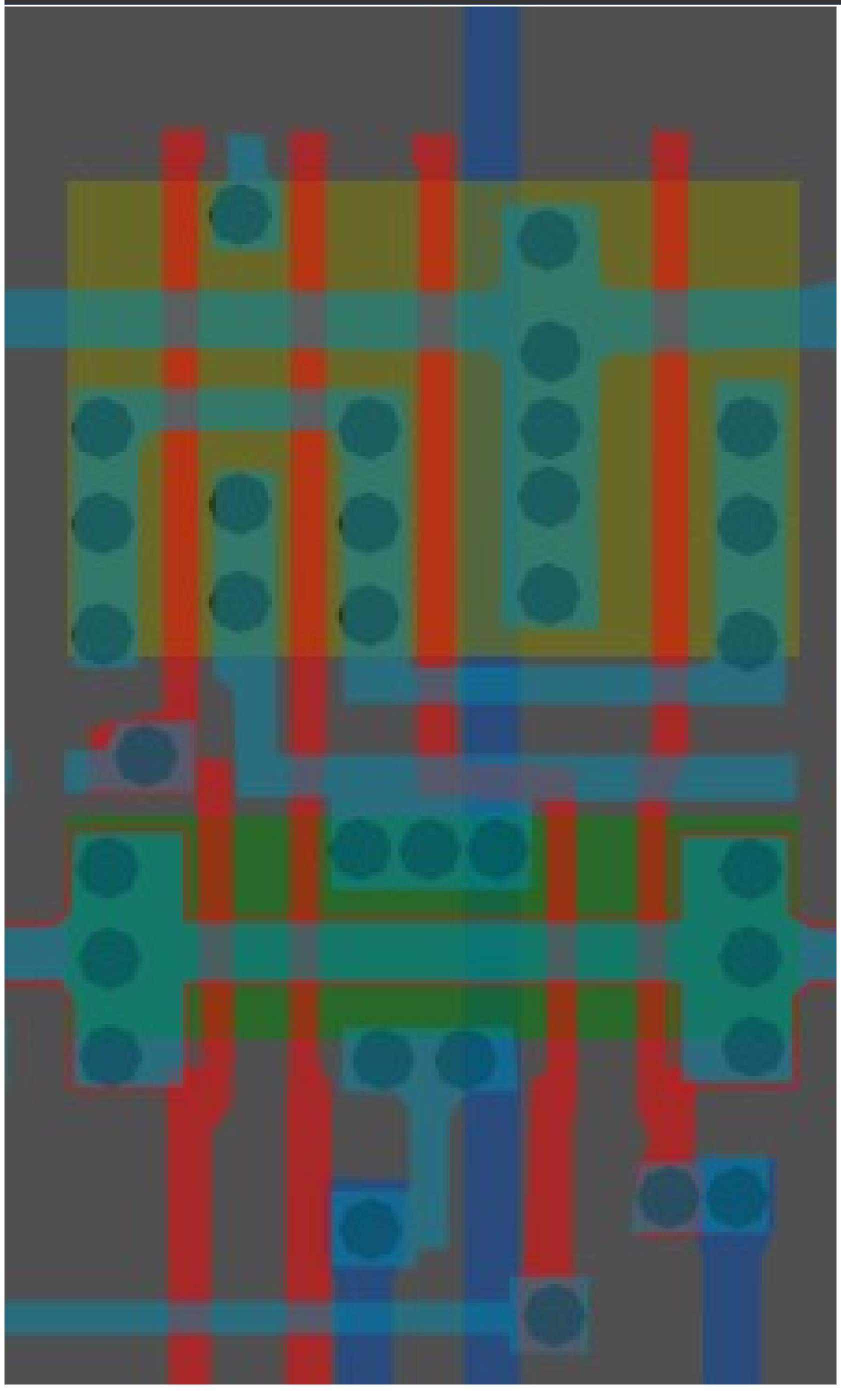


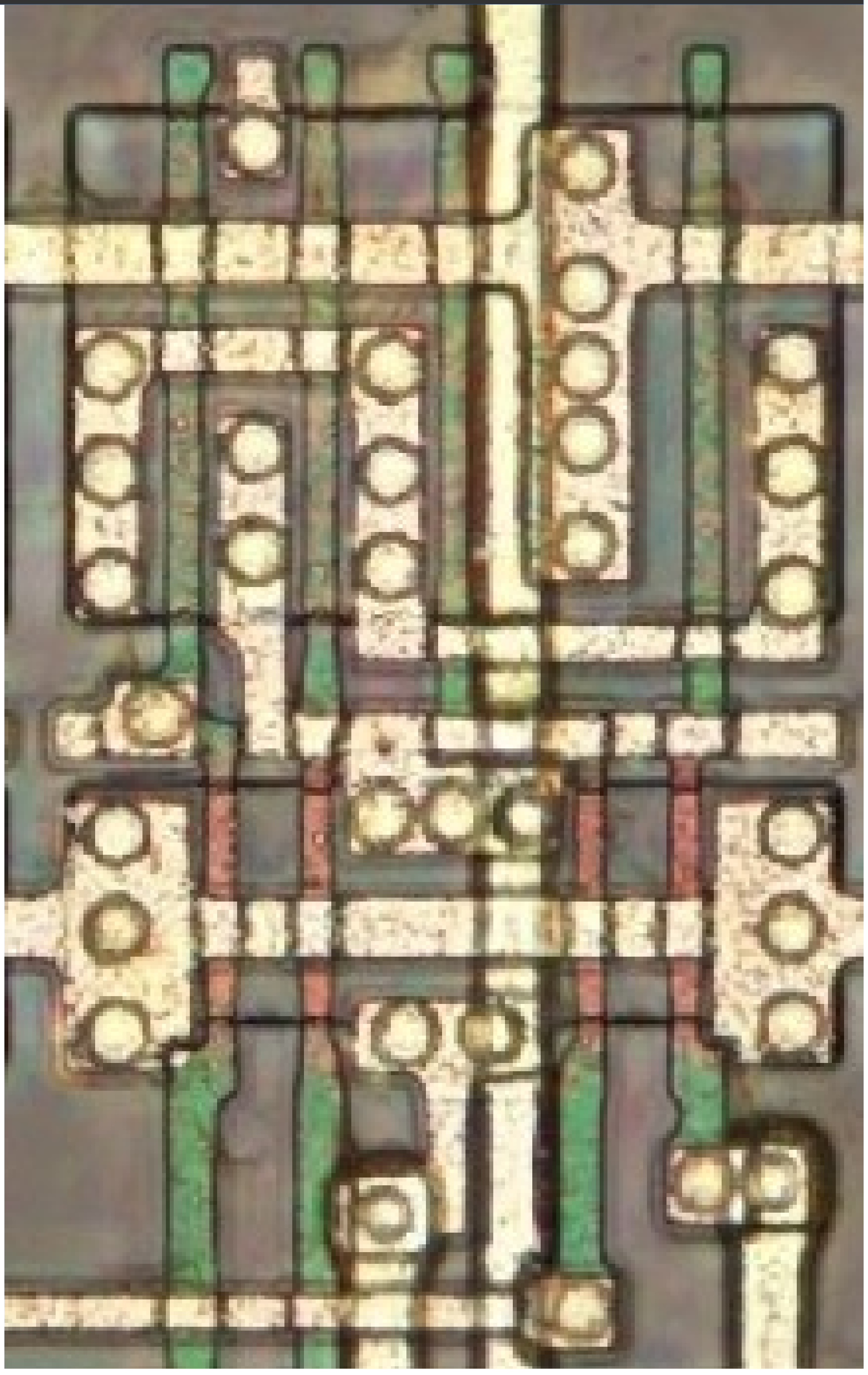
Same cell on
LA32
(MB87136A)



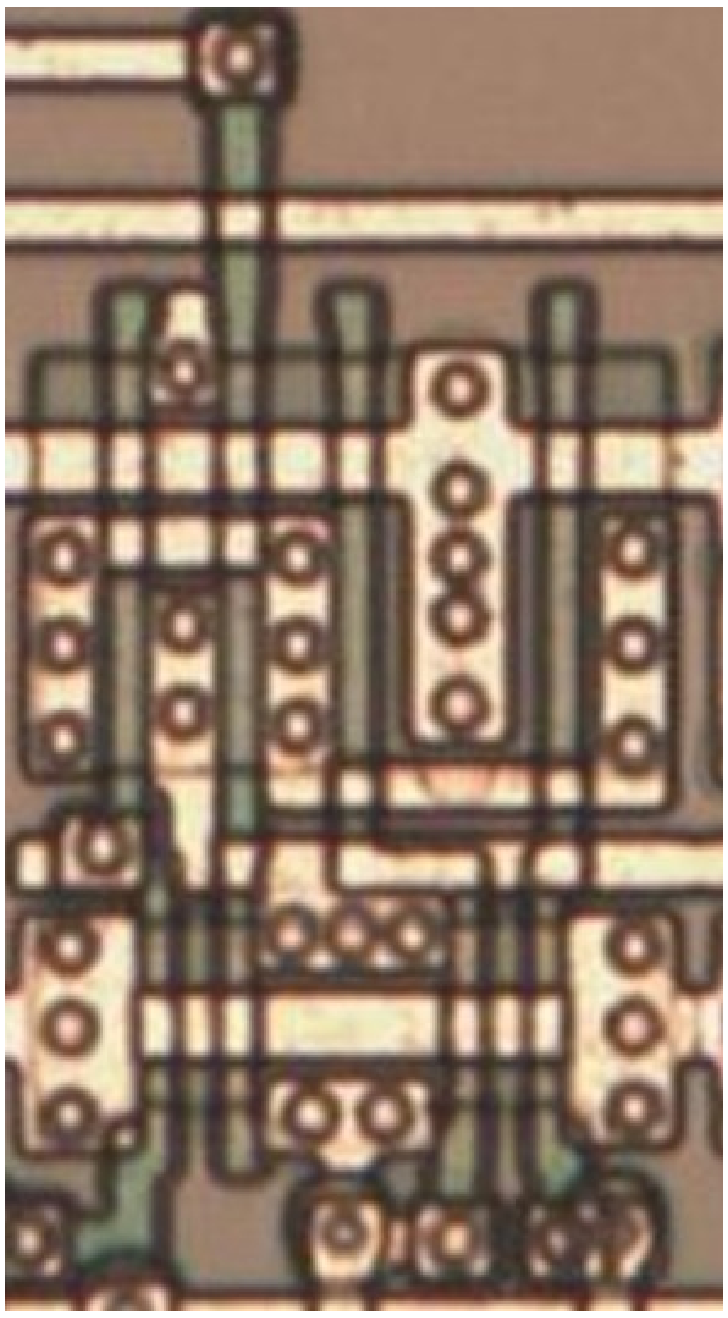
#ffb0ffff

Cell type 16



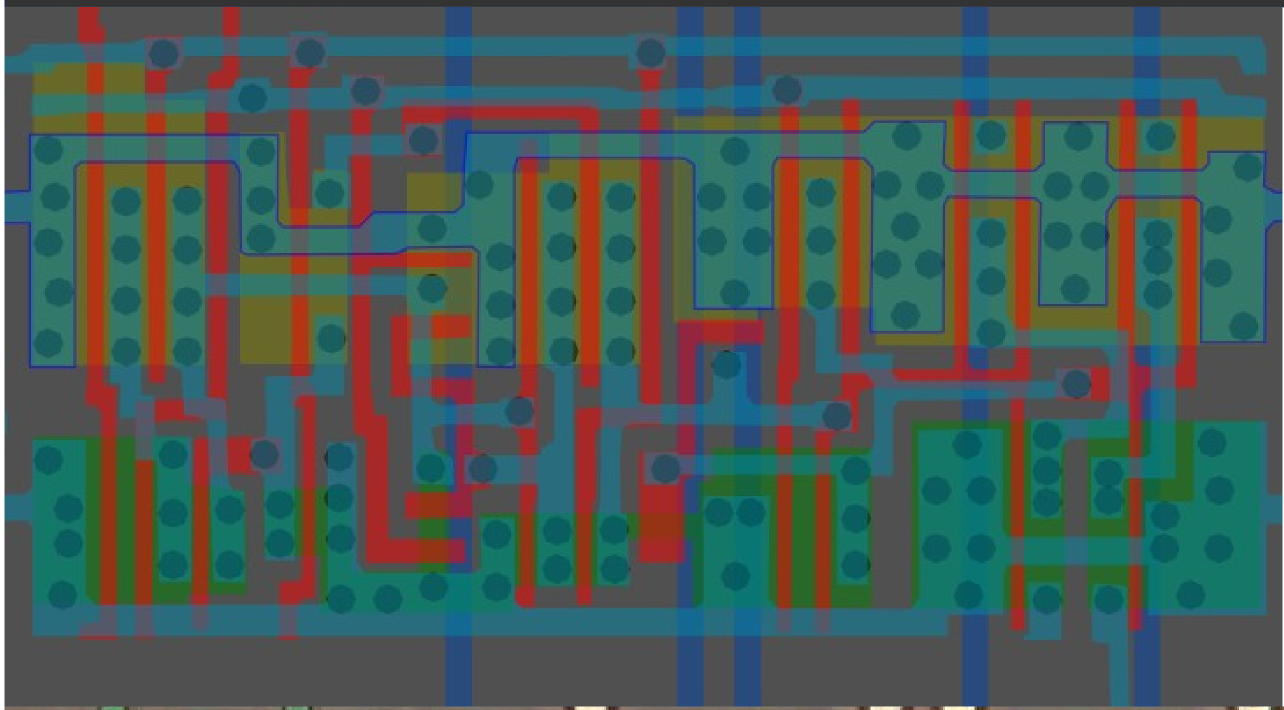


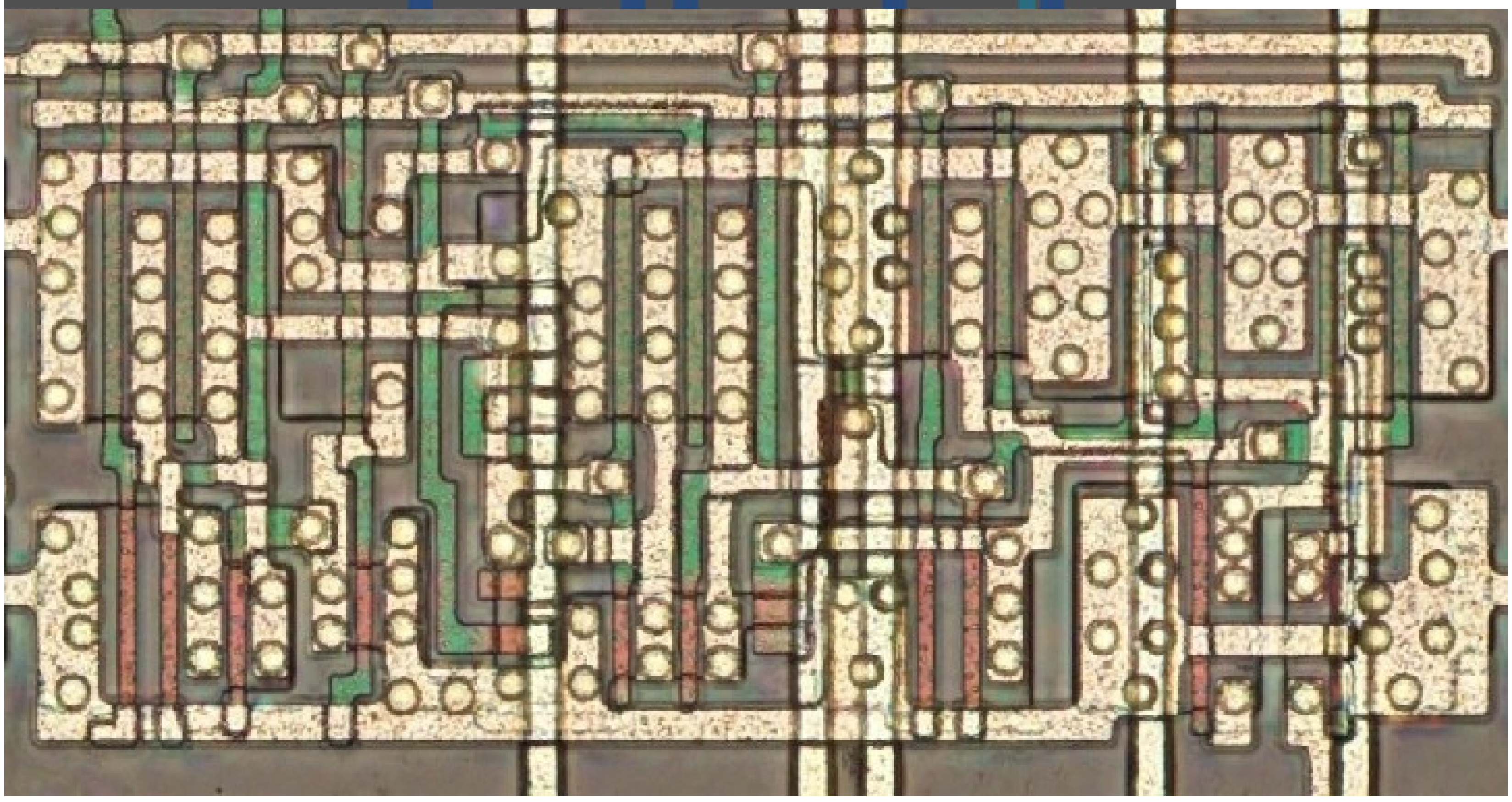
Same cell on
LA32
(MB87136A)

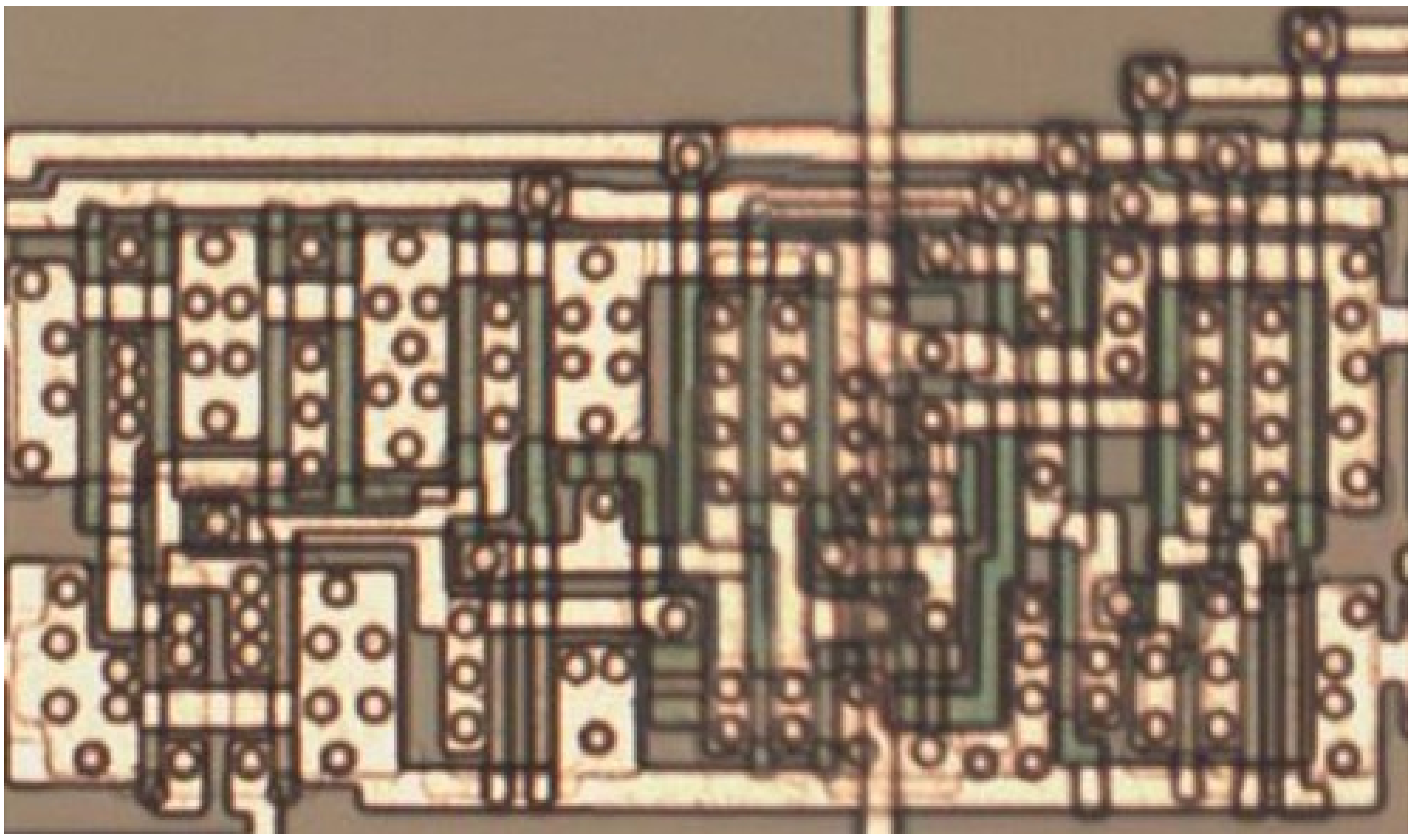


#0000ffff

Cell type 17

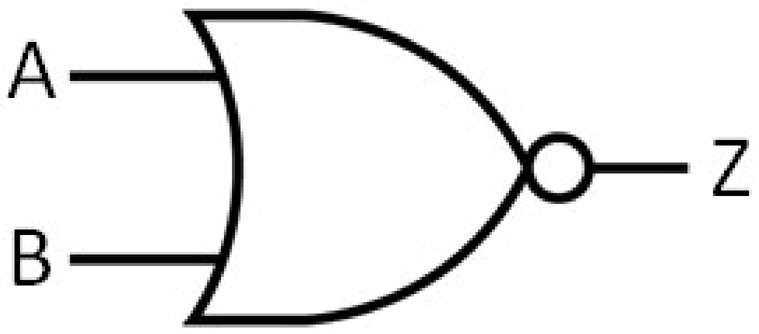




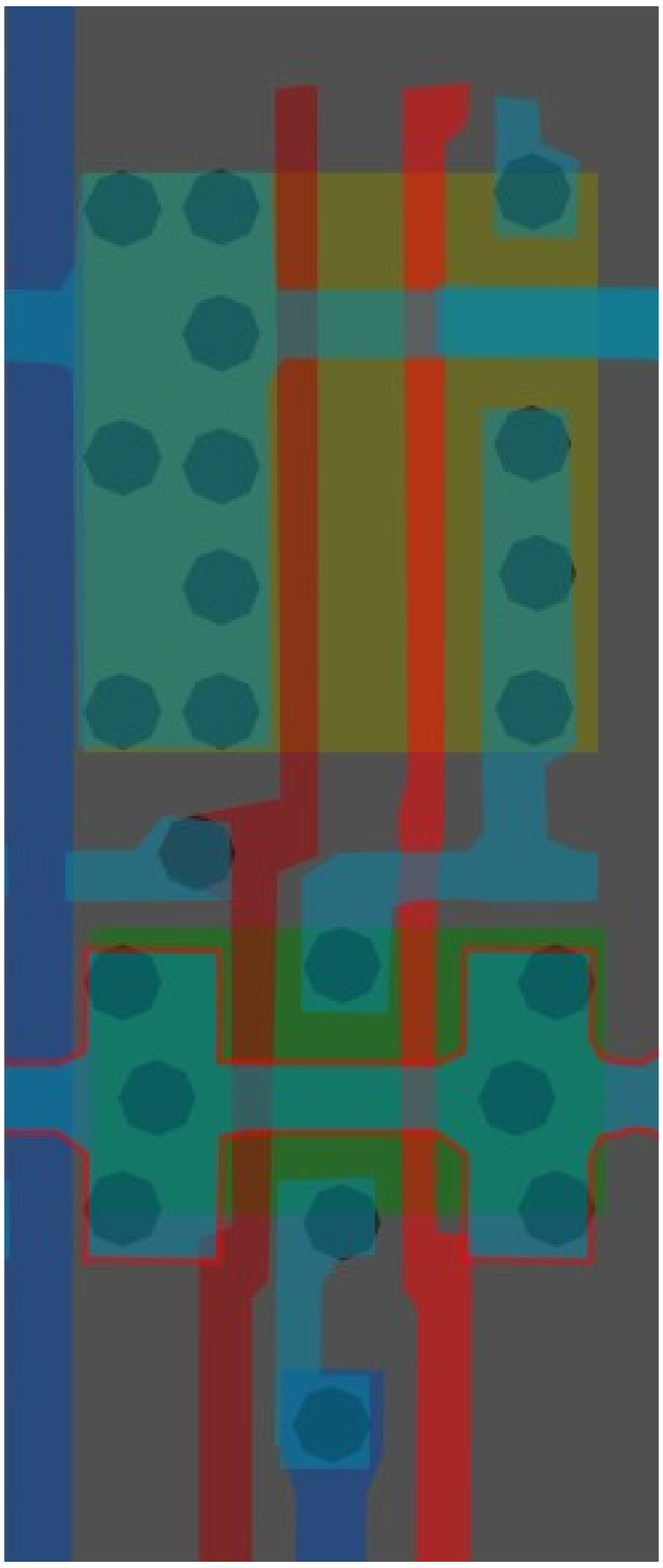


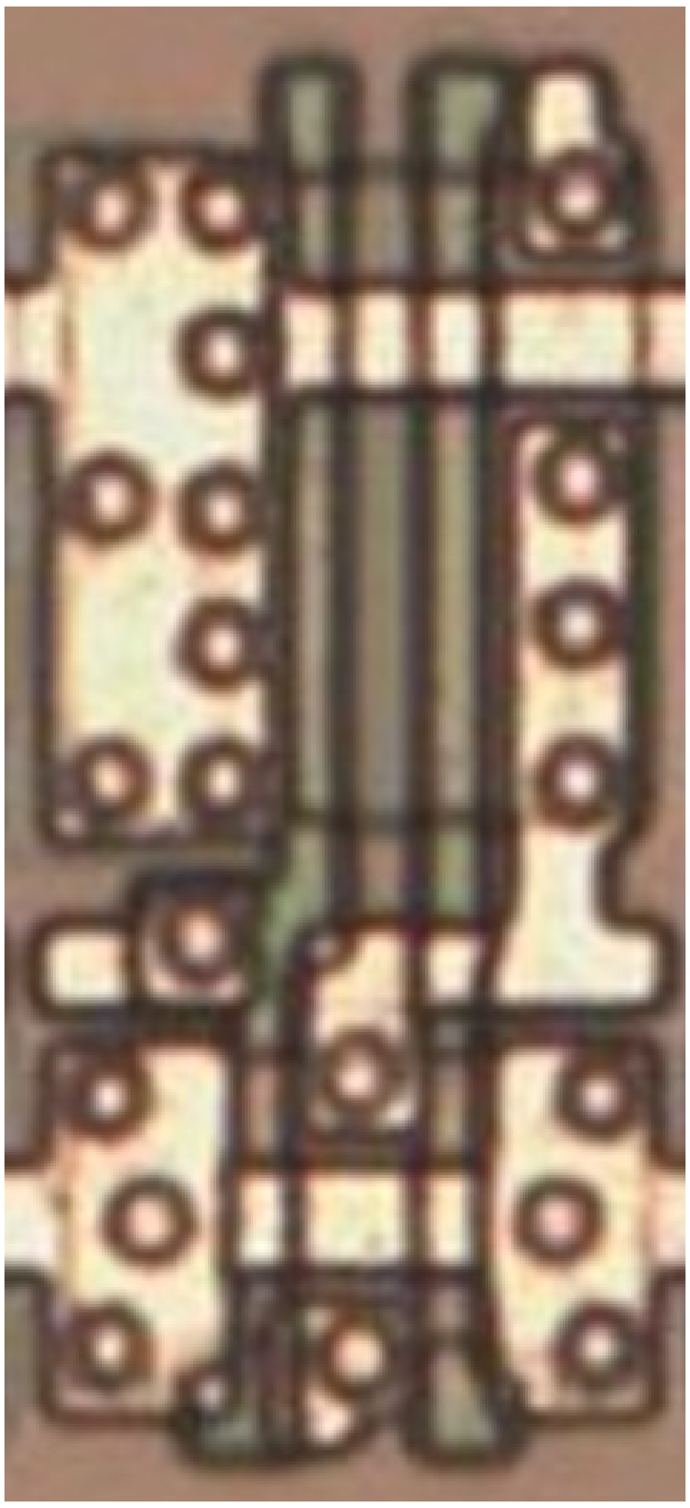
Same cell on LA32 (mirrored) (MB87136A)

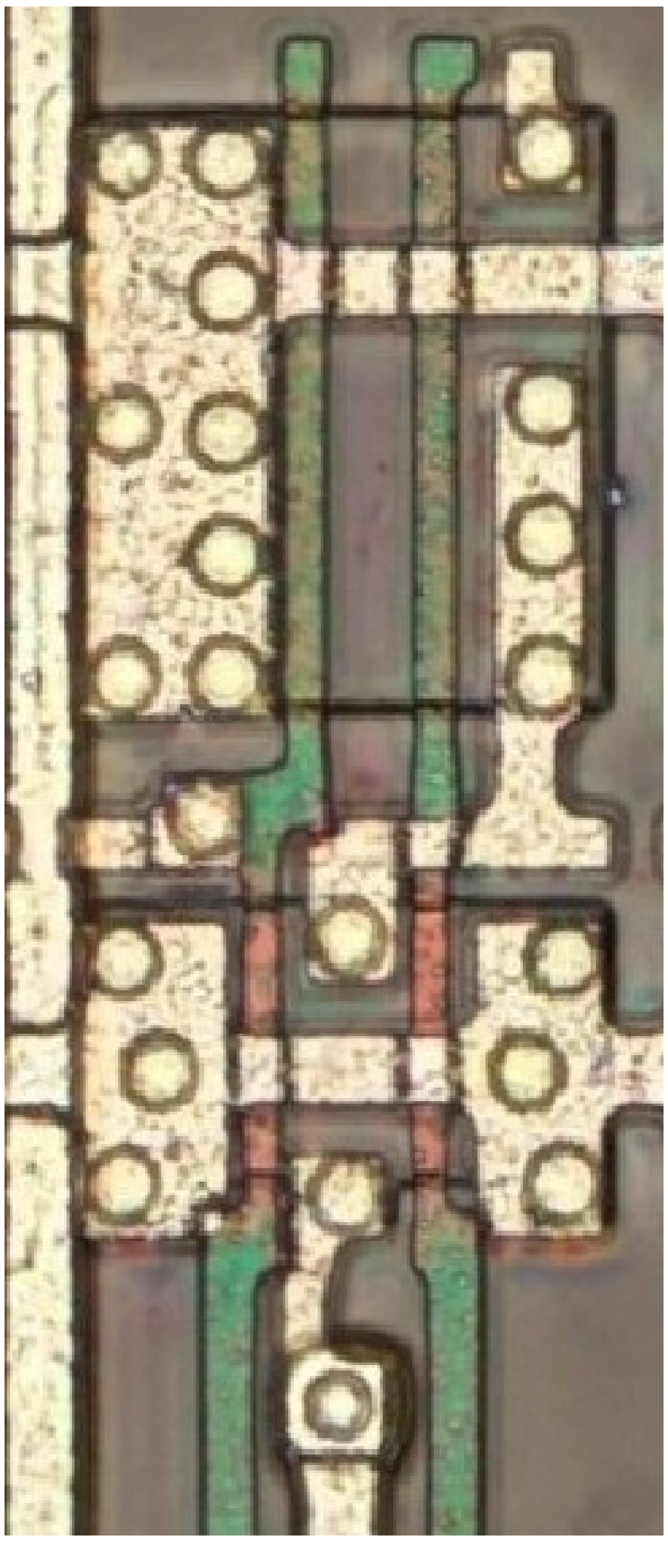
#005600ff | 2-Input NOR | Cell 18



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

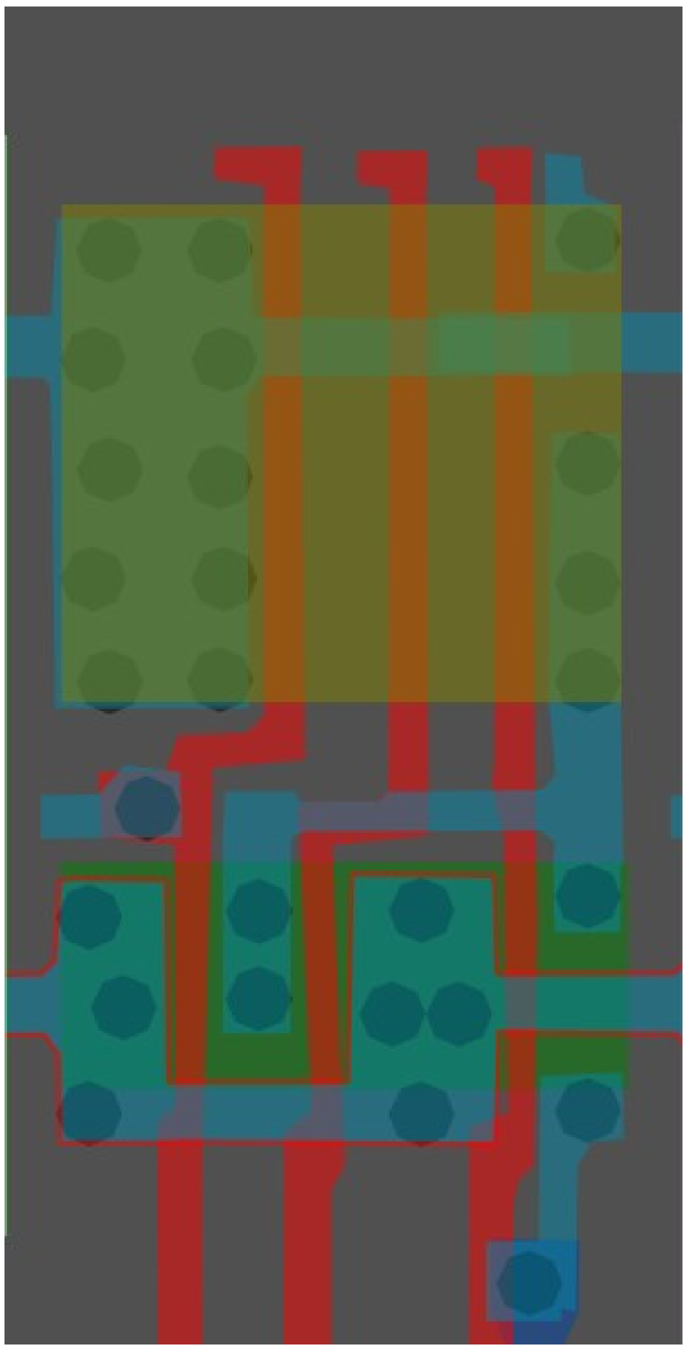


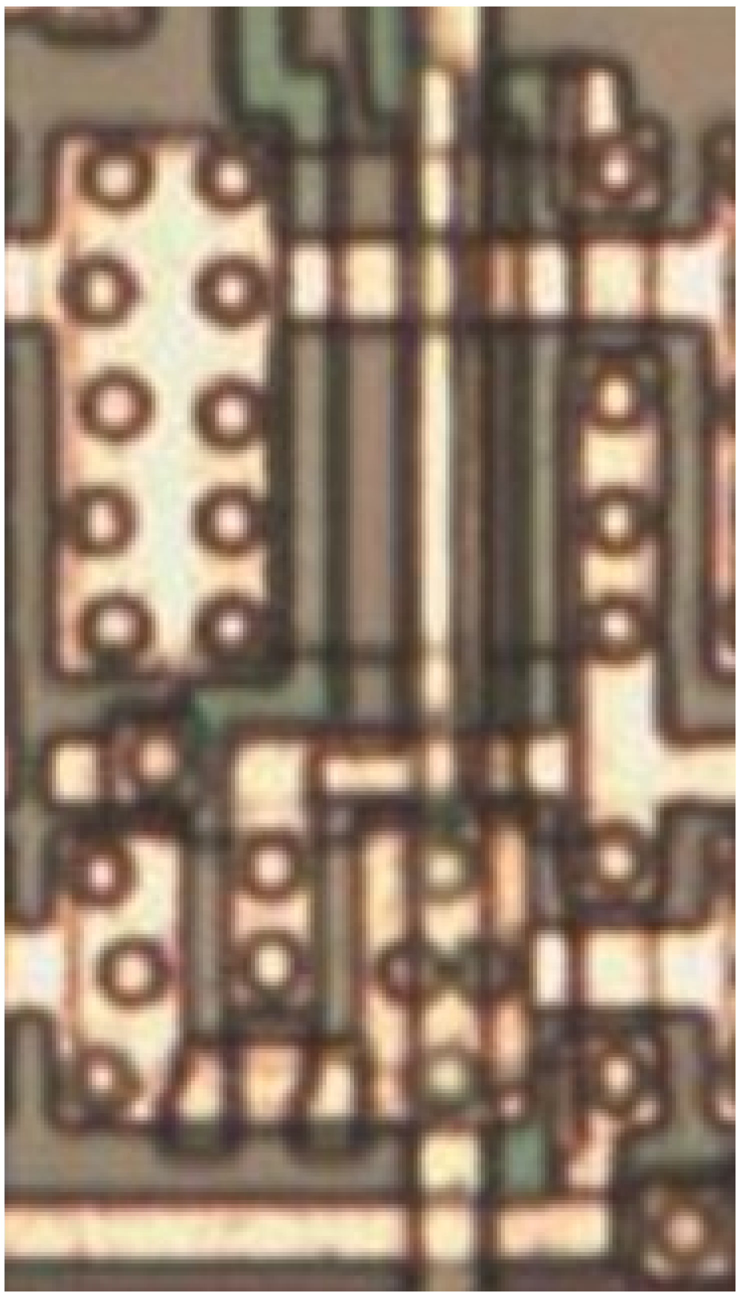




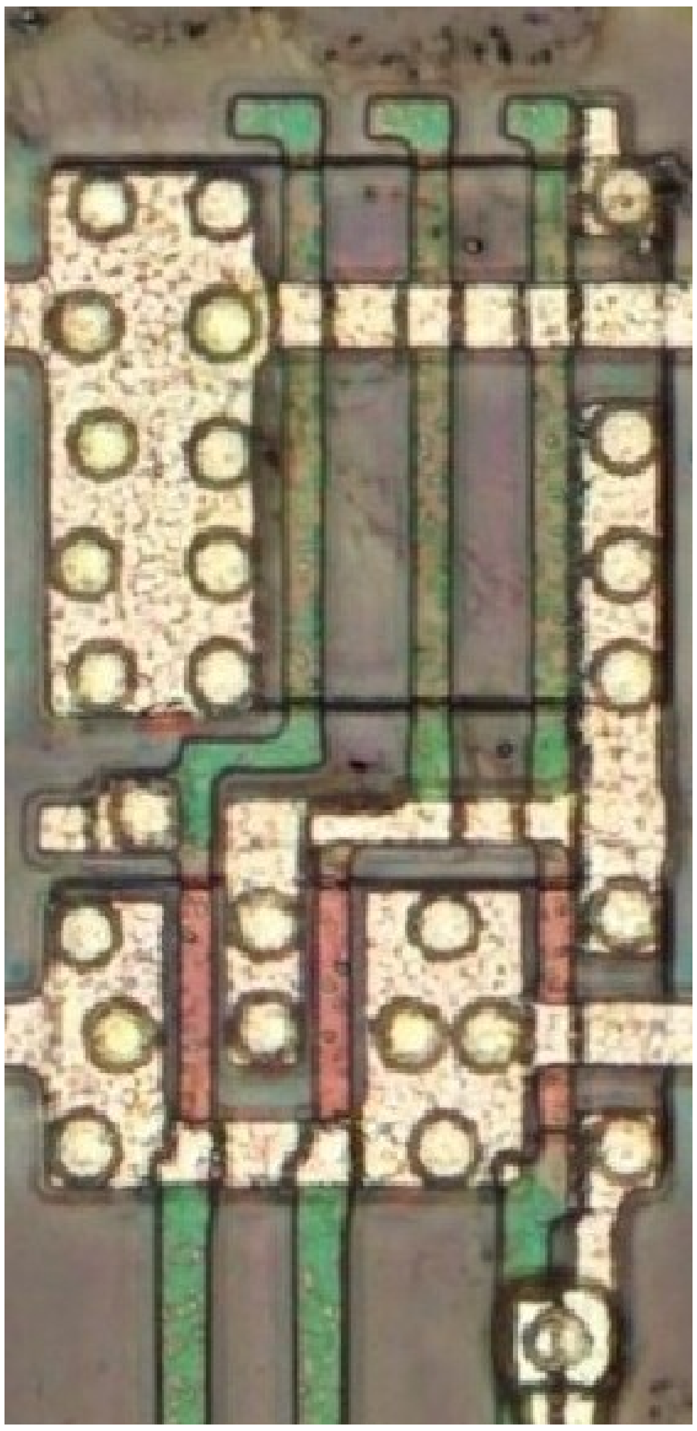
Same cell on
LA32
(MB87136A)

#970000ff **3-Input NOR** Cell 19



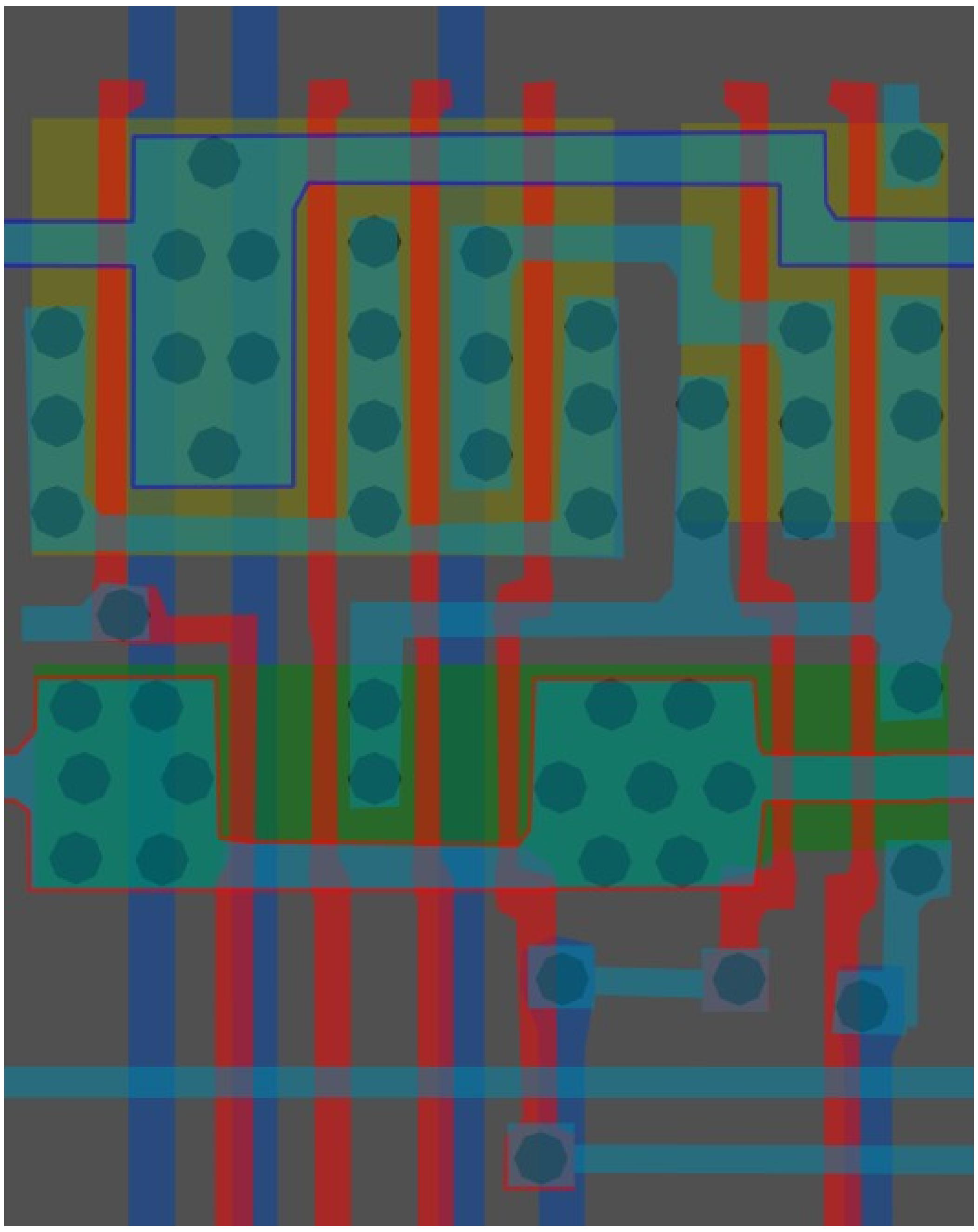


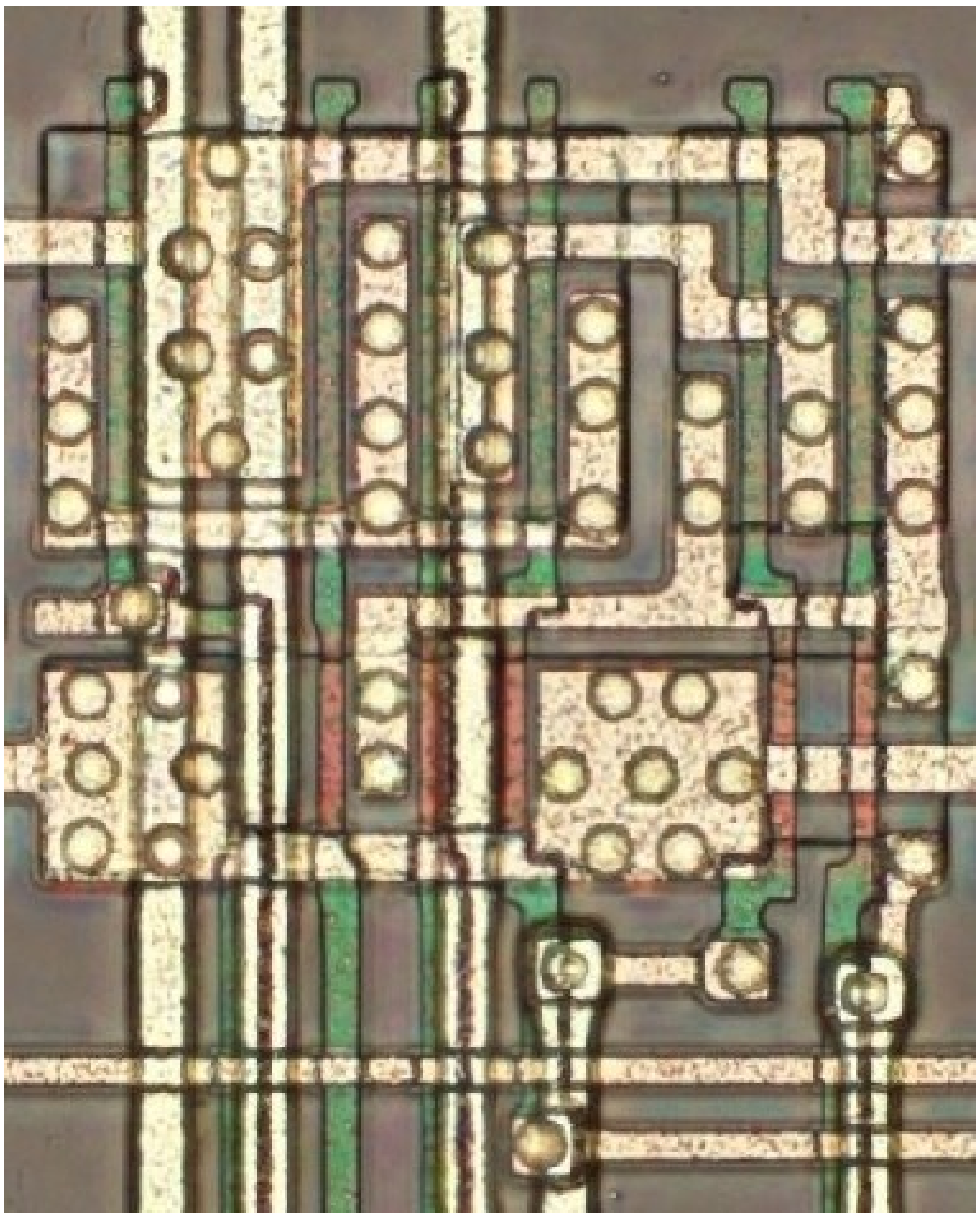
Same cell on
LA32
(MB87136A)



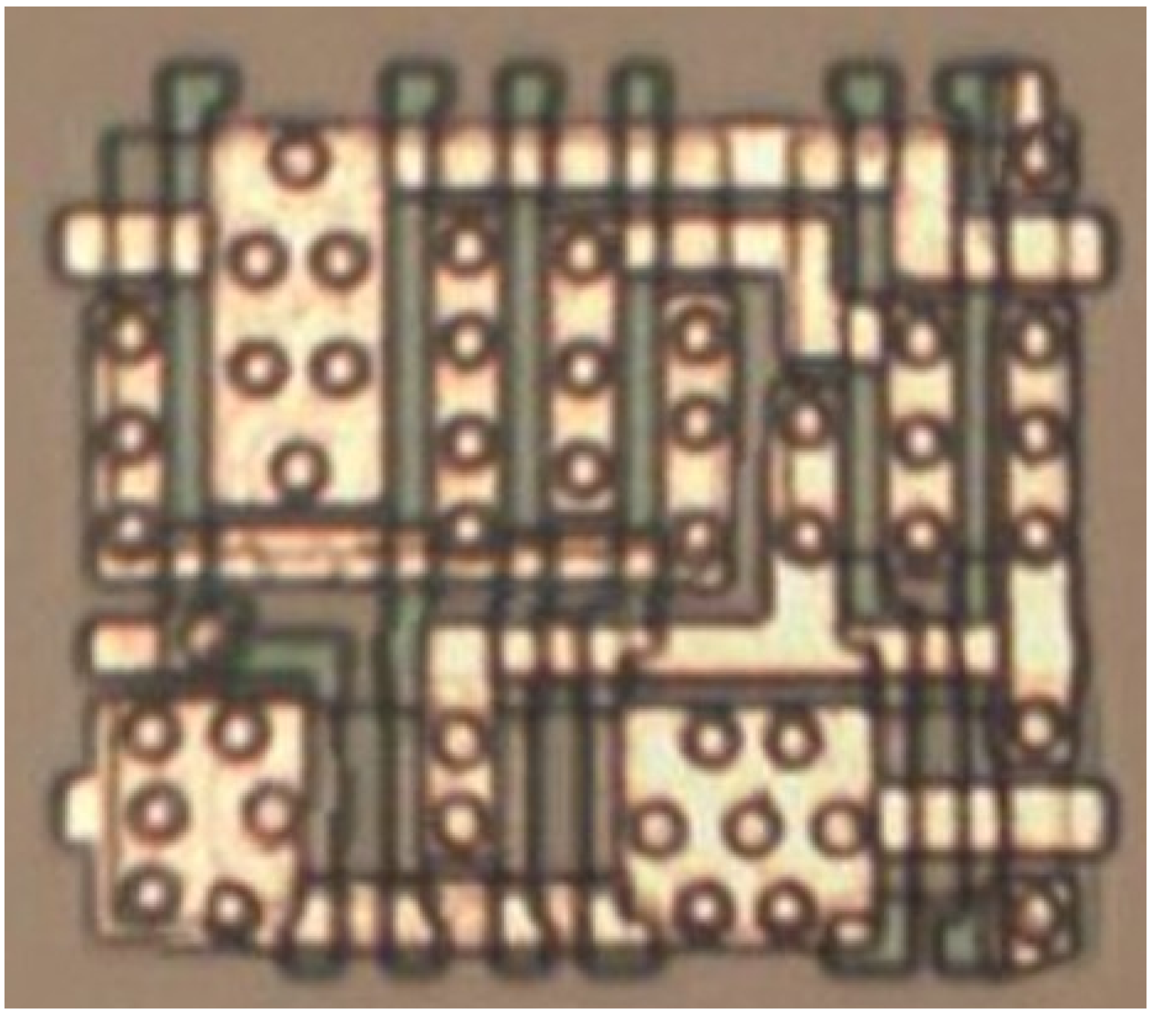
#ff0000ff

Cell type 20

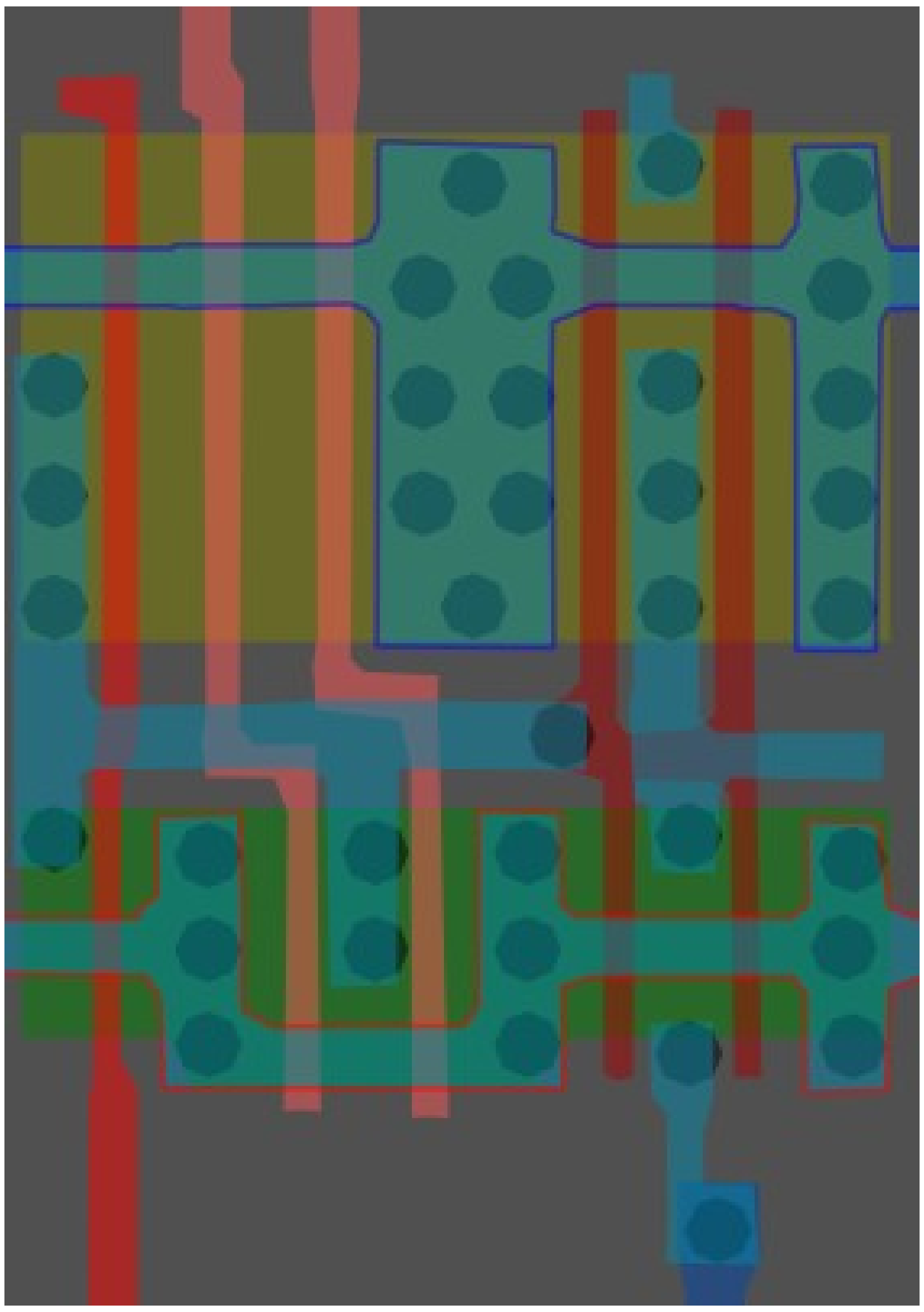


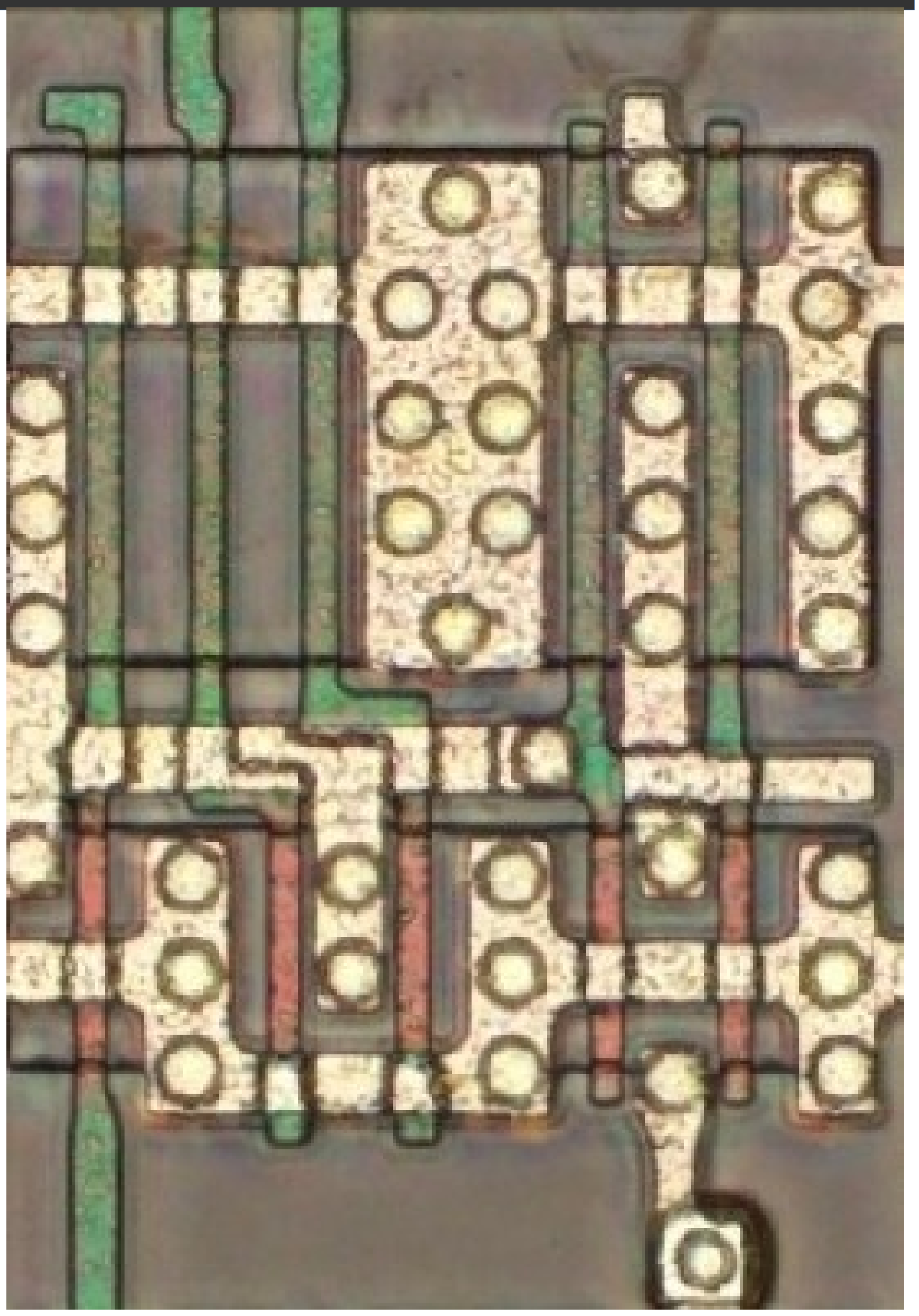


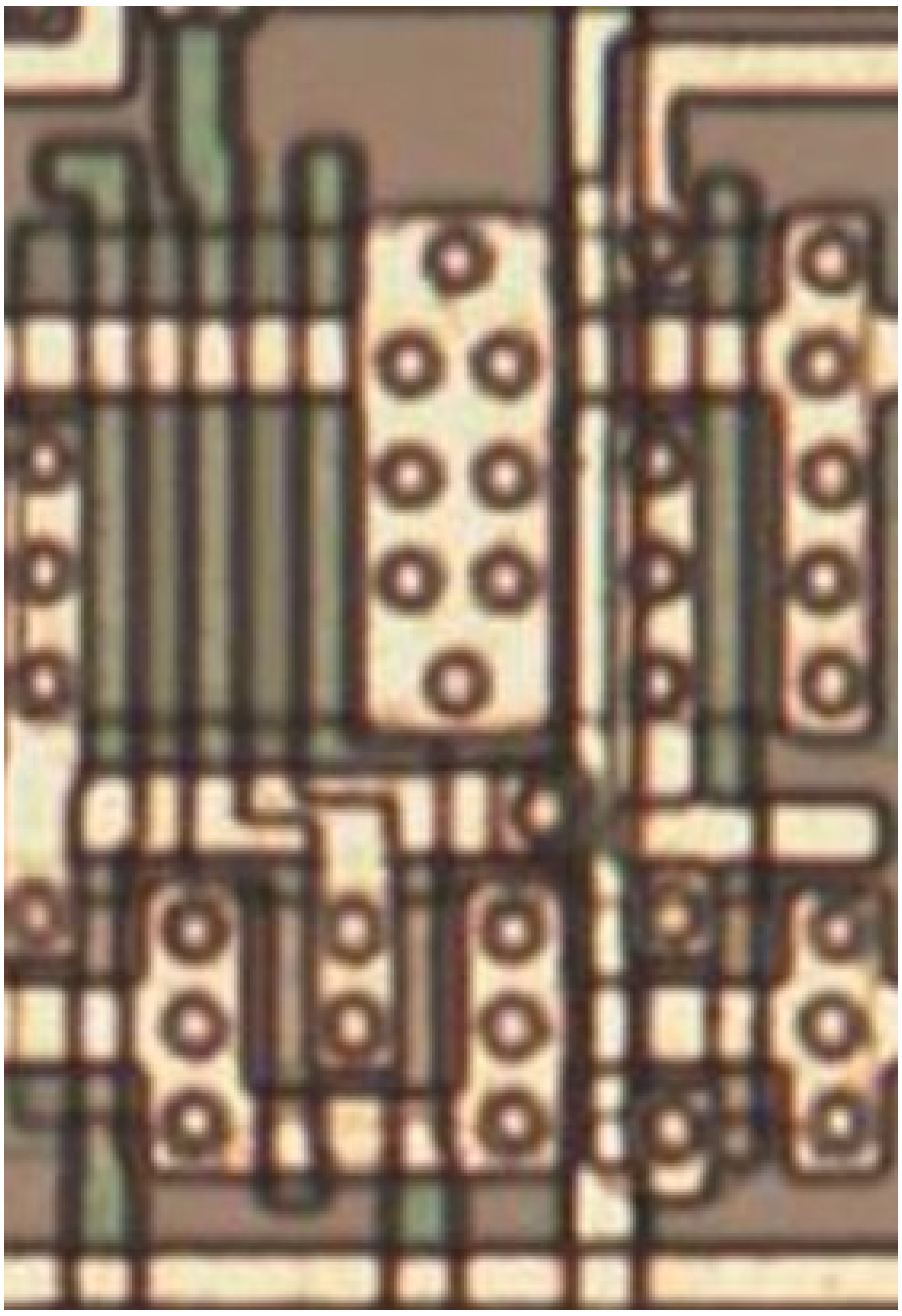
Same cell on LA32 (MB87136A)



#218268ff Cell type 21a



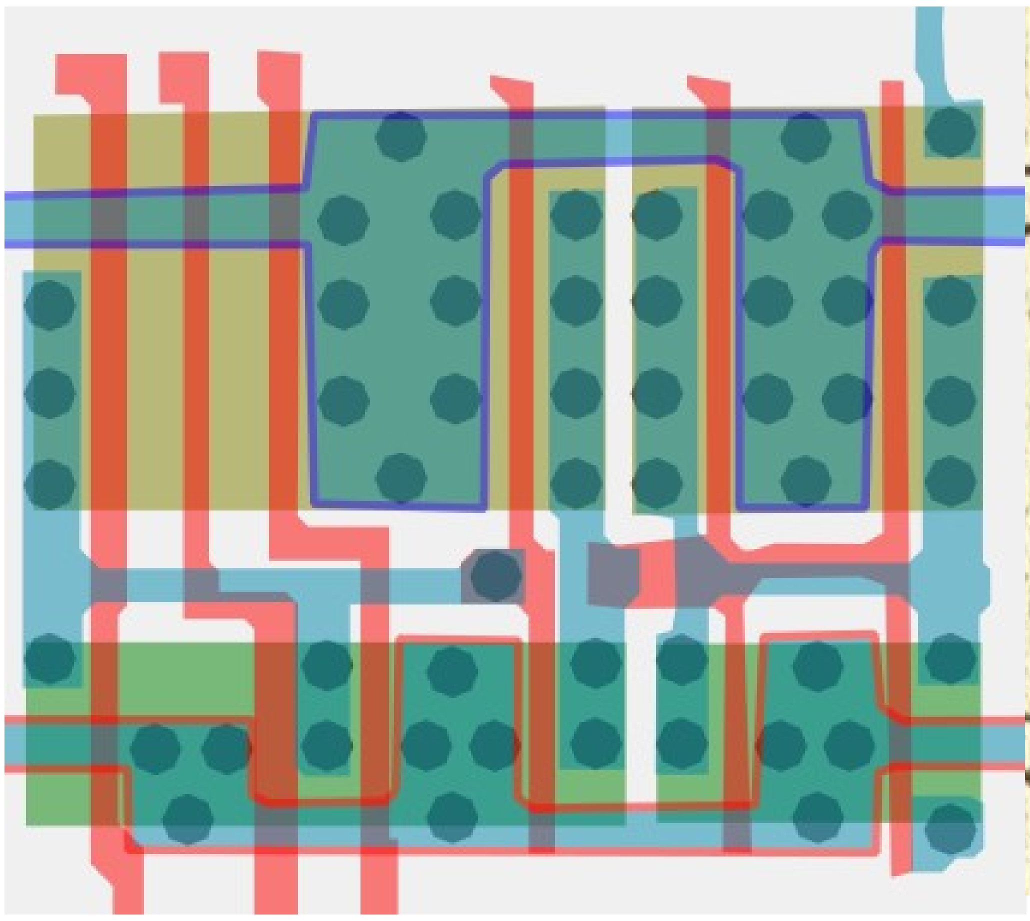
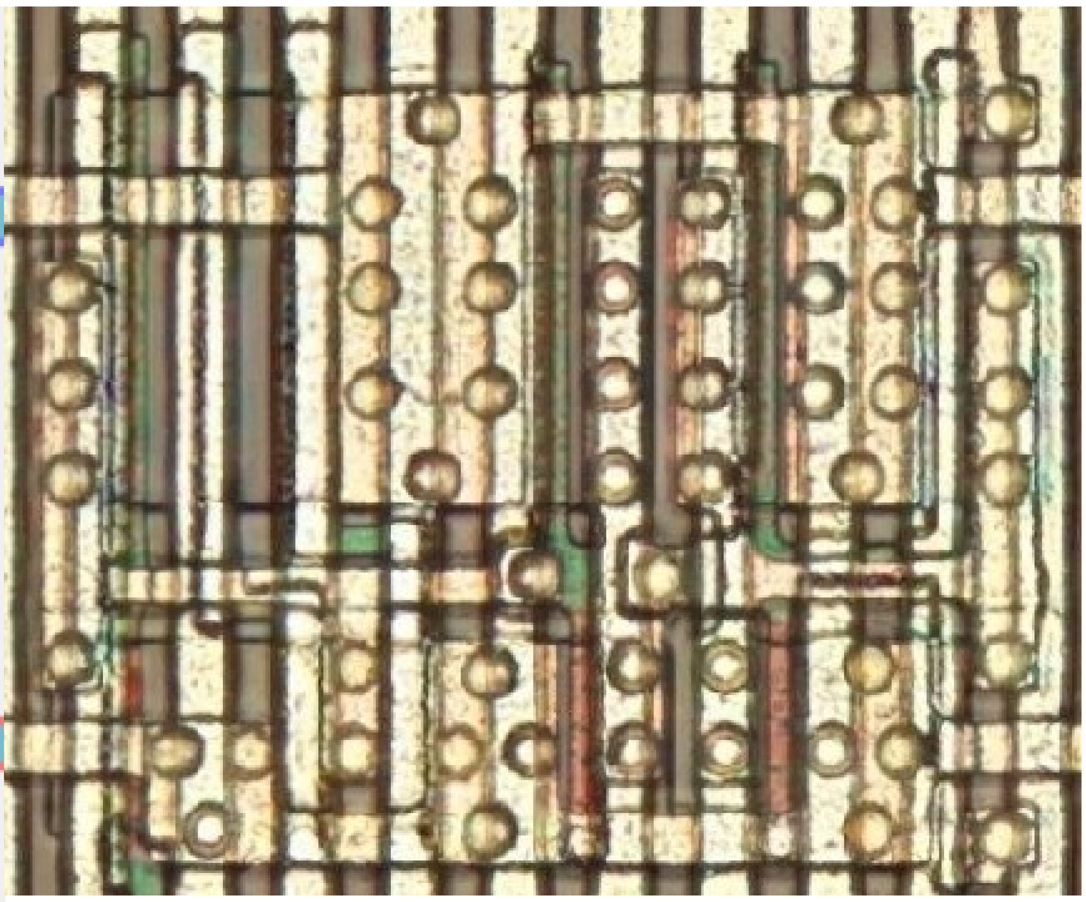




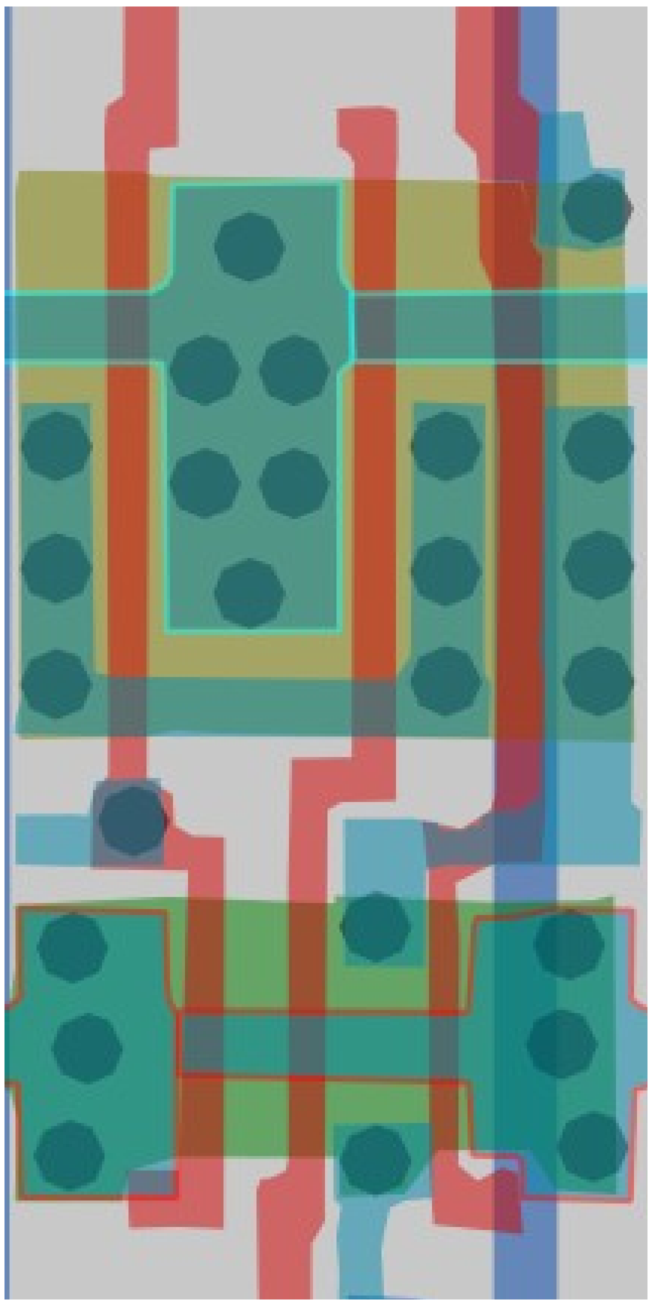
Same cell on
LA32
(MB87136A)

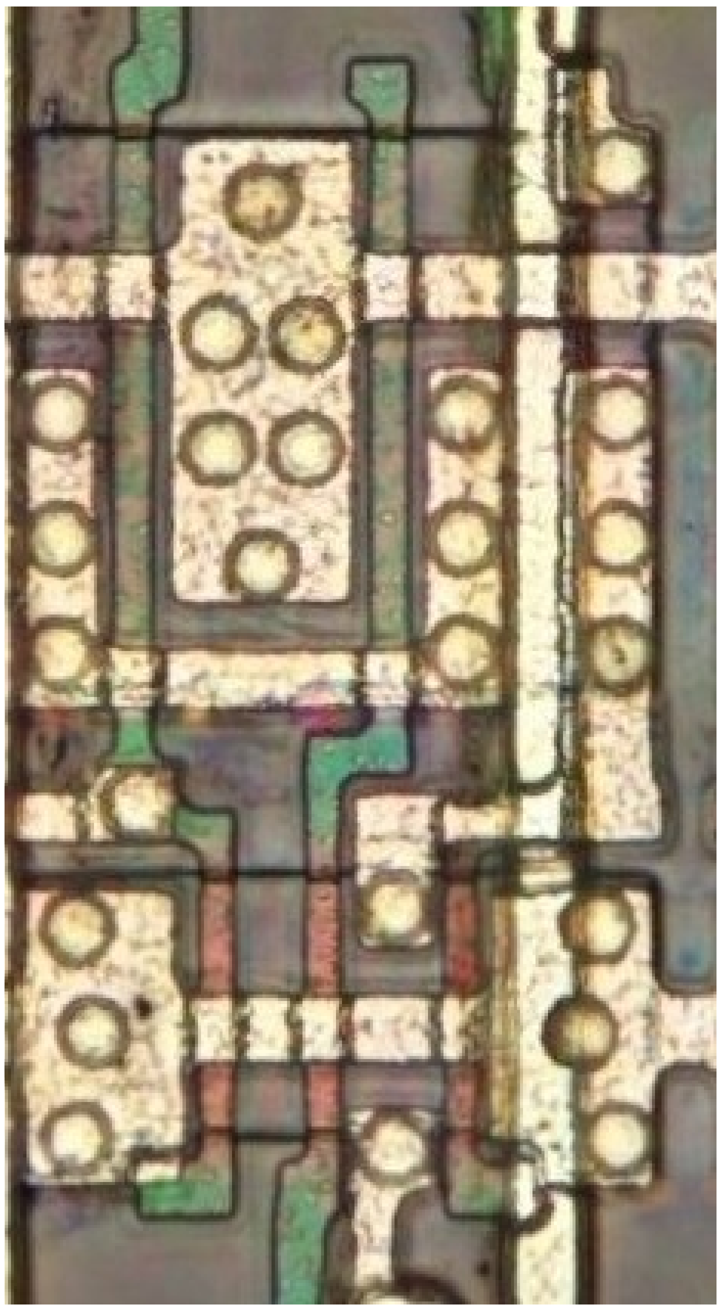
#115140ff

Cell type 21b

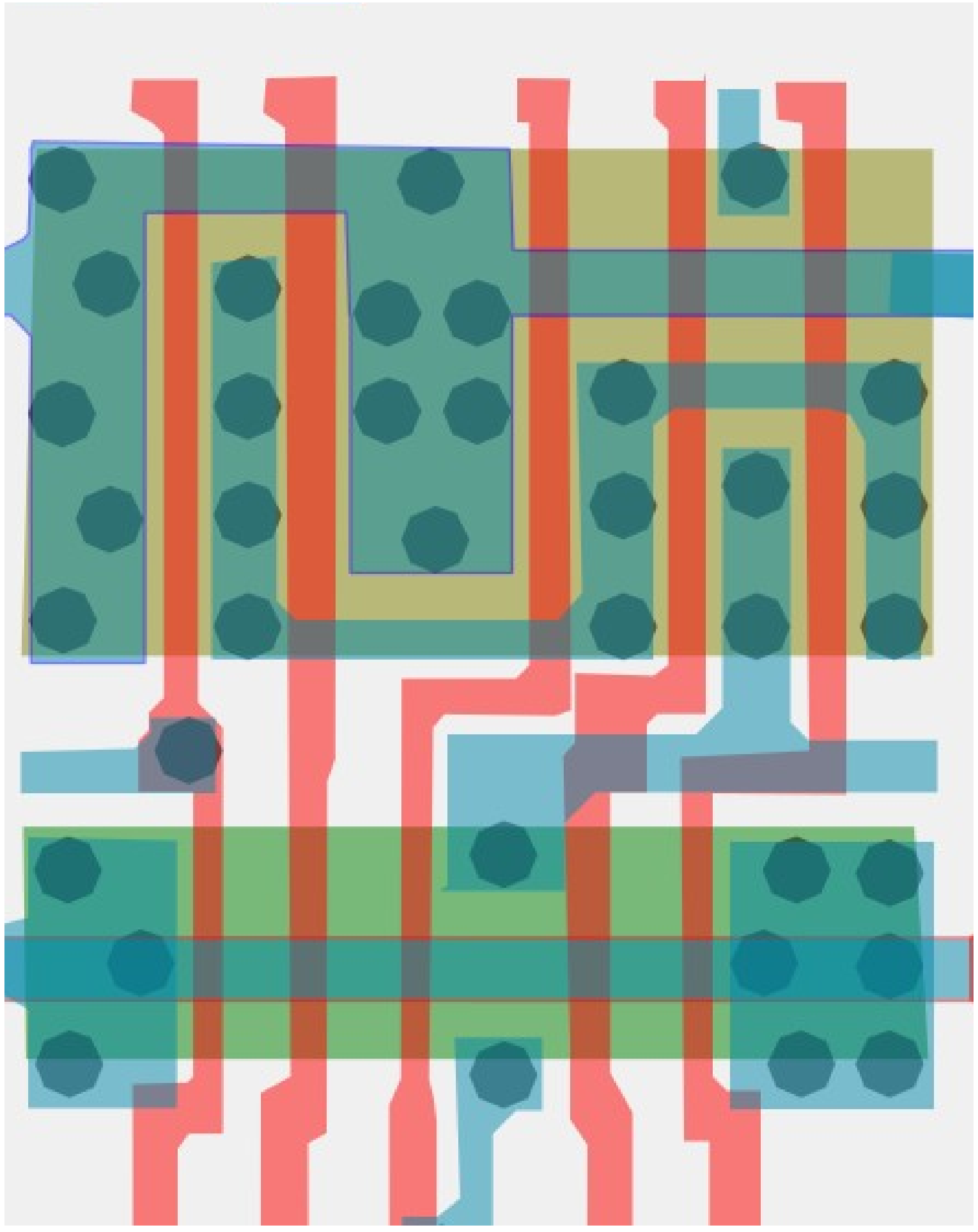
#d1796bff Cell type 24a

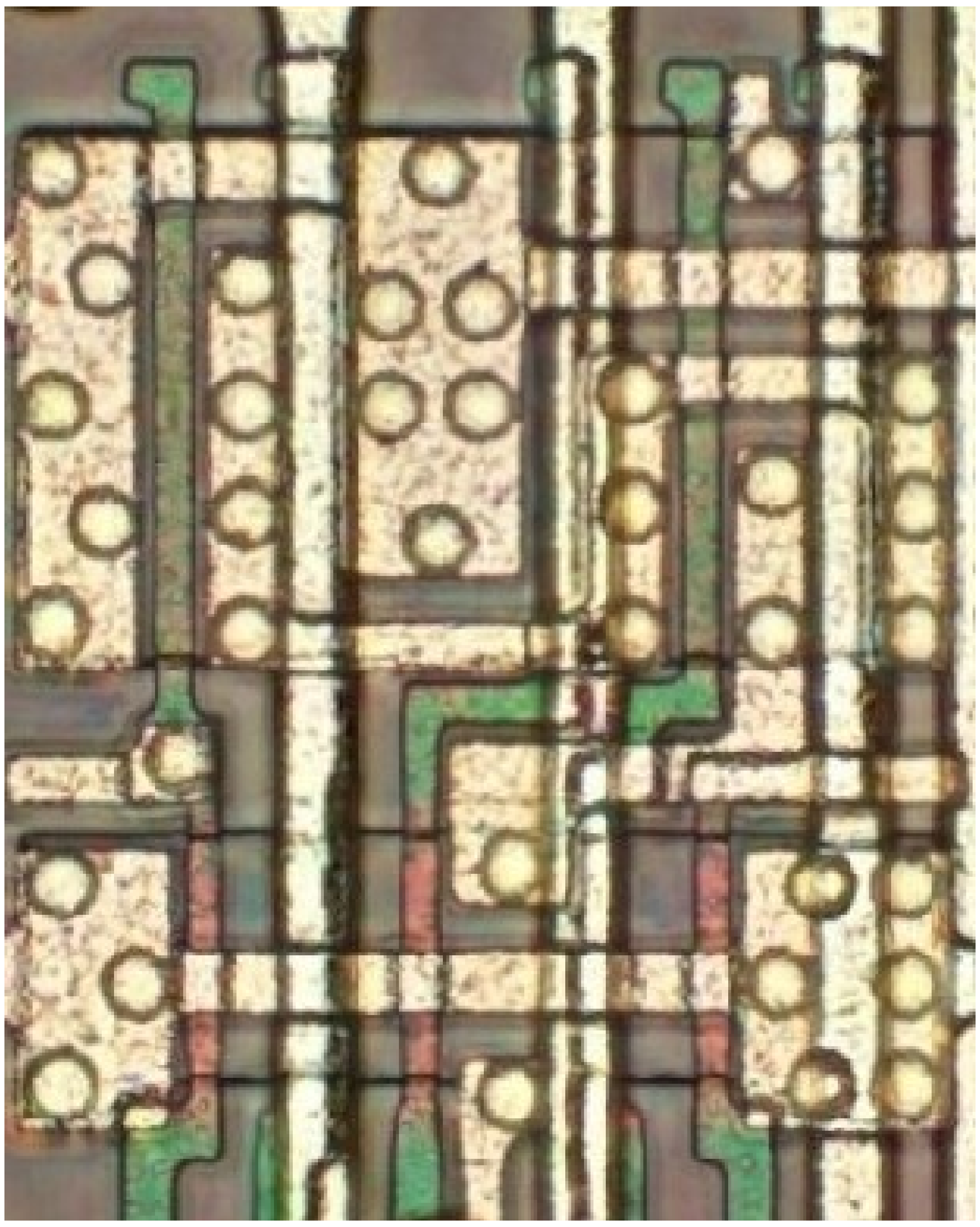




#7f8157ff

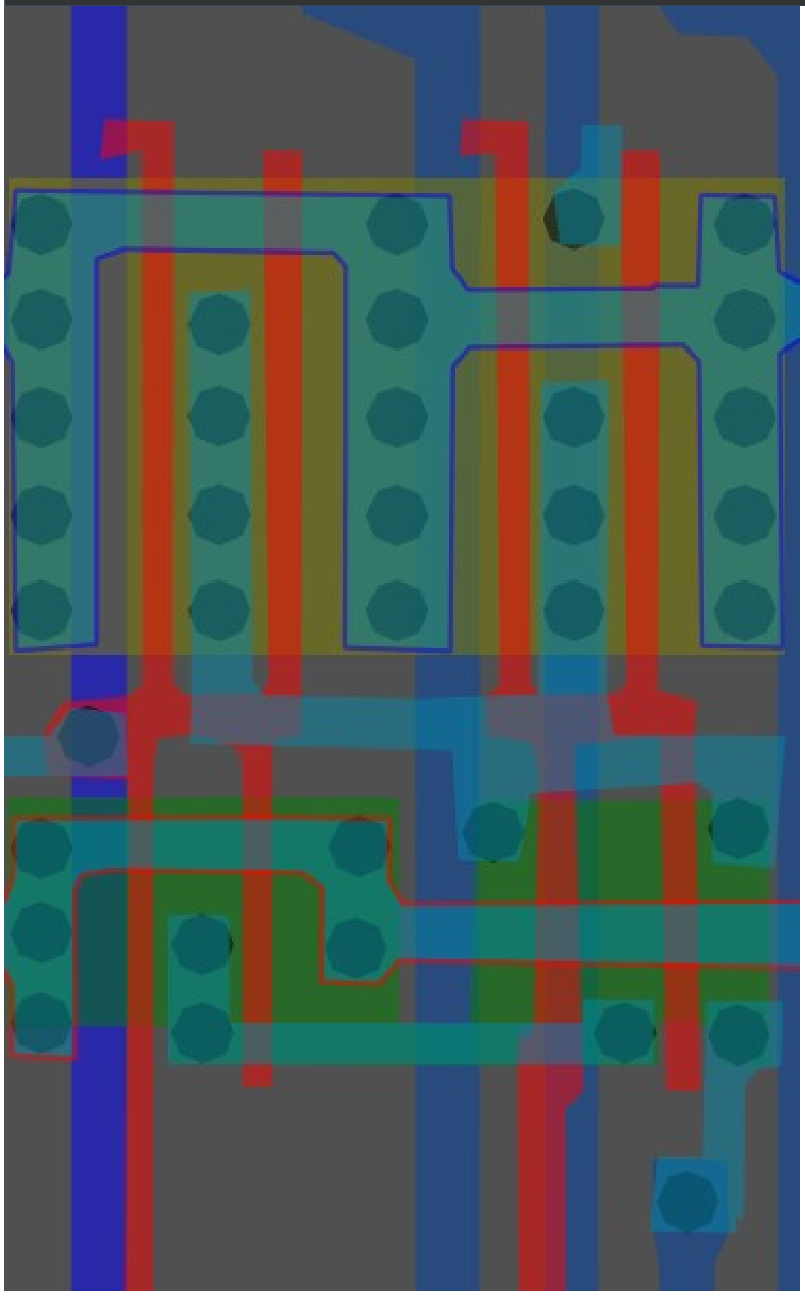
Cell type 24b

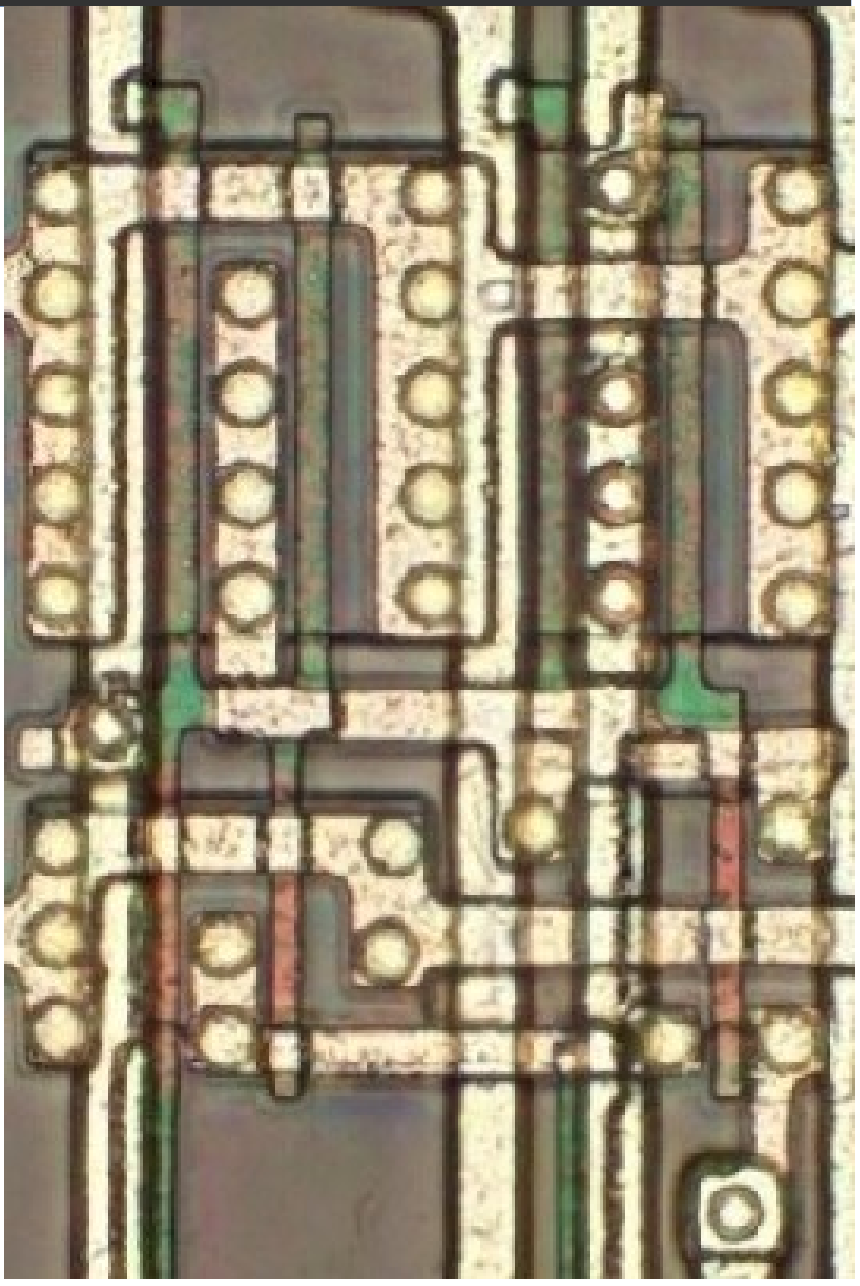




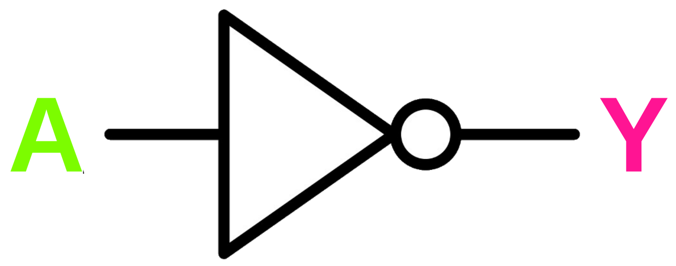
#ff9972ff

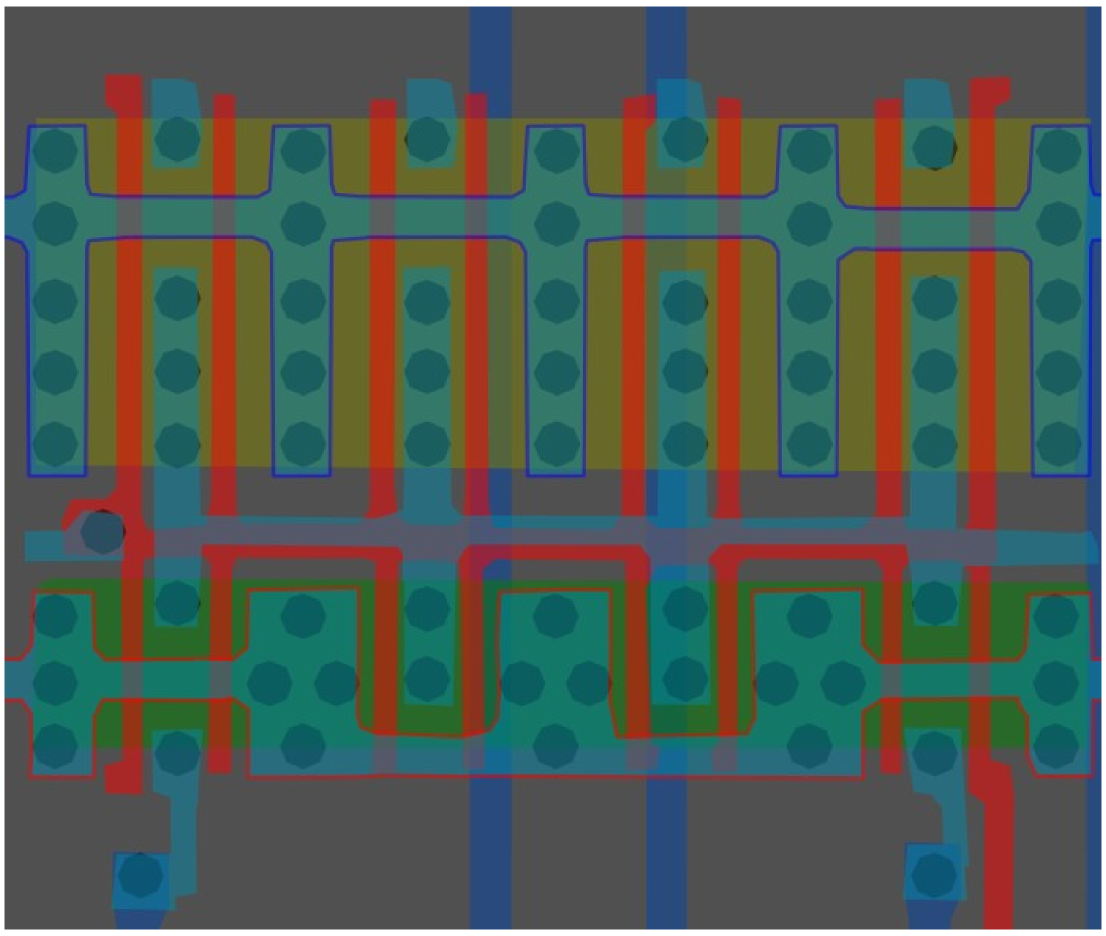
Cell type 25

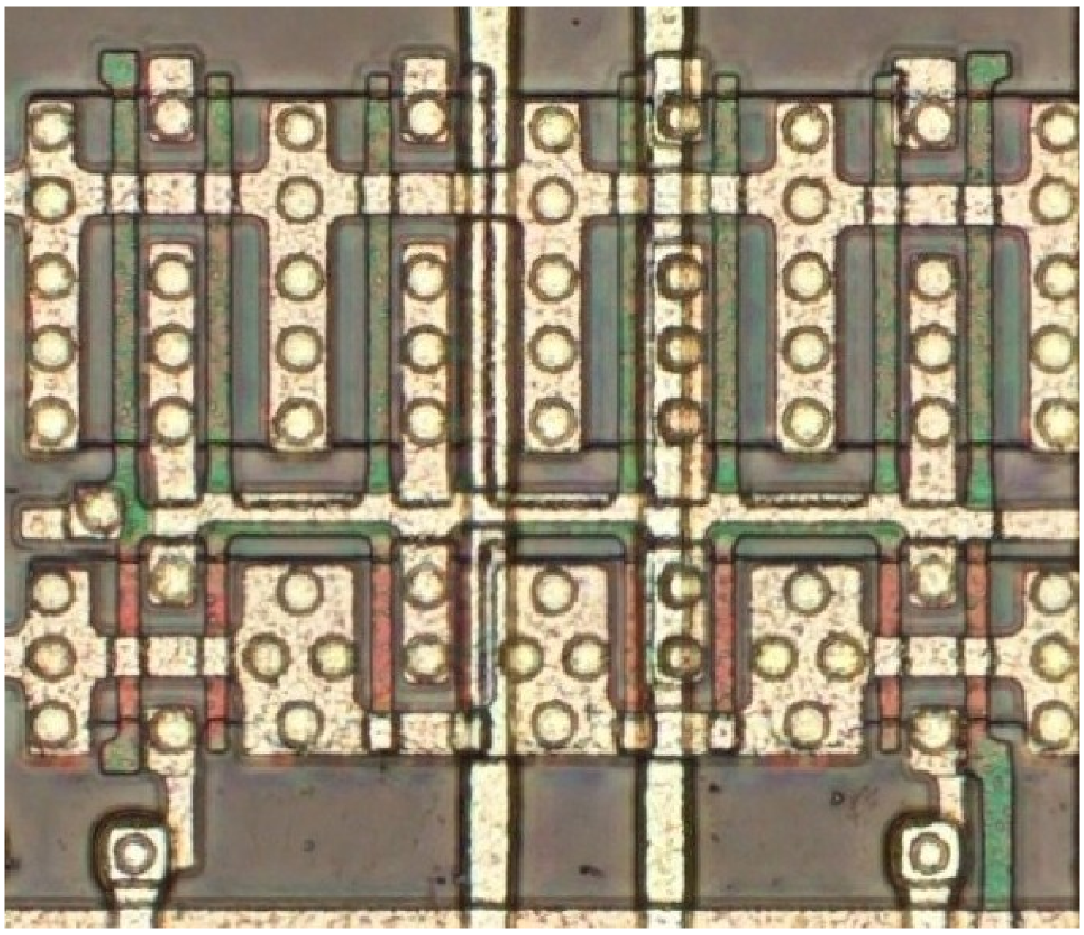


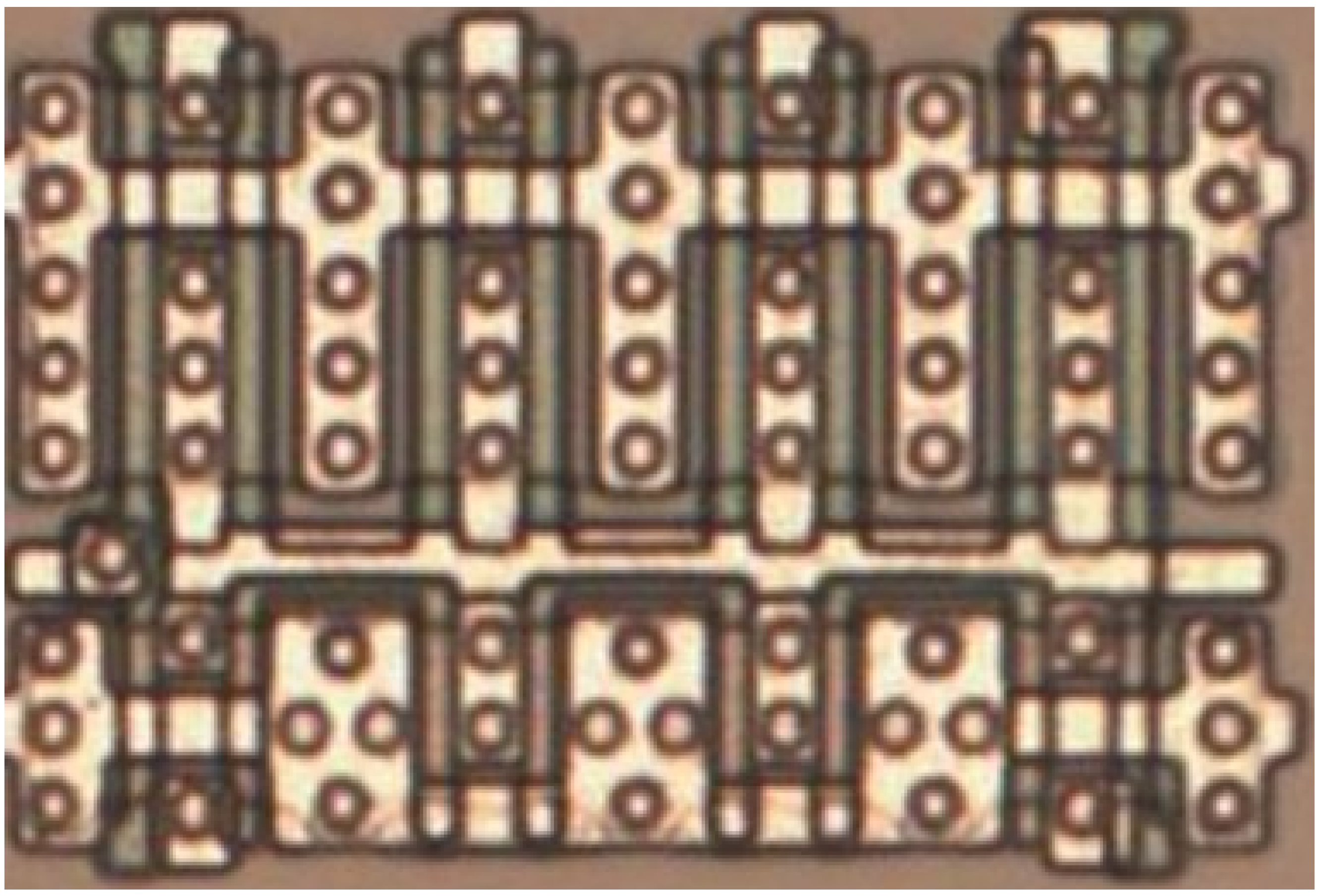


#ffb94cff **Inverter/NOT x8** **Cell 26**





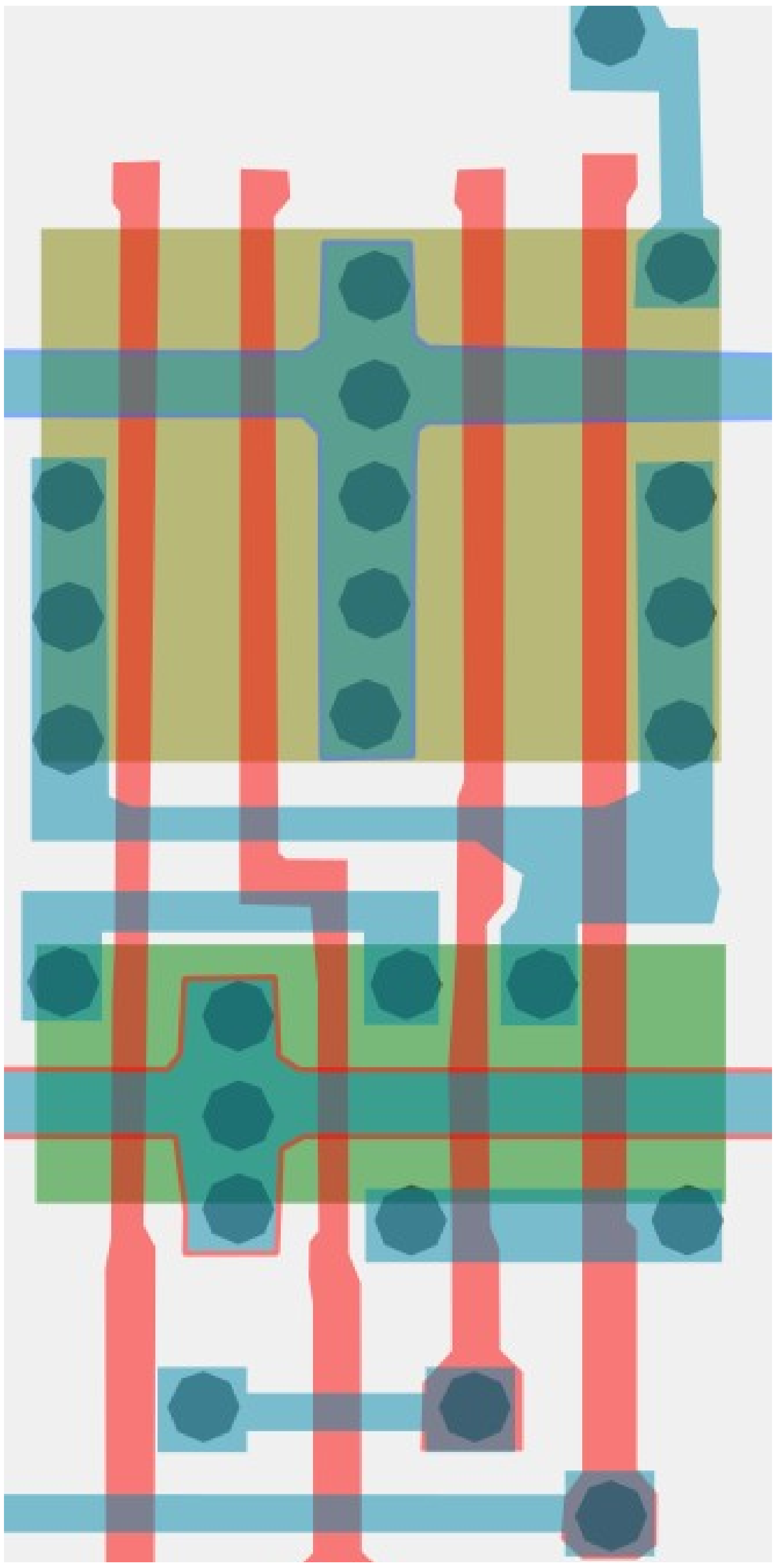


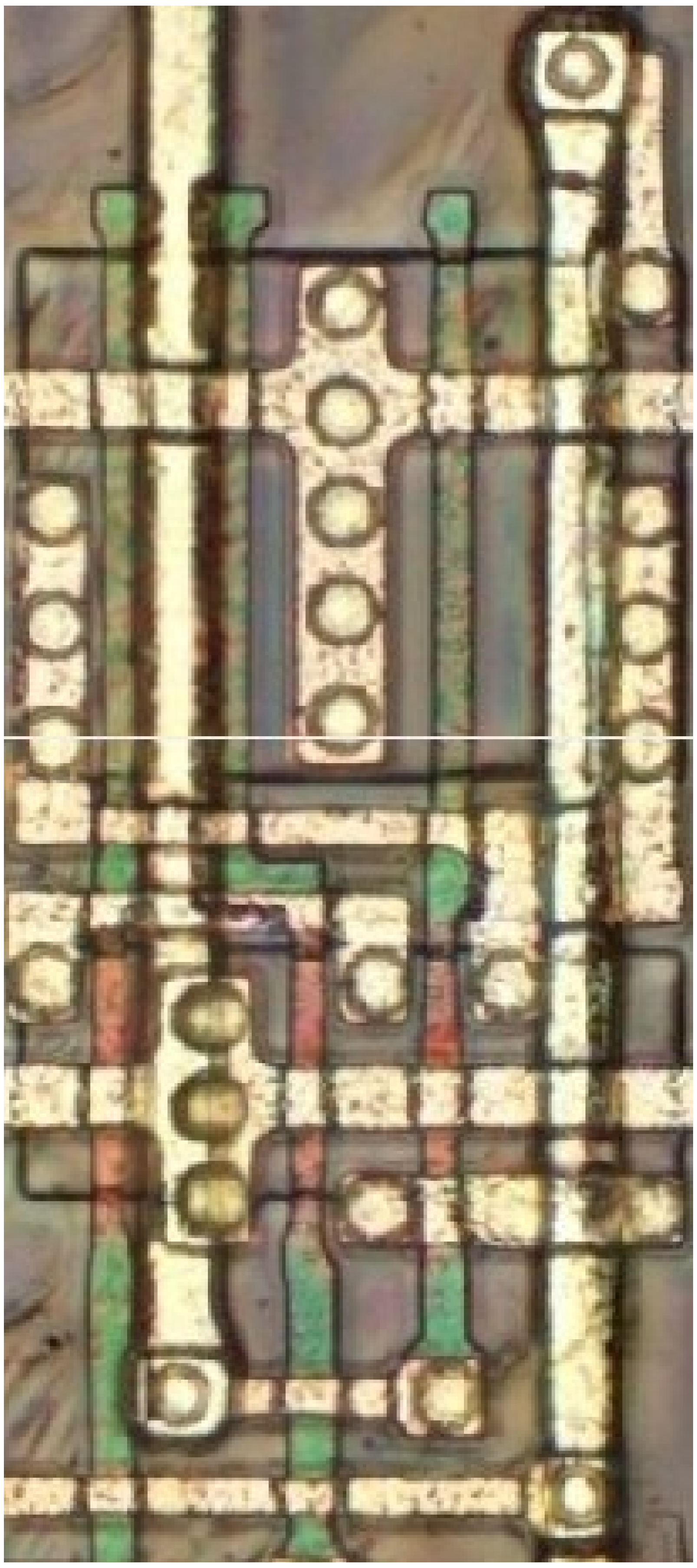


Same cell on
LA32
(MB87136A)

#bec1c3ff

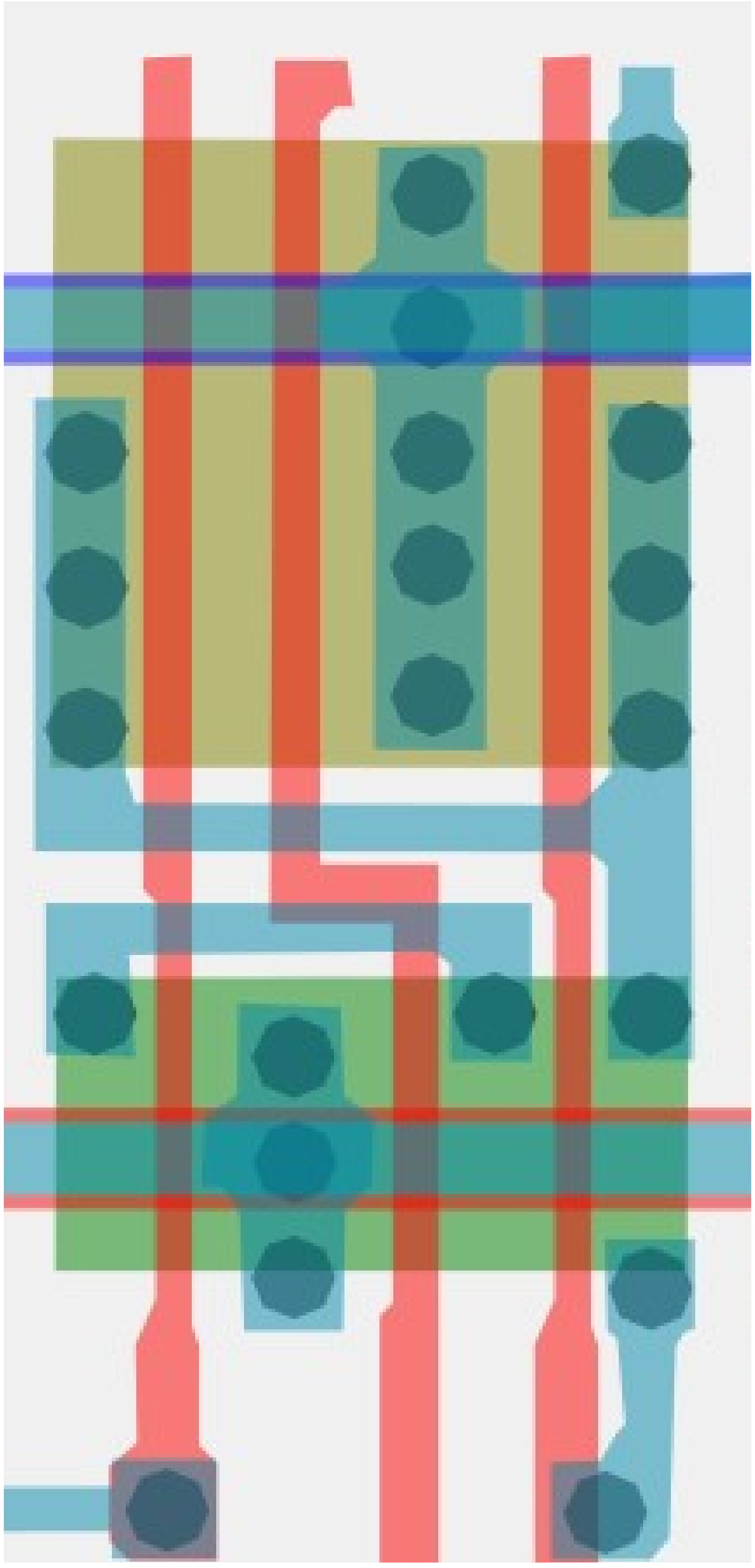
Cell type 27a

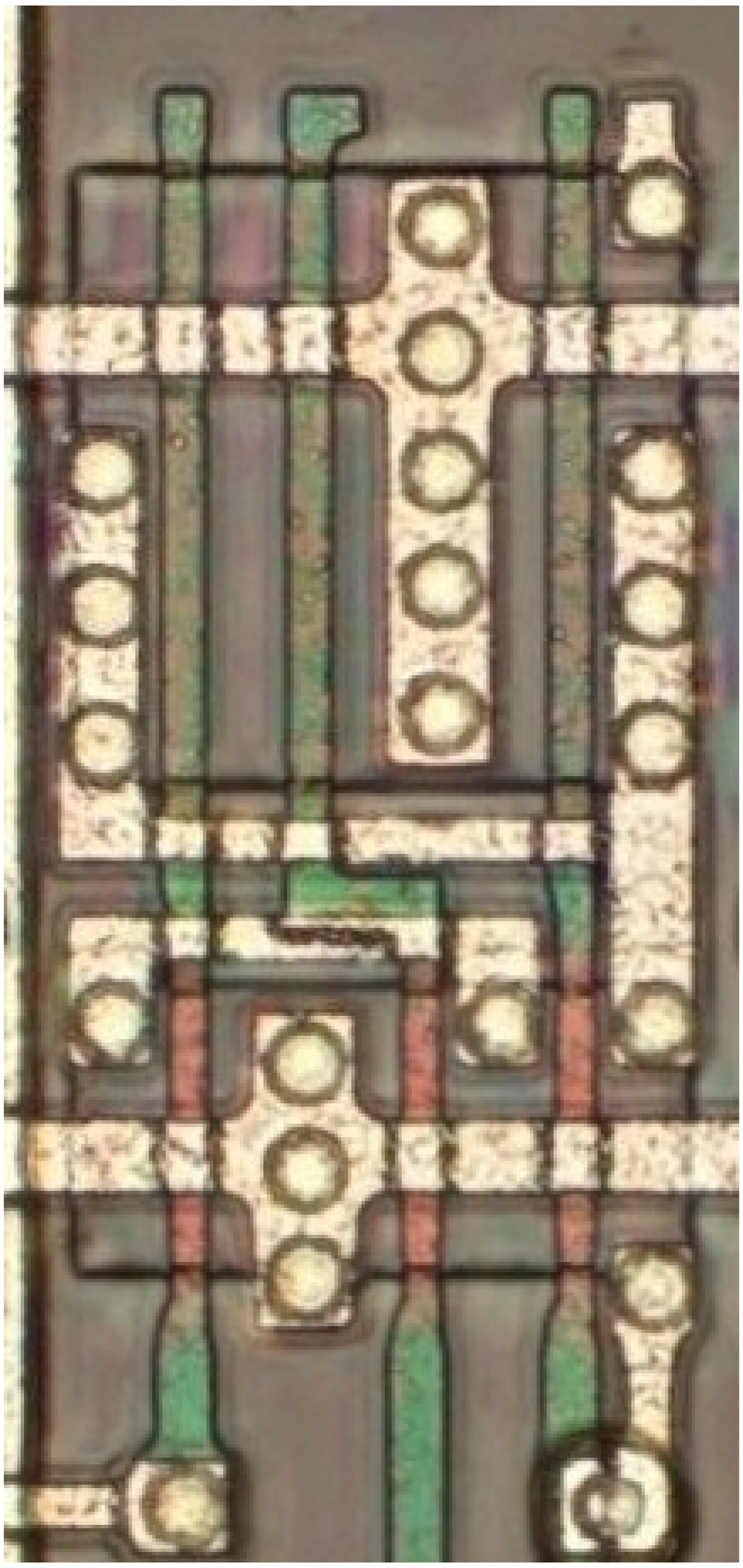




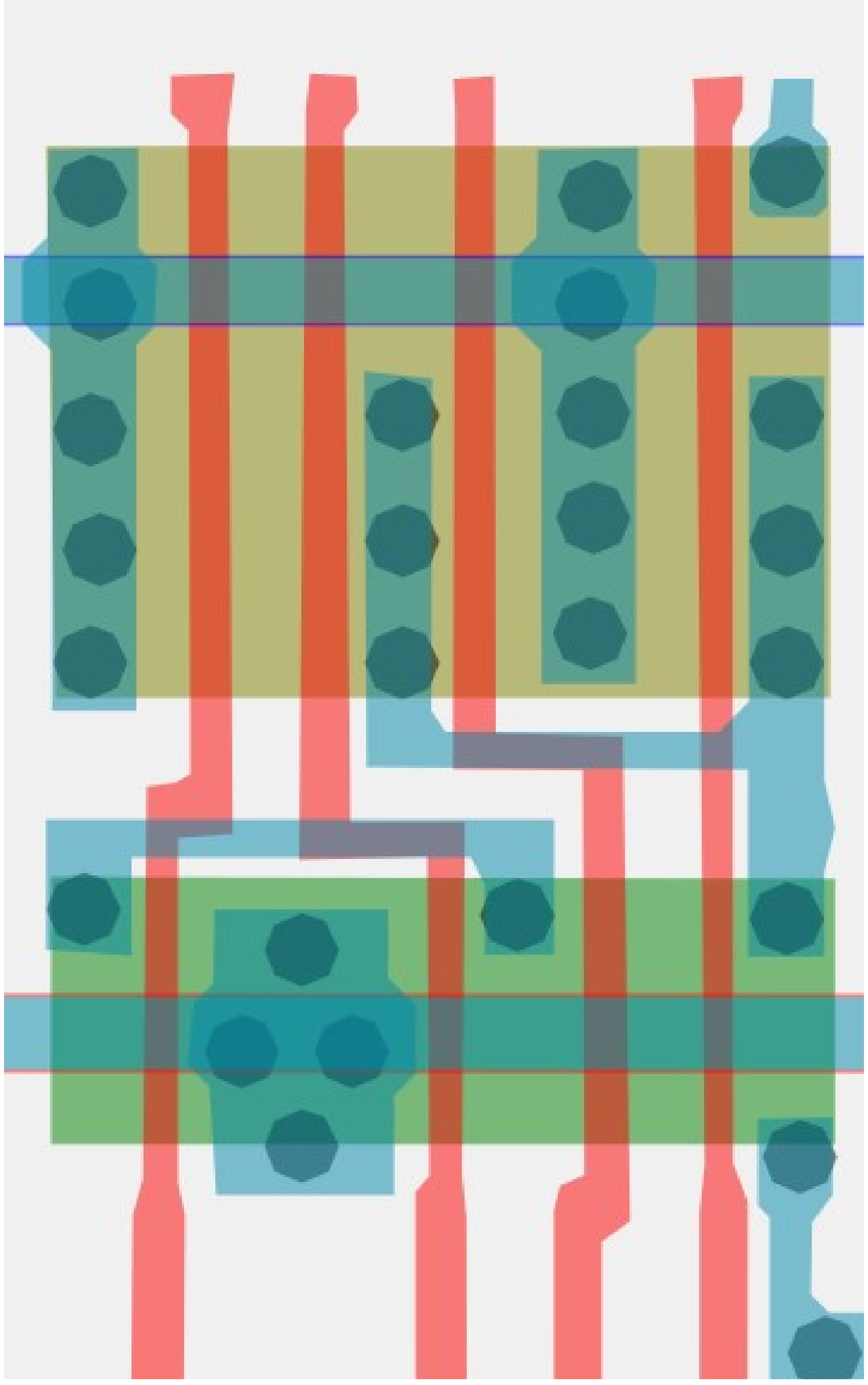
#787878ff

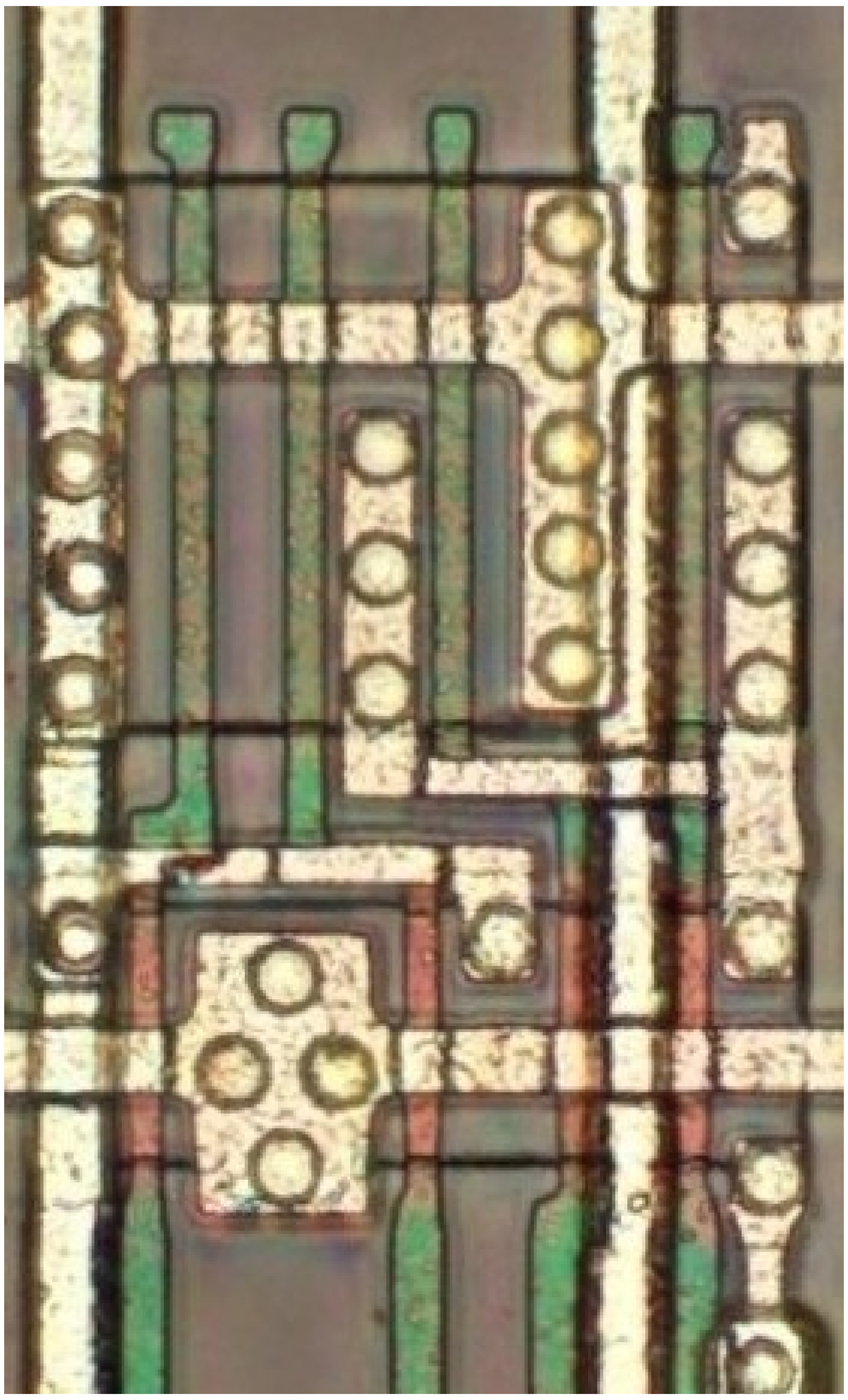
Cell type 27b





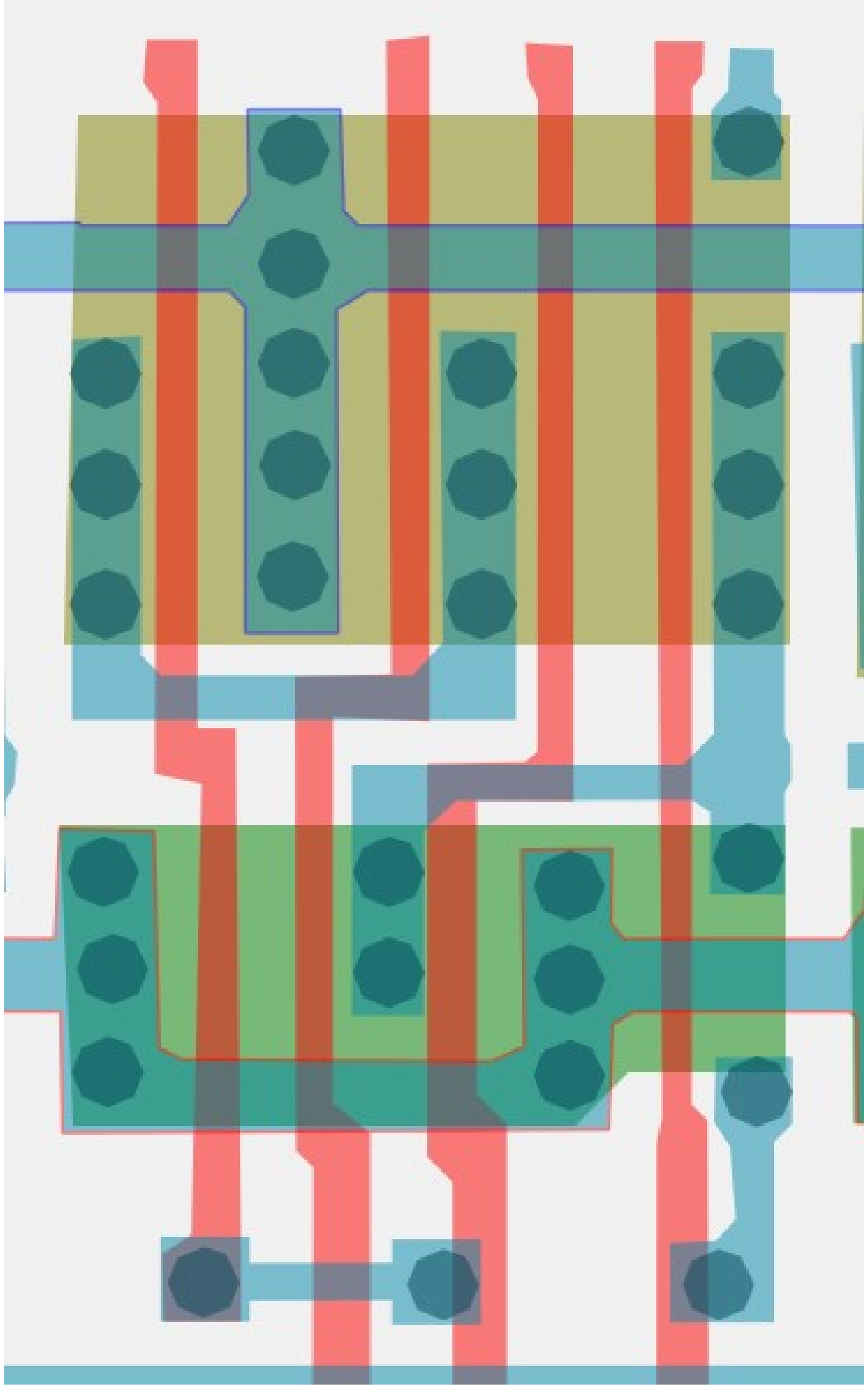
#5d5d5dff Cell type 27C

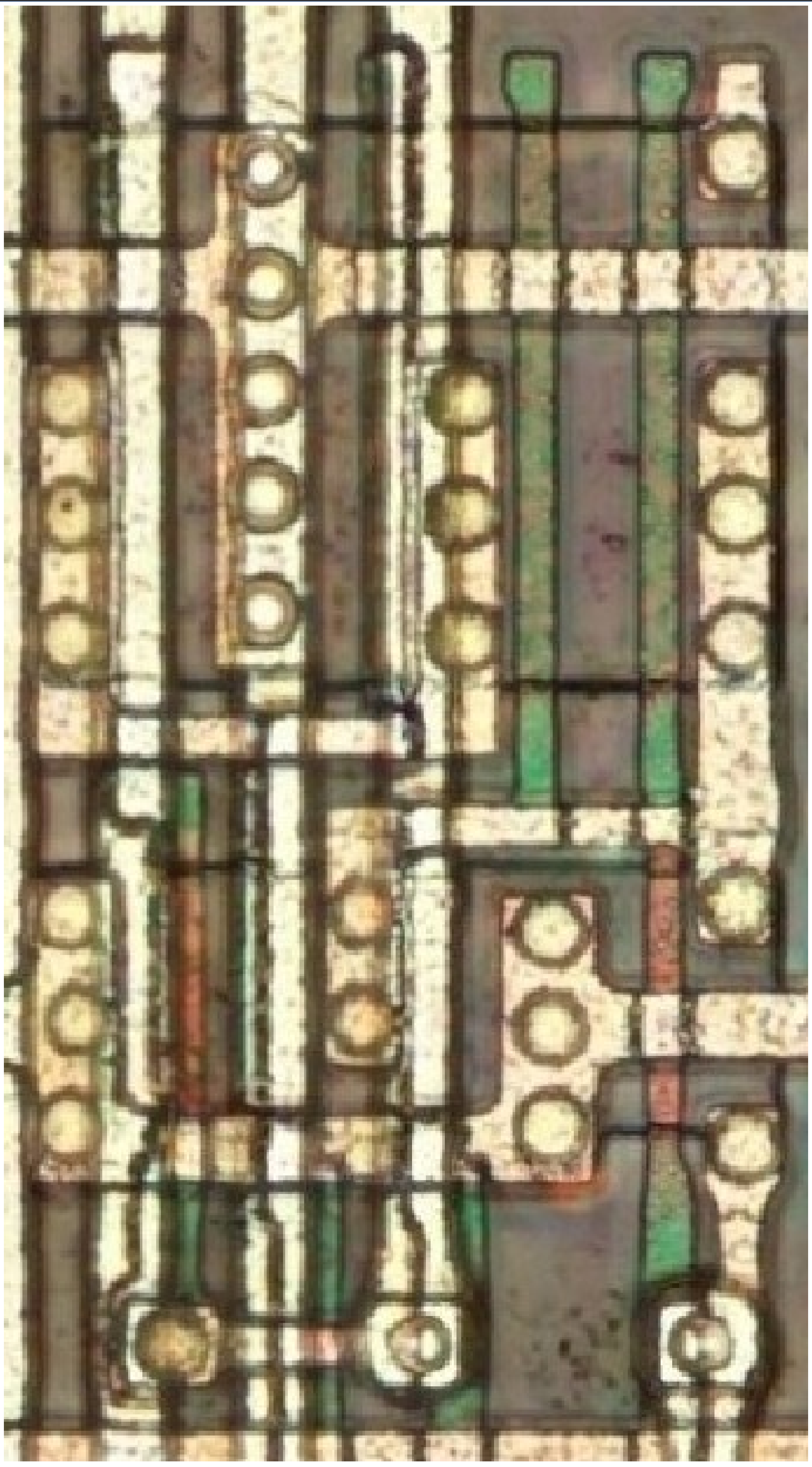




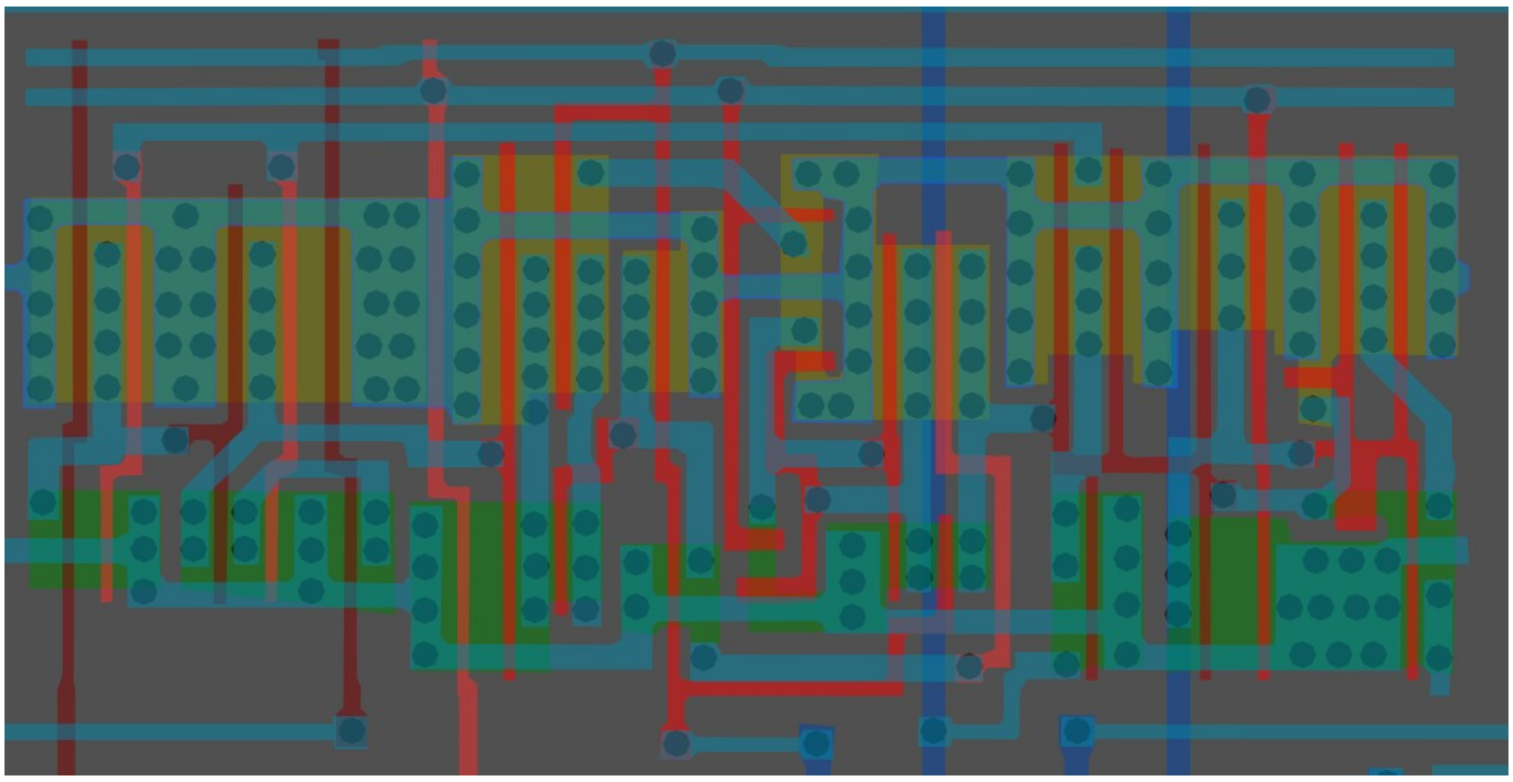
#668573ff

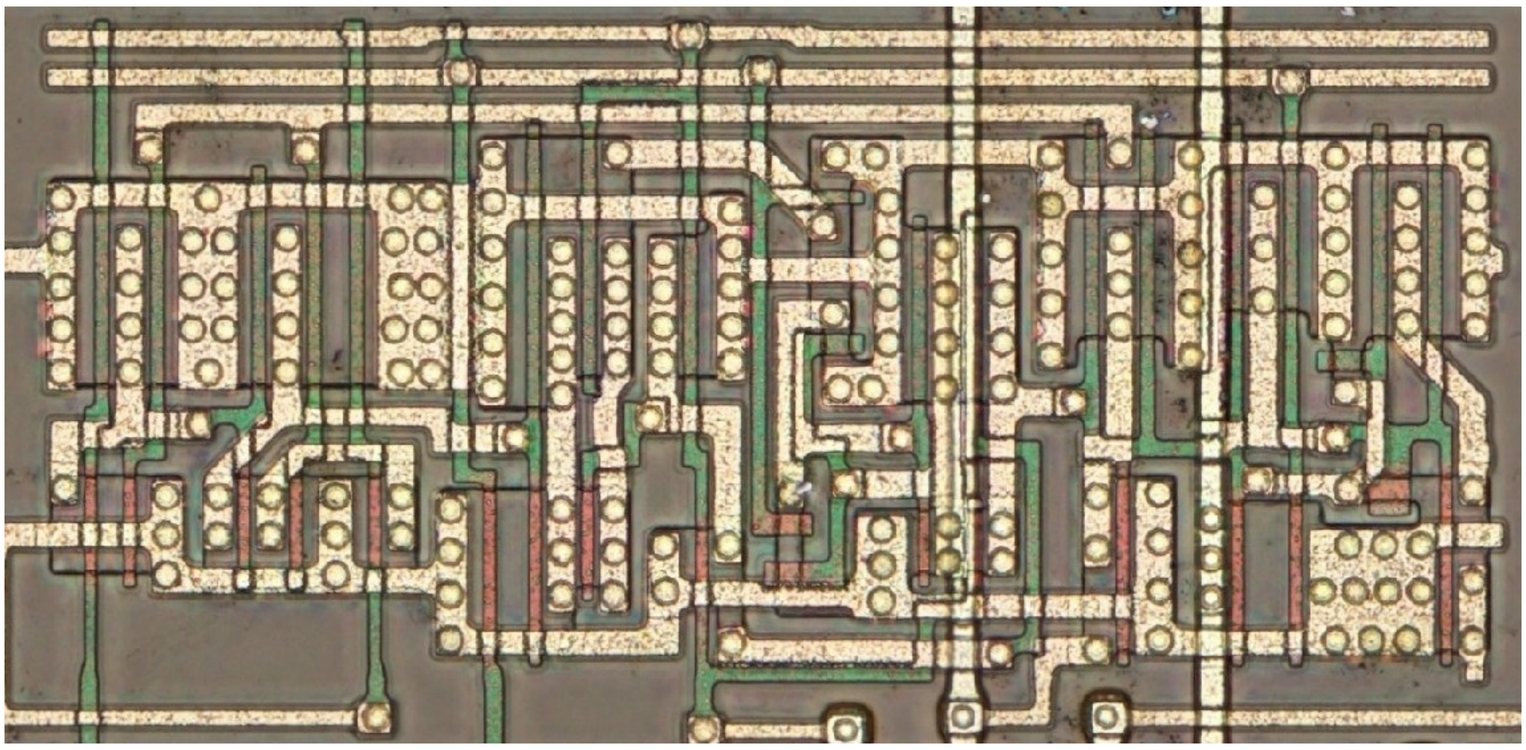
Cell type 27d



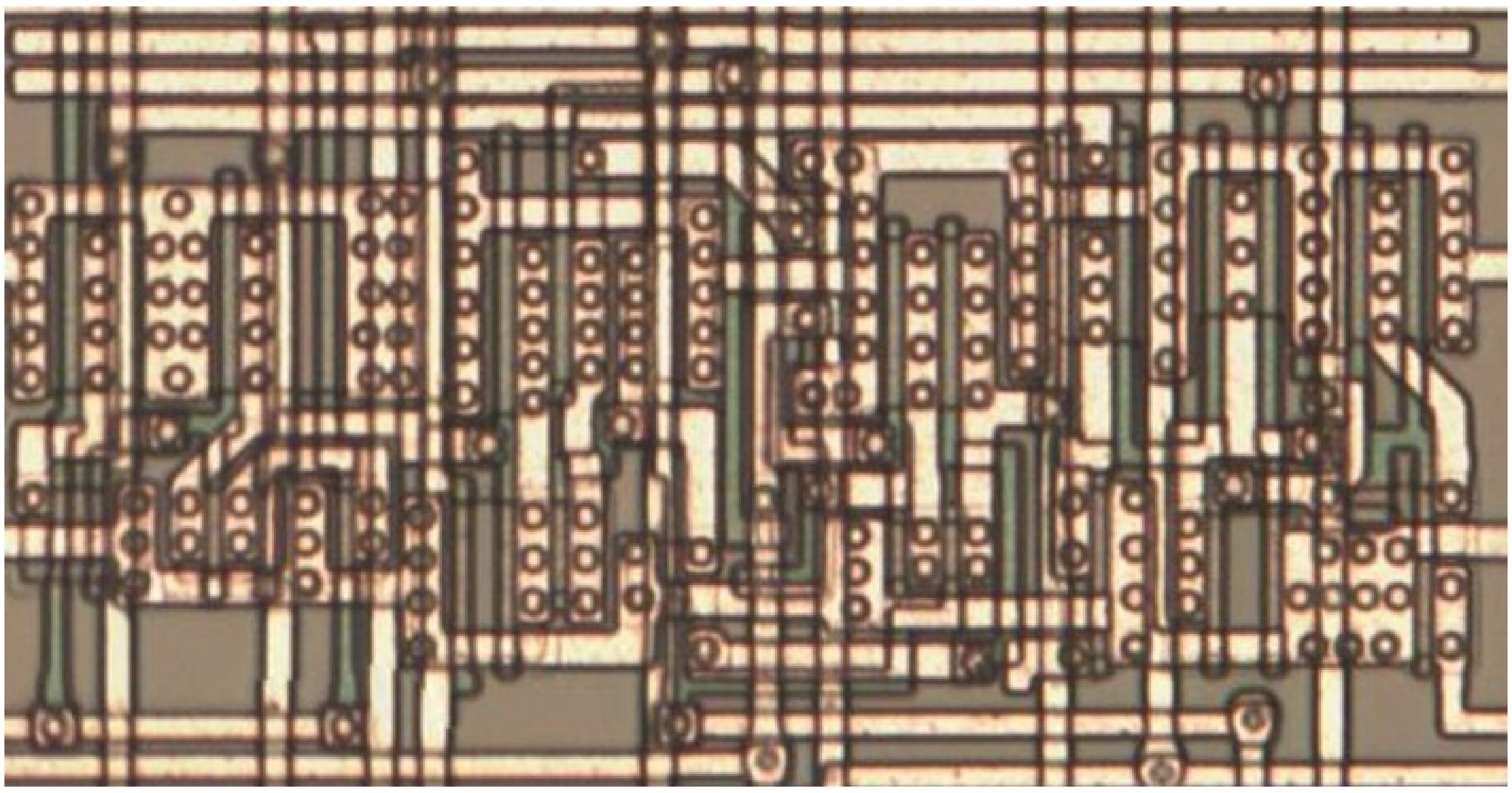


#00b75aff Cell type 28





Same cell on LA32 (MB87136A)

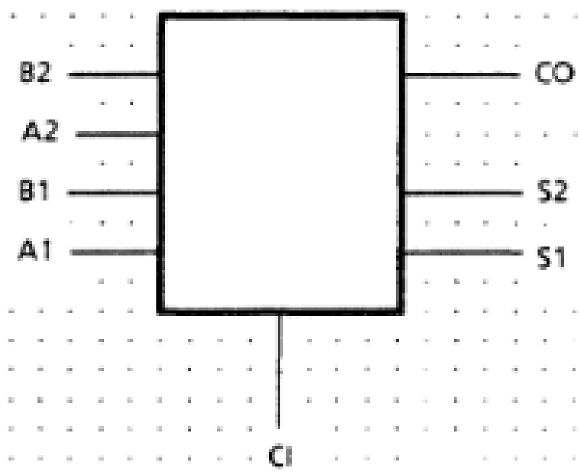


#d5546dff

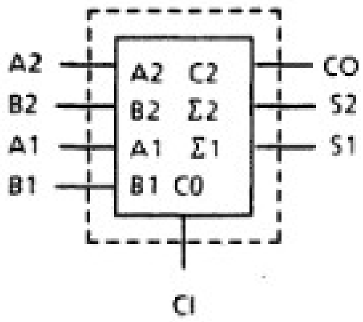
2-bit Full Adder

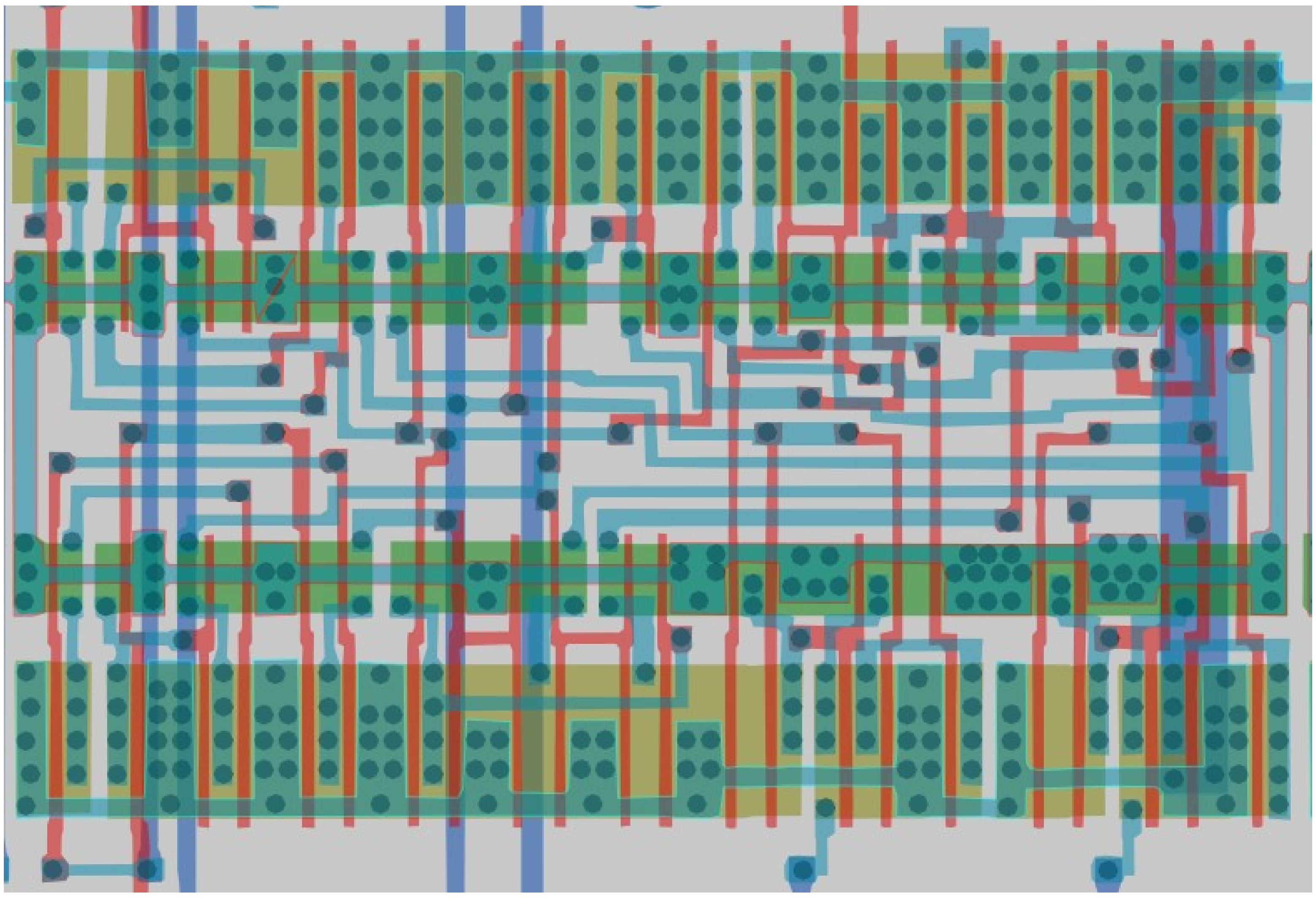
Cell 29

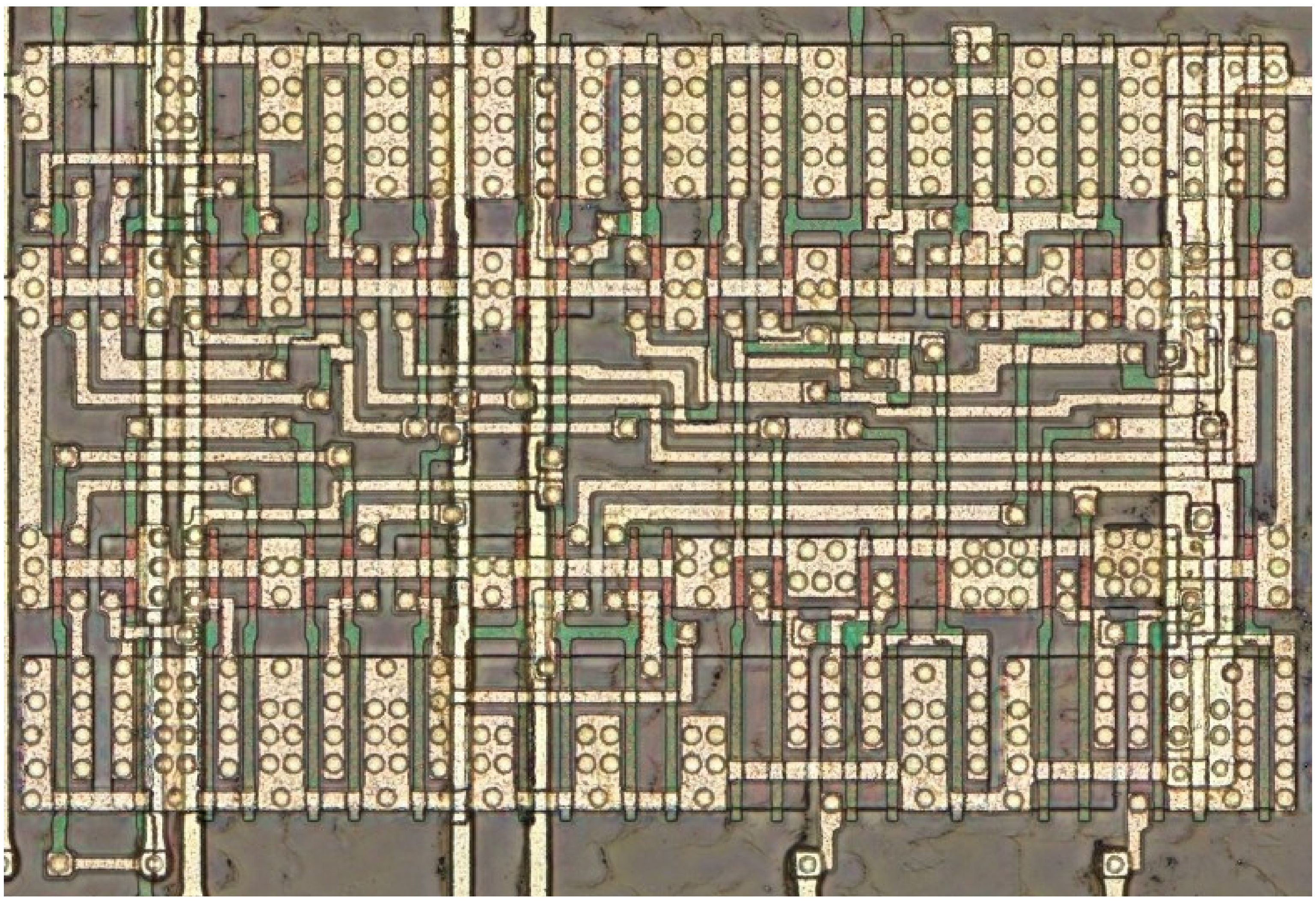
Inputs				Outputs					
				CI = L			CI = H		
A1	B1	A2	B2	S1	S2	$\overline{CO}$	S1	S2	$\overline{CO}$
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H



7482







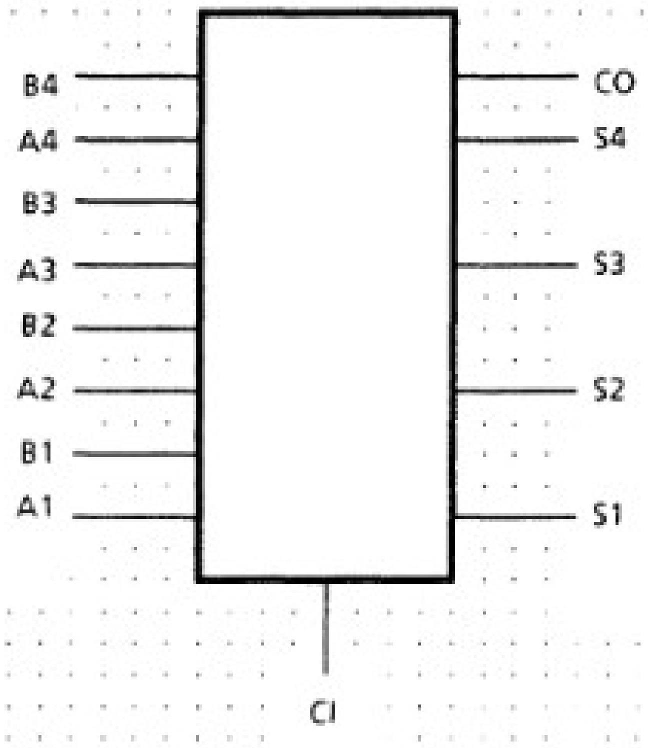
#a9546dff

4-bit Full Adder

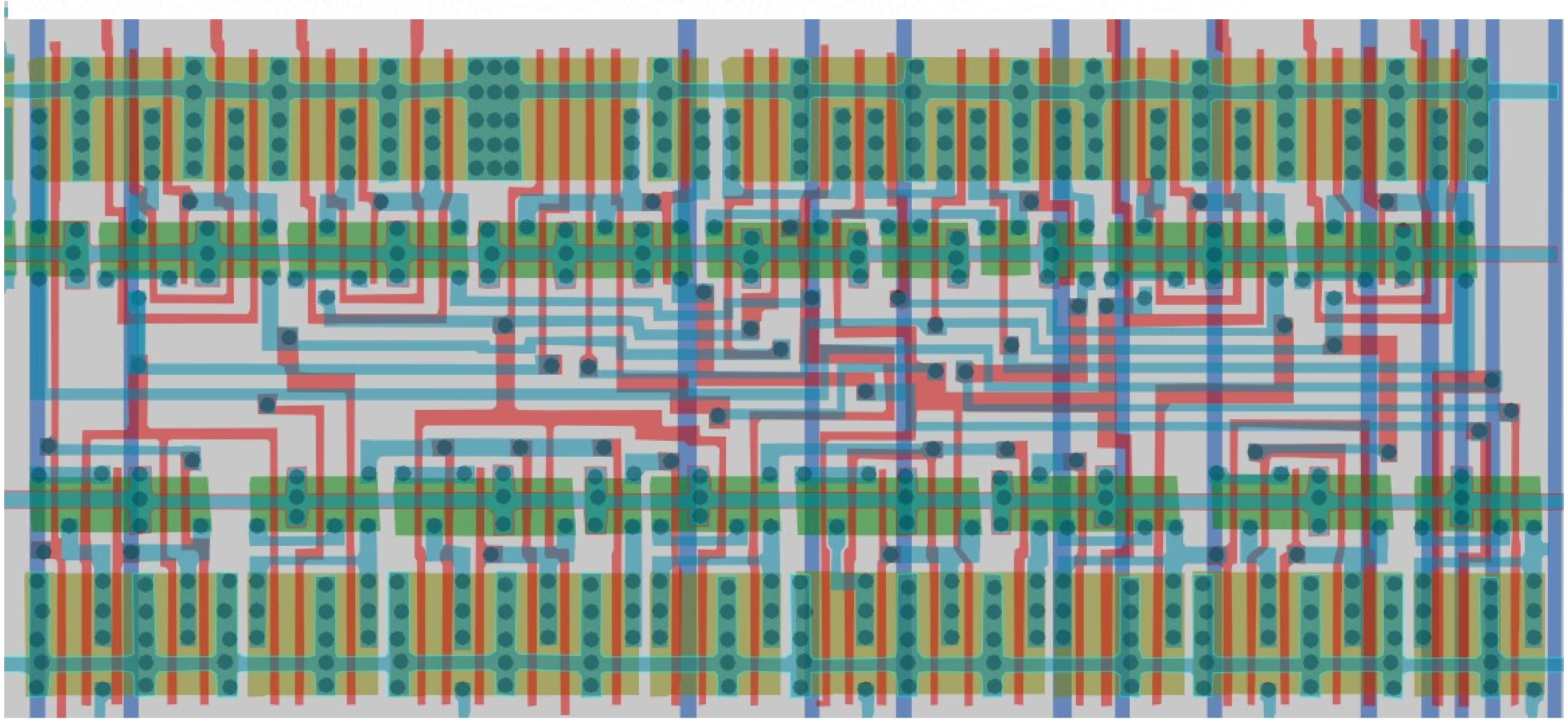
Cell 30

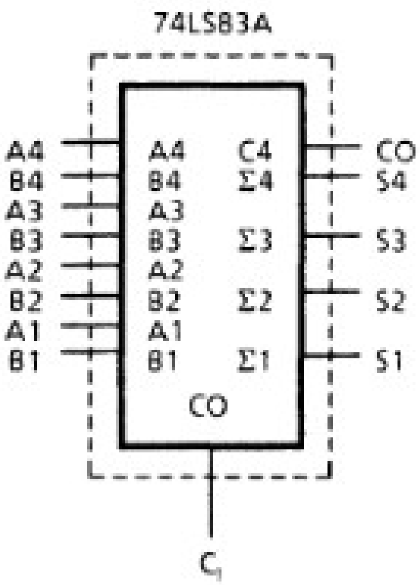
Standard cell only found in adder section of chip (as of now)

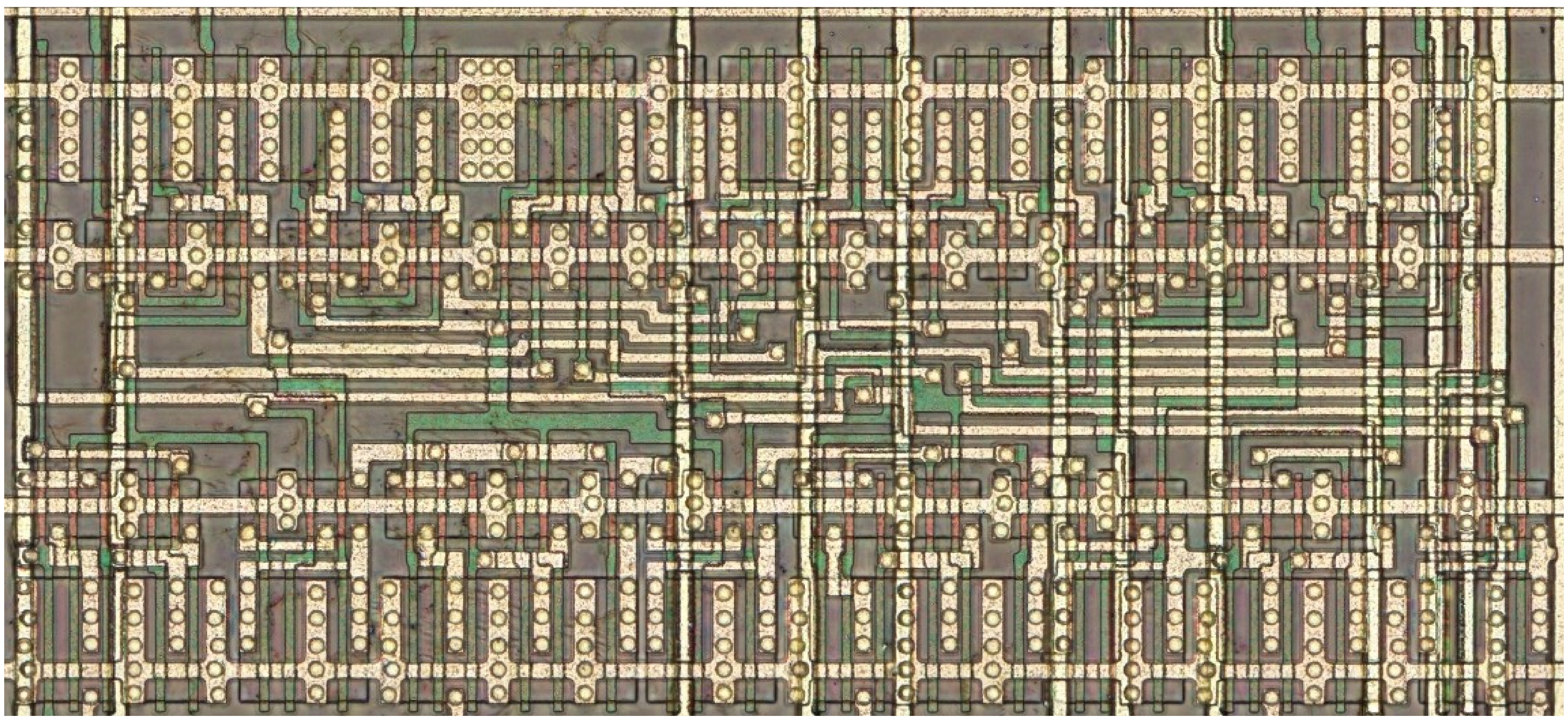
Inputs				Outputs					
				$C_I = L$ / $C_2 = L$			$C_I = H$ / $C_2 = H$		
A1 / A3	B1 / B3	A2 / A4	B2 / B4	S1 / S3	S2 / S4	C2 / CO	S1 / S3	S2 / S4	C2 / CO
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H



Note:
Input conditions at A1, B1, B2 and CI are used to determine outputs S1 and S2 and the value of the internal carry C2. The values at C2, A3, B3, A4 and B4 are then used to determine outputs S3, S4, and CO.

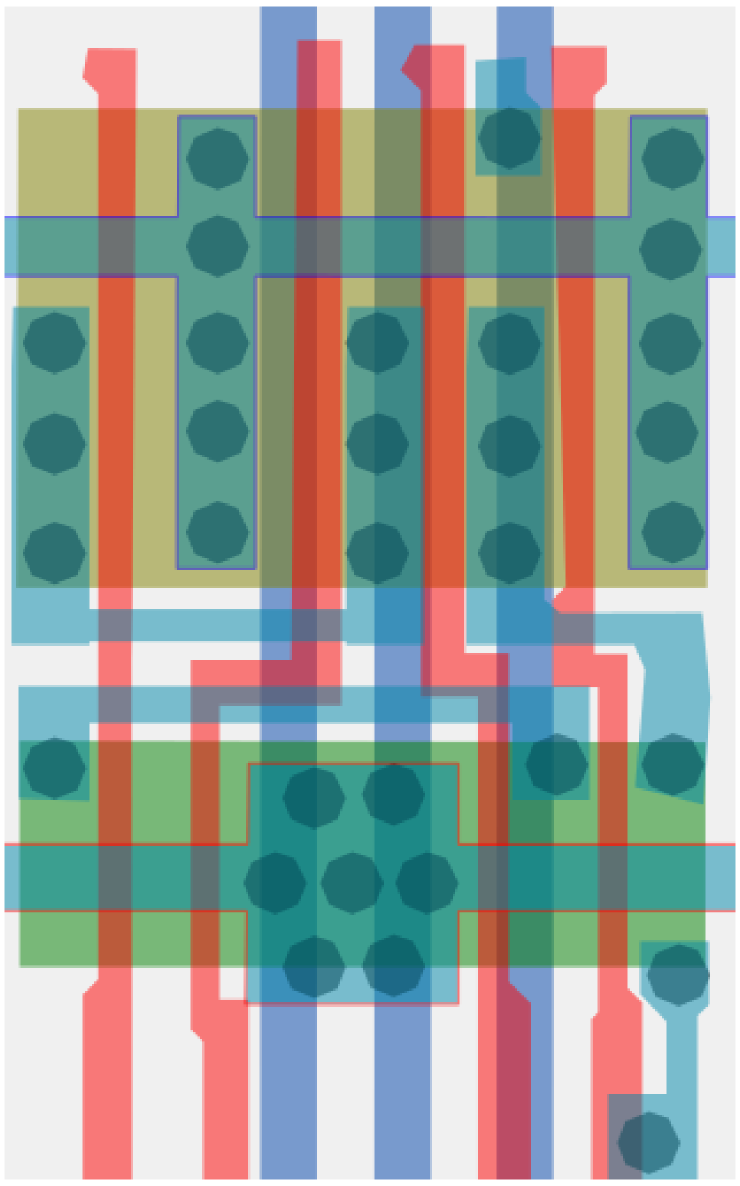


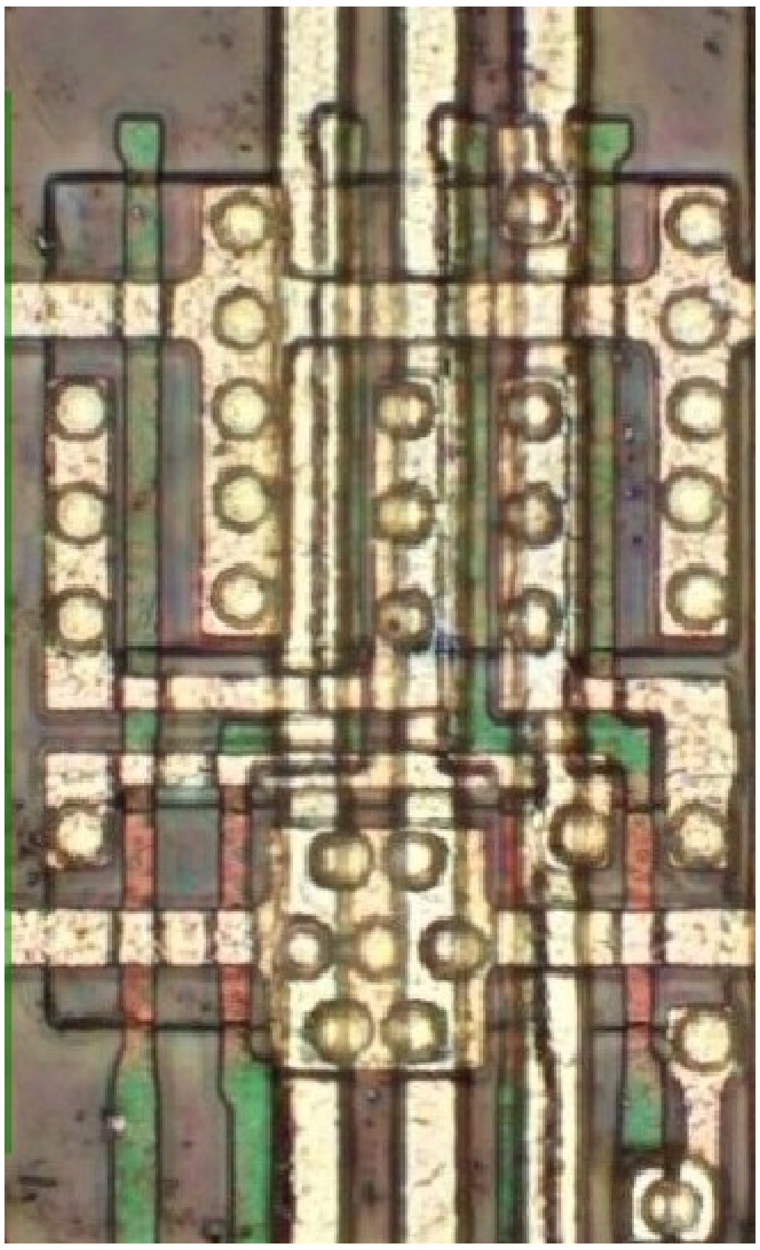




#ff4af8ff

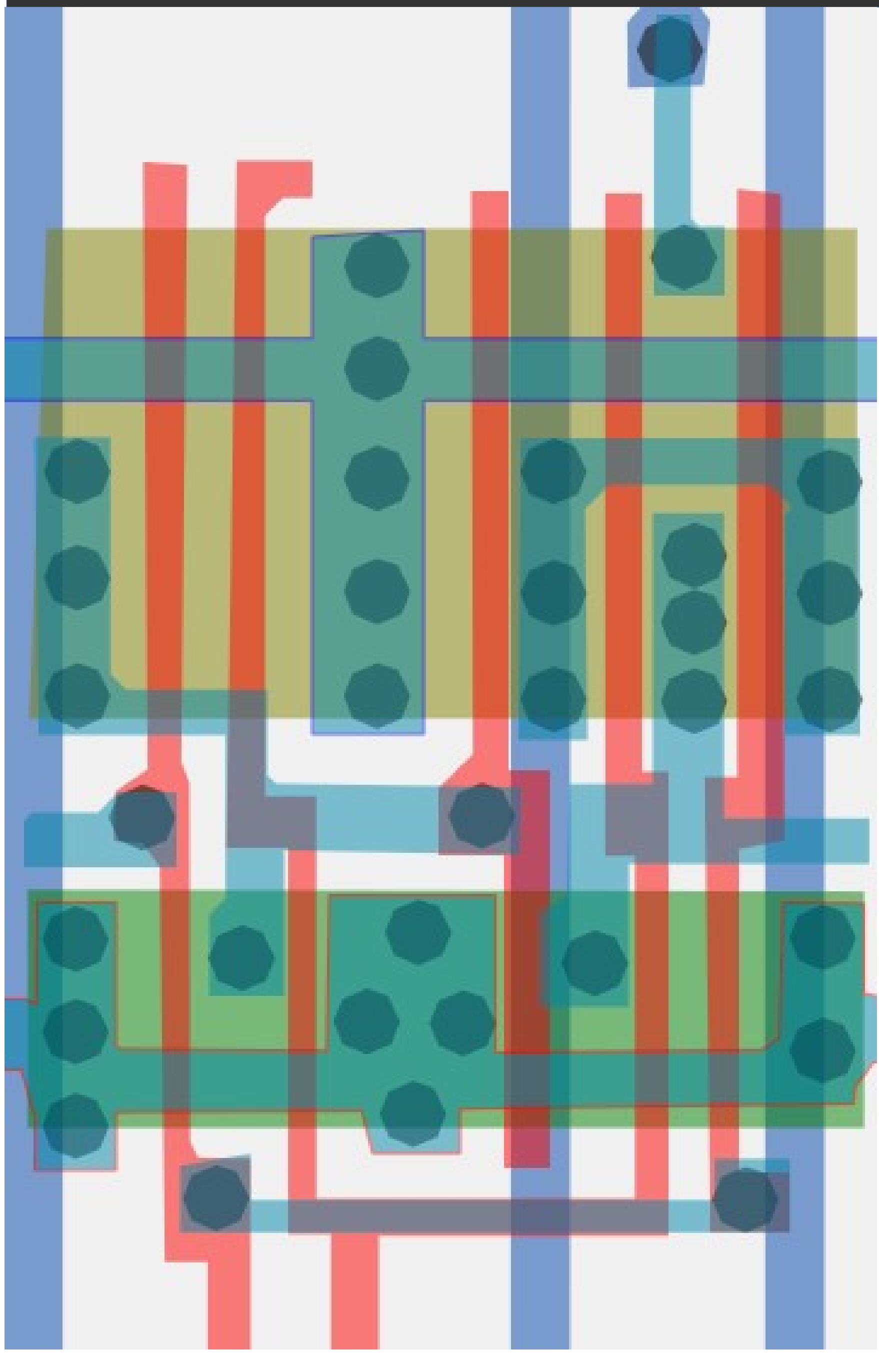
Cell type 31

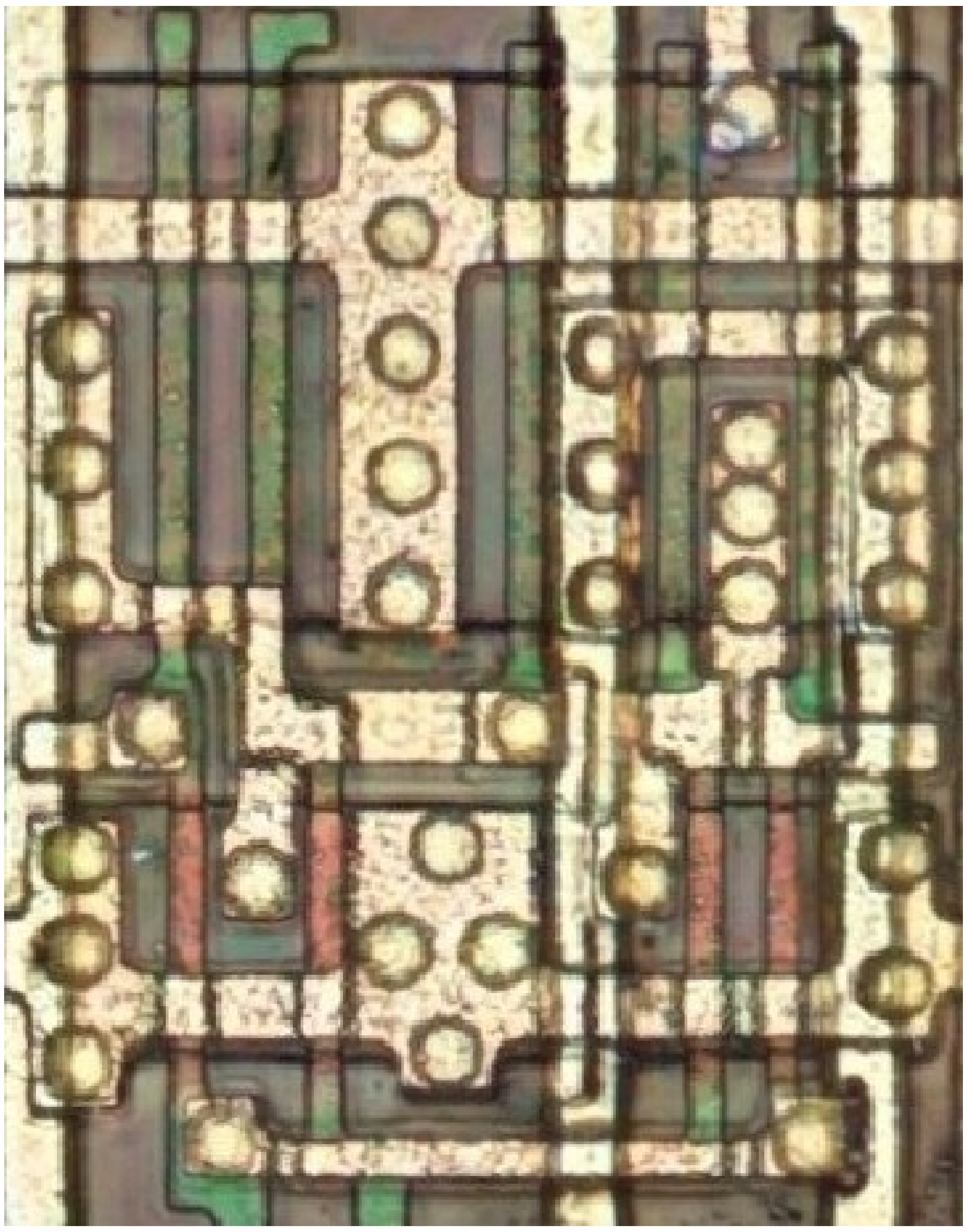




#90a05aff

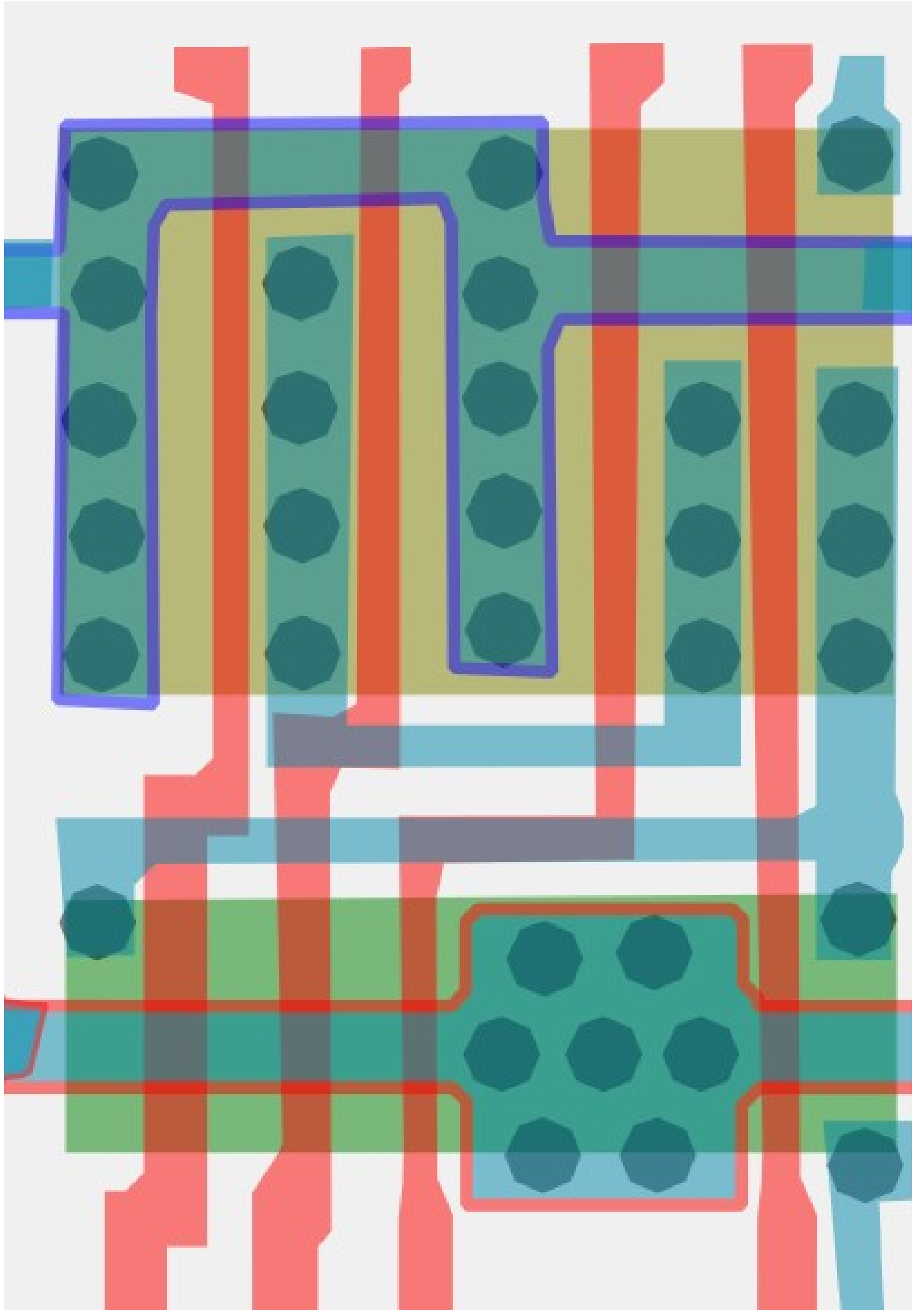
Cell type 32

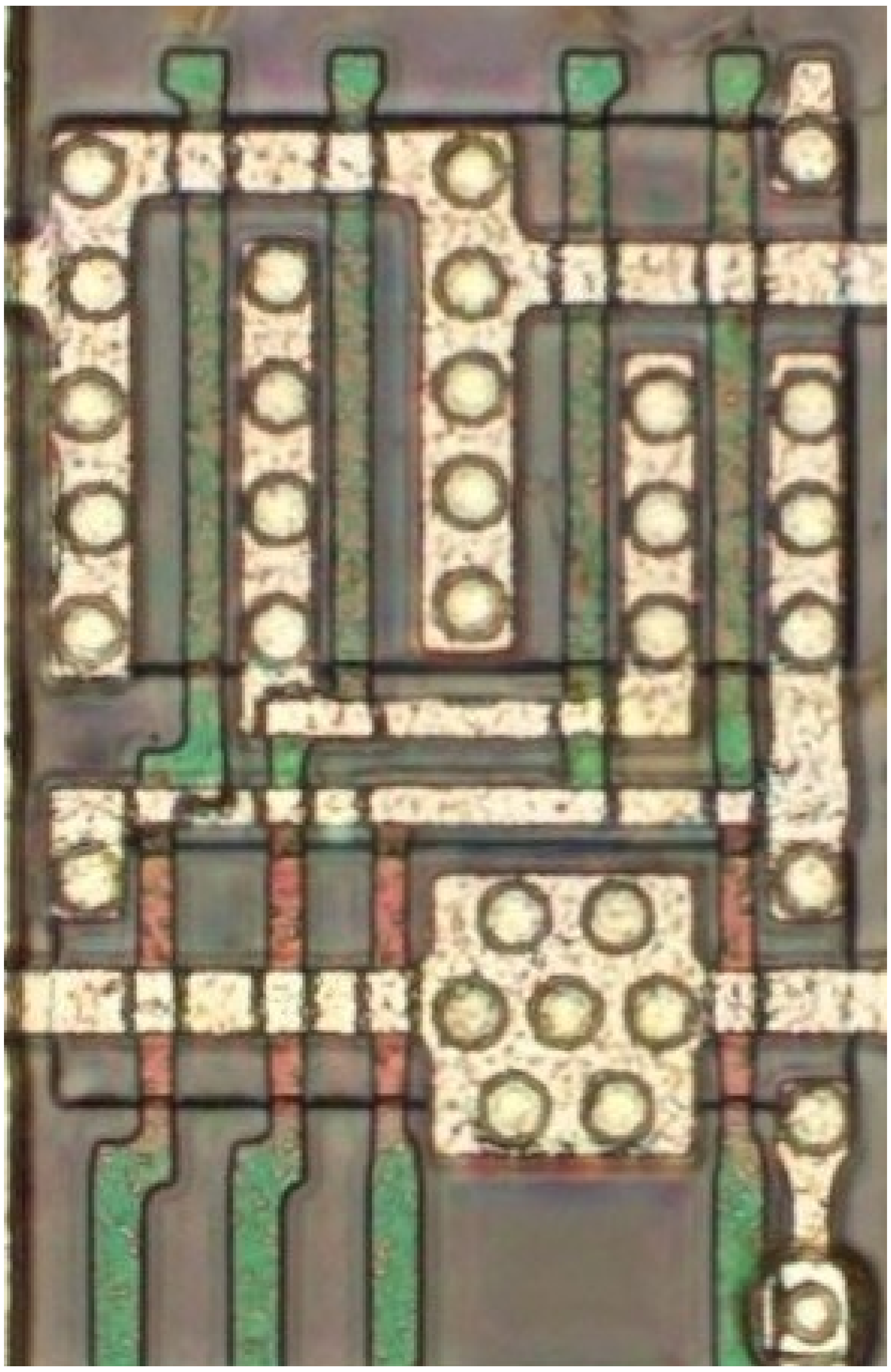




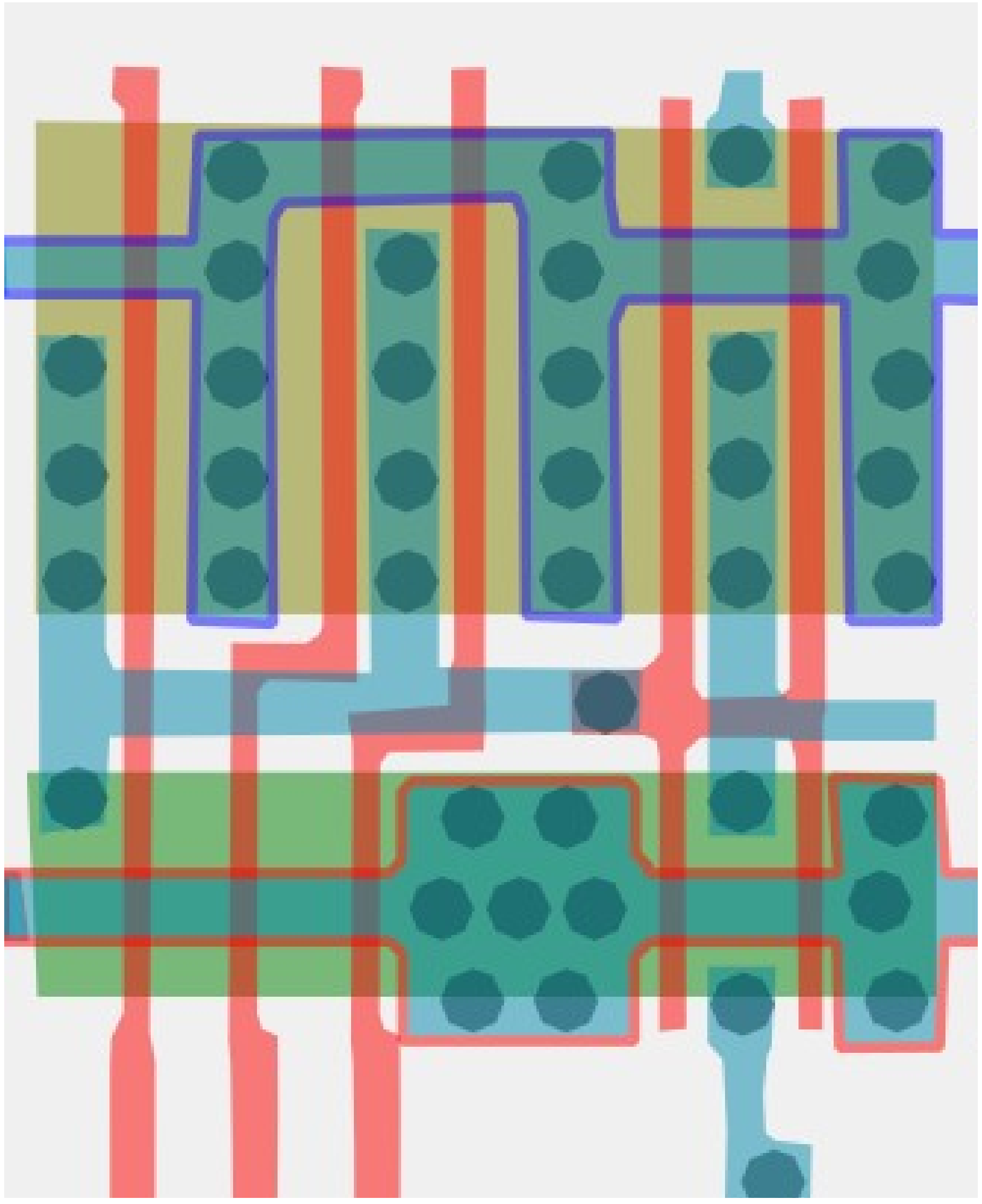
#57f330ff

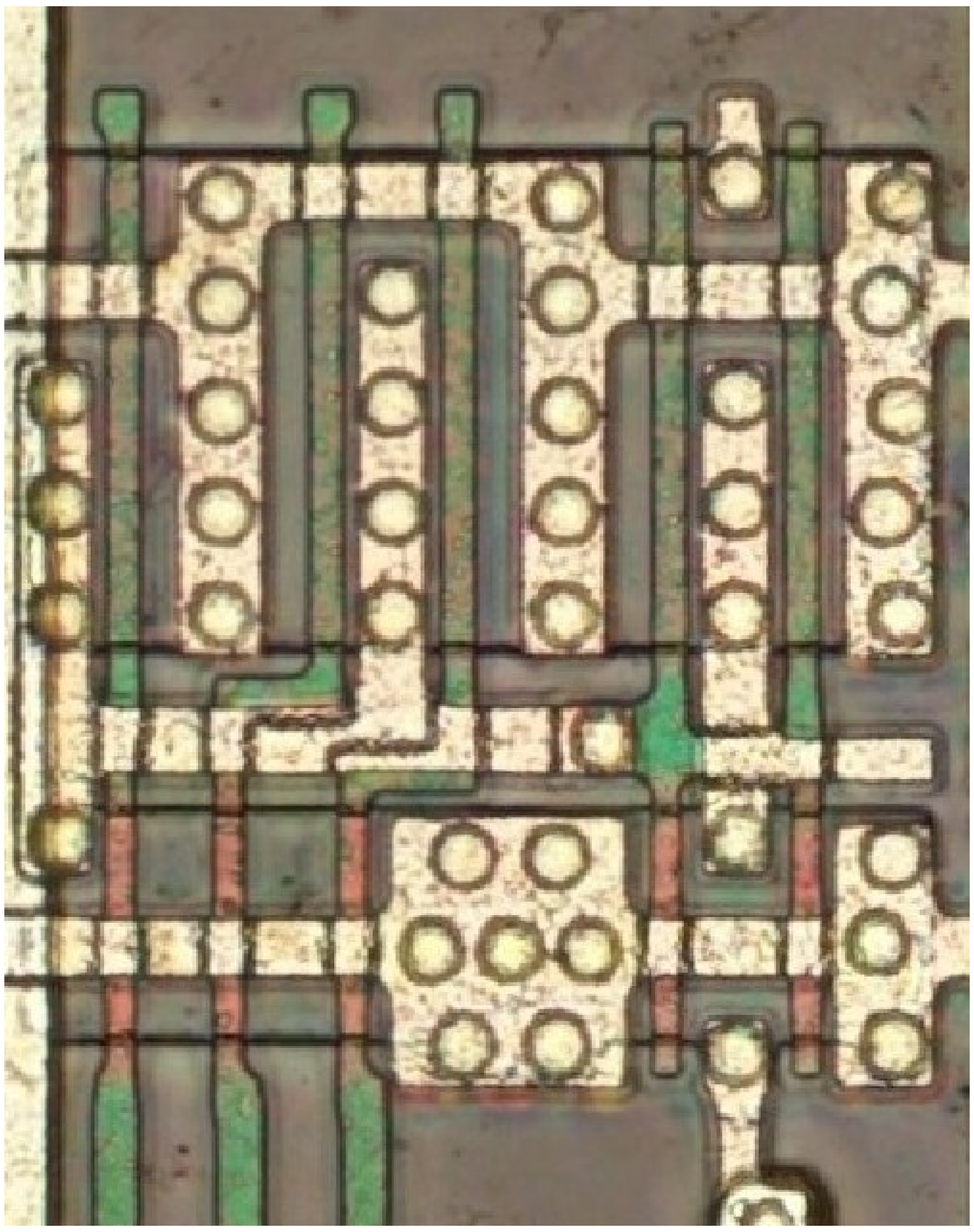
Cell type 33a



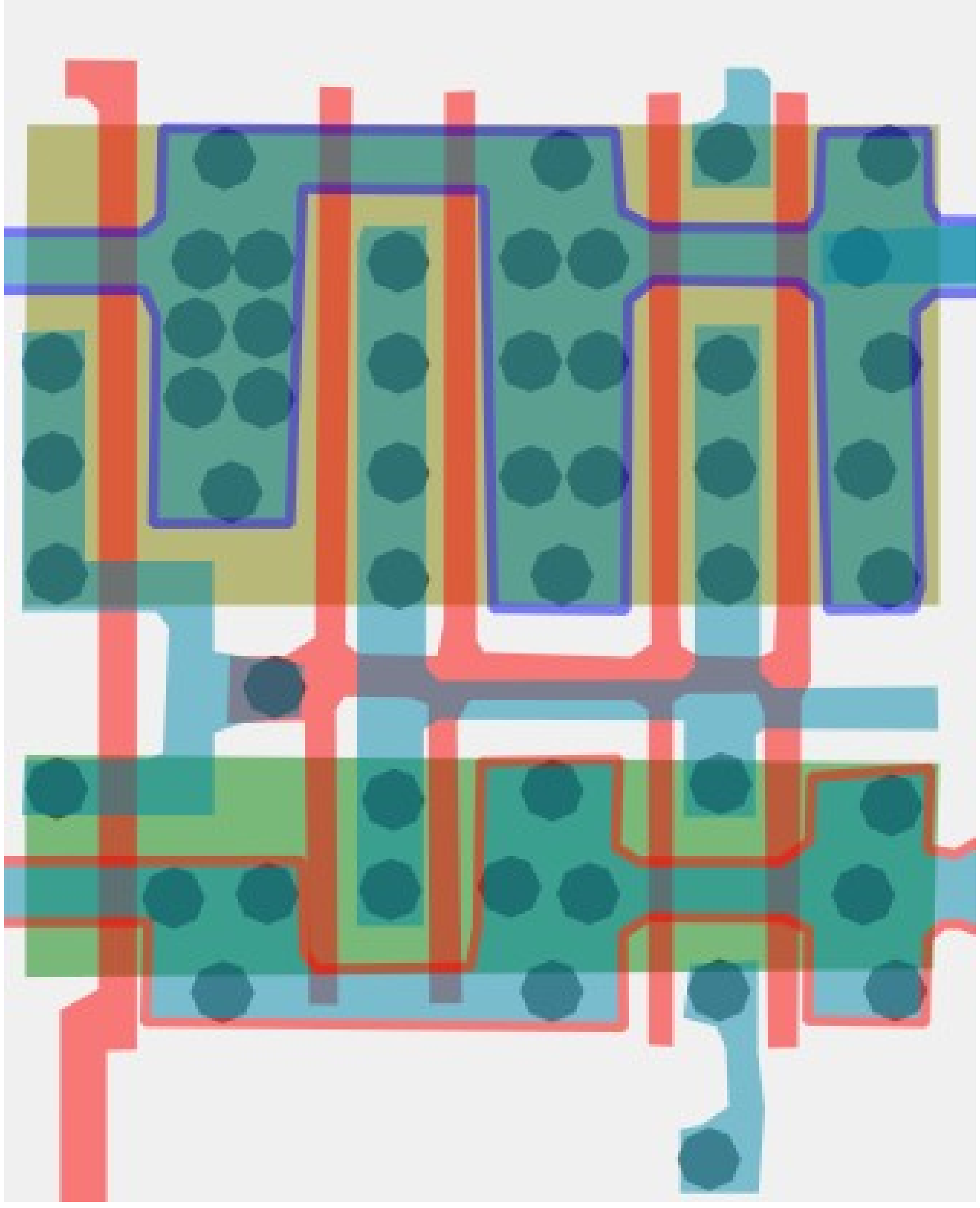


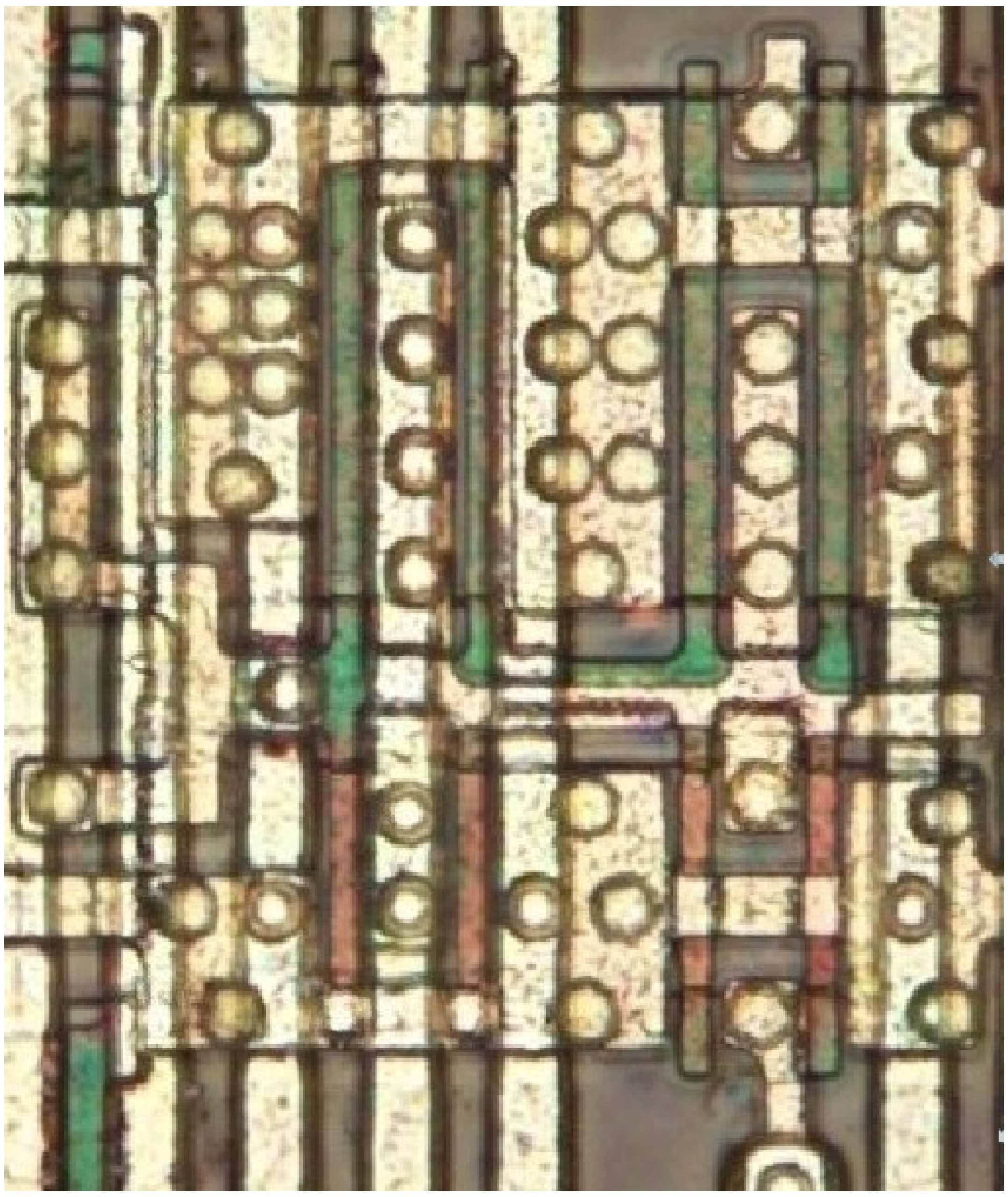
#77bd31ff Cell type 33b





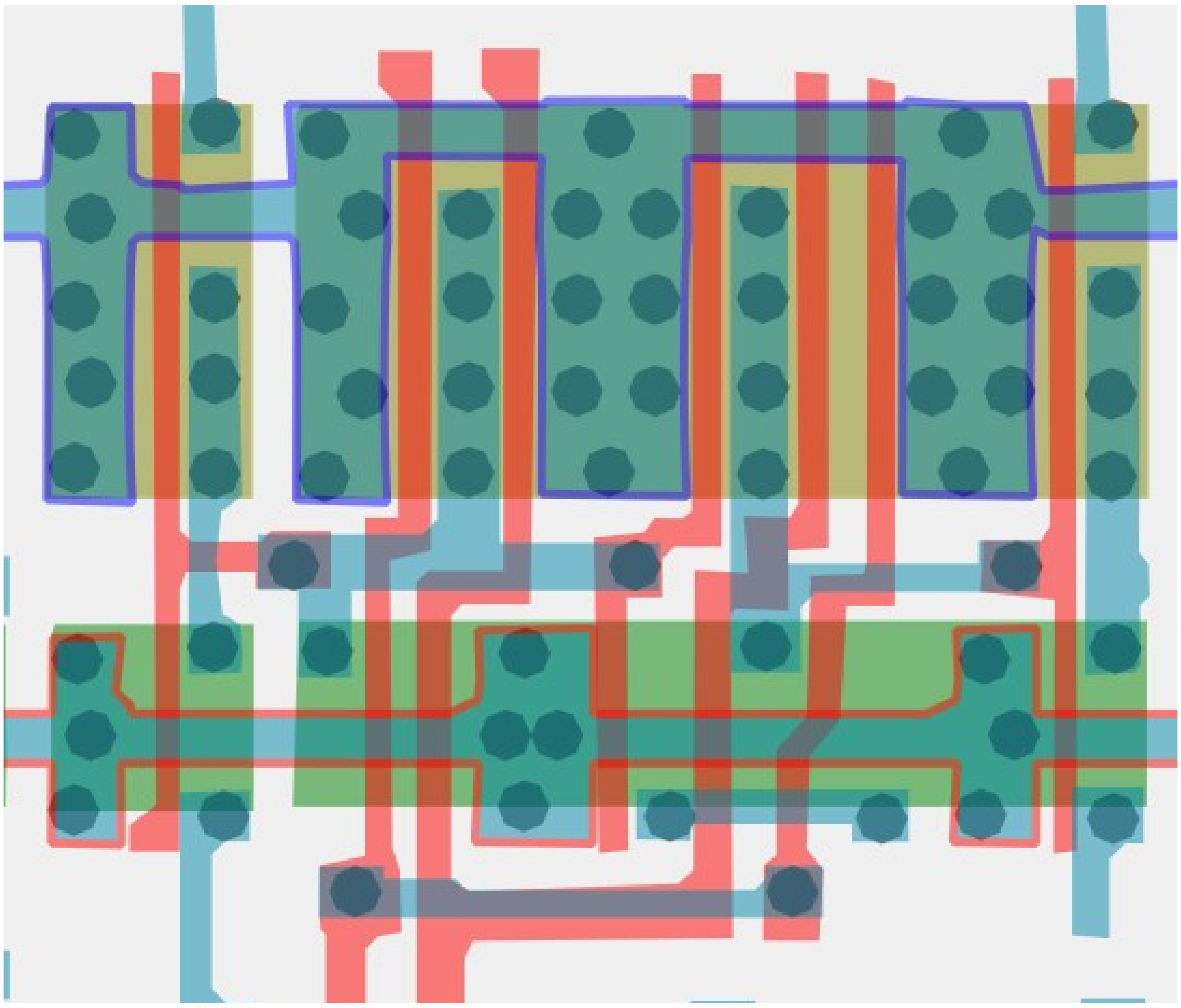
#5e58adff Cell type 34

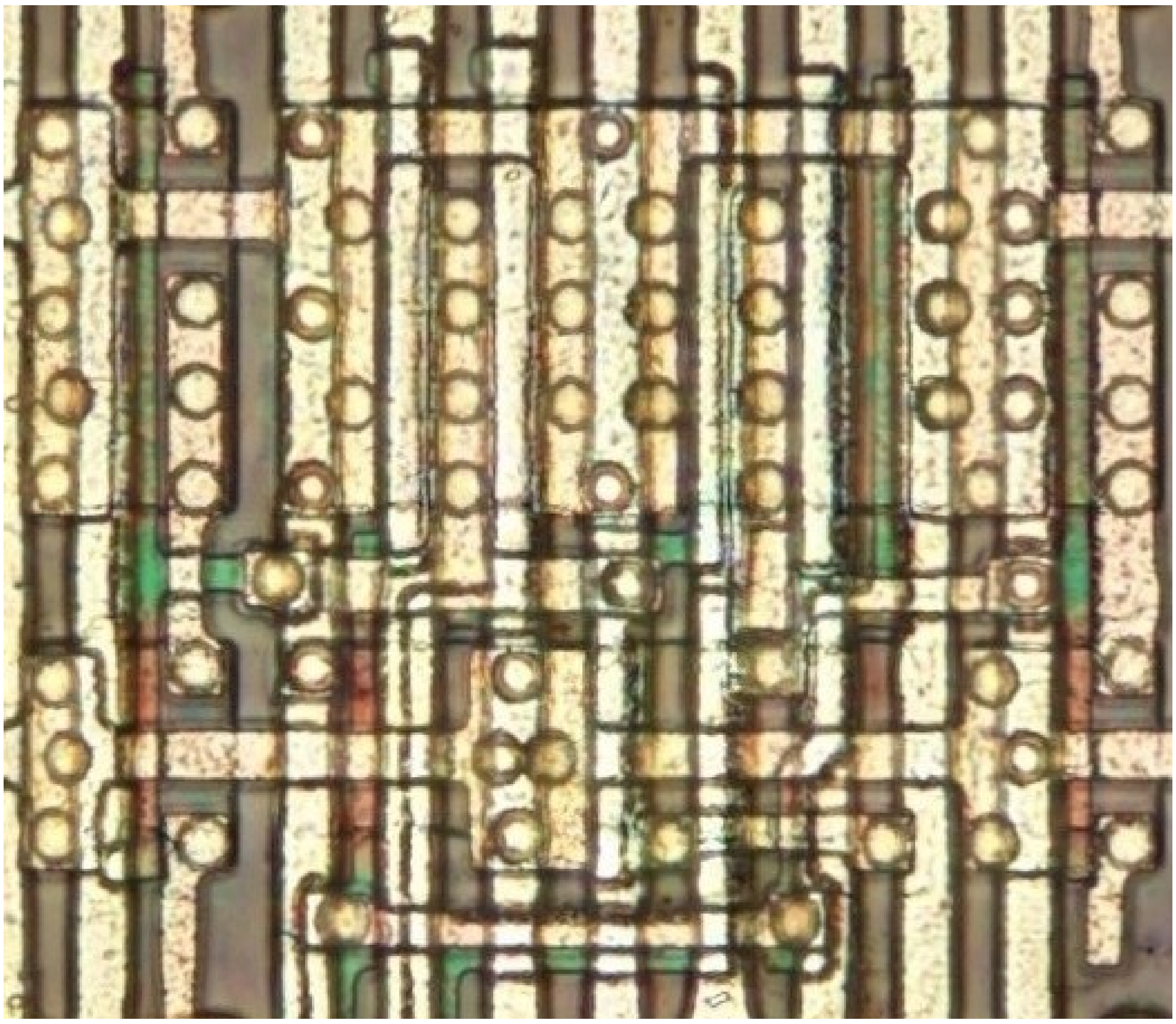




#8581c6ff

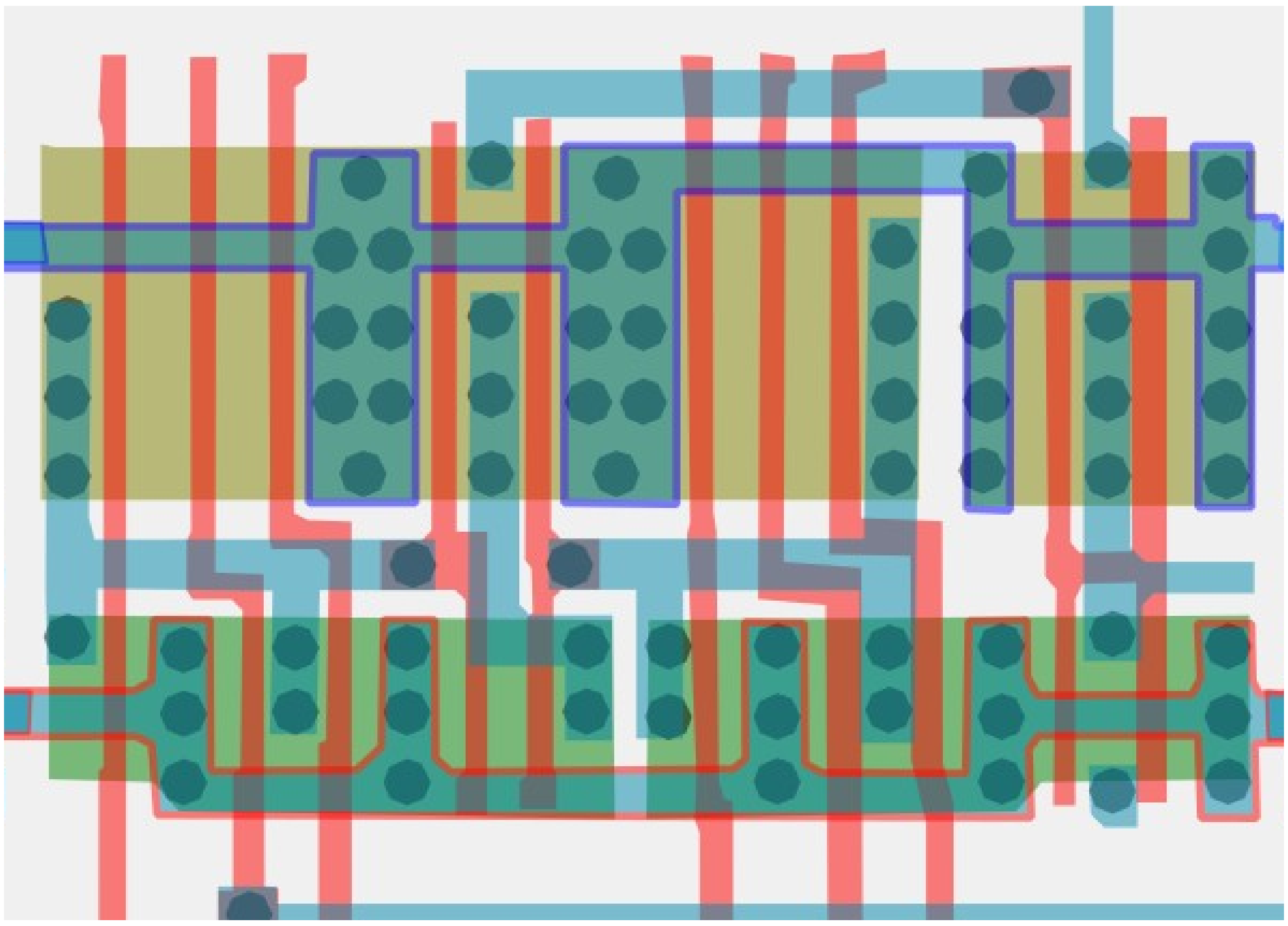
Cell type 35

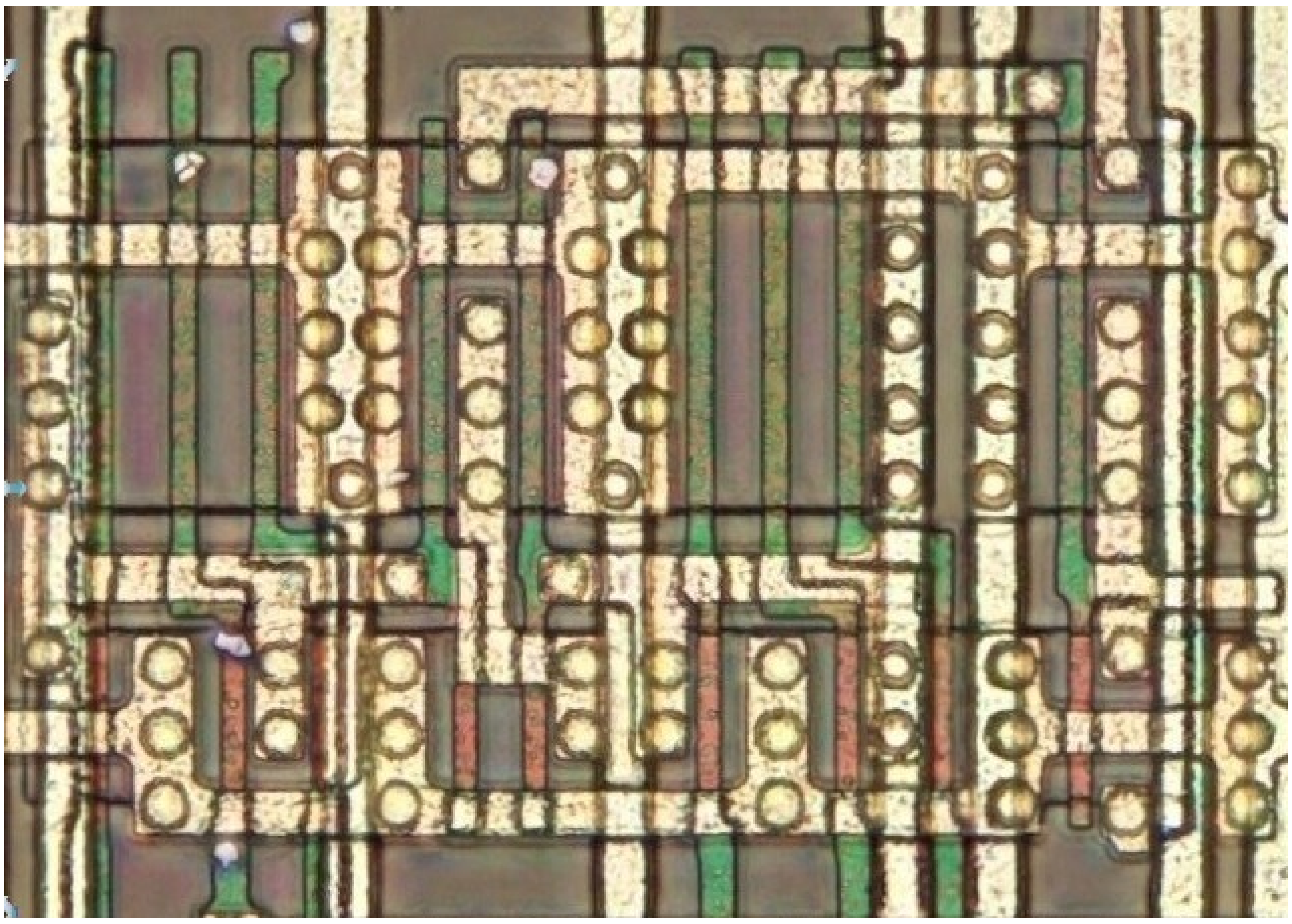




#a09dd4ff

Cell type 36a





#b0aedbff Cell type 36b